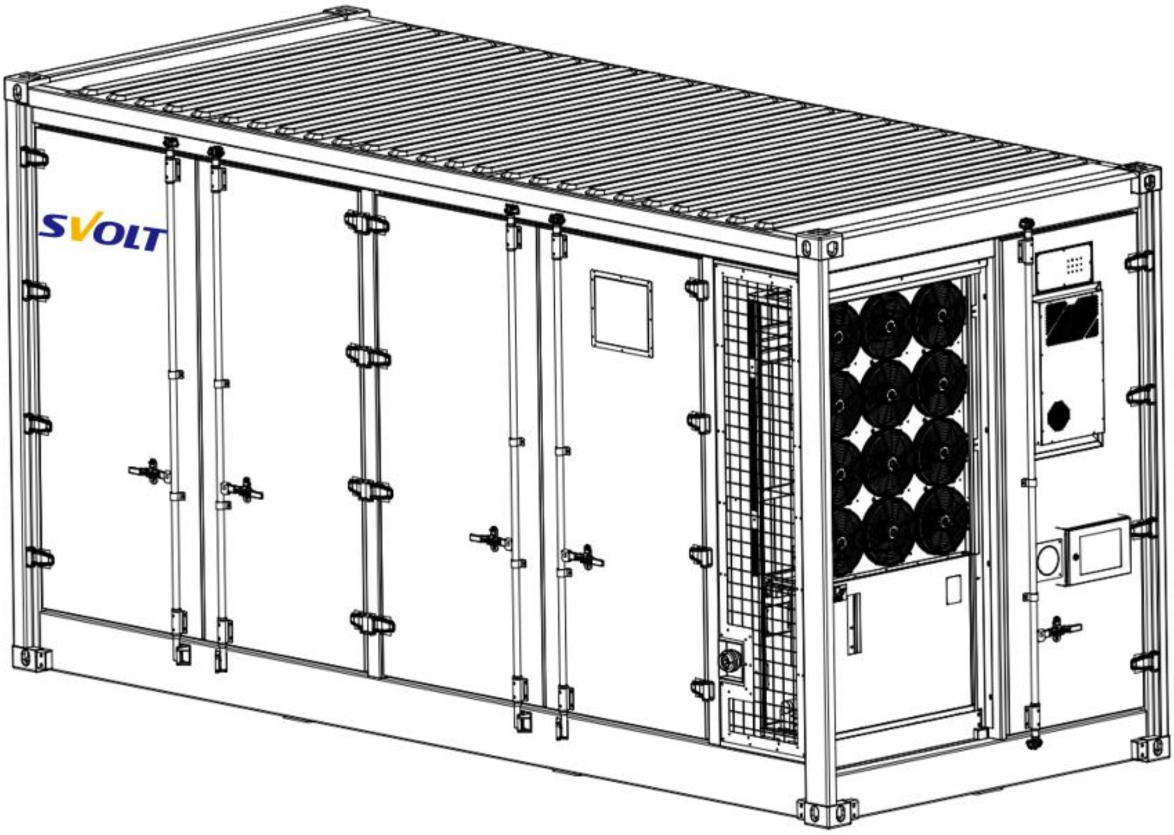


CE-L-2580/5160-B

Liquid-cooled energy storage container system

User Manual

Version number：V2.1.1

Release date：2025-12-10

SVOLT Energy Storage

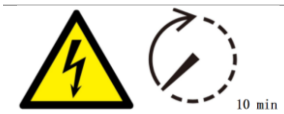
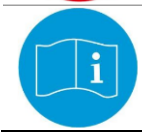
Catalogue

1. Safety Instructions

1.1 Safety Symbol Description

Safety symbols are used to indicate the safety matters that operators should observe when installing, operating and maintaining equipment. Common identifiers are as follows:

Table 1-1 Safety Symbols

Safety symbol	Meaning
	**Dangerous Voltage Warning:** Warn of the presence of high voltage, which may cause personal injury or equipment damage.
	**General Warning:** Warn of non-electrical factors that may cause personal injury or equipment damage.
	**Warning of electrostatic sensitive equipment:** Warning of electrostatic discharge that can cause equipment damage.
	**Grounding:** Grounding mark outside the machine, which needs to be firmly grounded to ensure the safety of operators.
	**Delayed discharge warning:** Do not touch them for 10 minutes after disconnecting the power supply.
	**Garbage disposal**: Do not treat this product as household garbage.
	**Prevent open flames:** Keep this product away from open flames, and no combustibles should be placed nearby.
	**Read the manual:** Please read the instructions before performing any operation on the product.

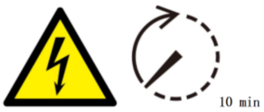
1.2 Safety Precautions

Read the manual!

- Operators should carefully read the relevant precautions and operation instructions before operating any power equipment, and strictly follow the safety instructions to avoid accidents. The safety precautions in this manual do not represent all the safety precautions to be observed, but only serve as a supplement to the safety precautions in various operations.
- Operators should operate according to the requirements of user manual when installing, operating and maintaining equipment. Power supply equipment can only be repaired by professional maintenance personnel. Our company shall not be liable for any consequence caused by violation of safe operation requirements or violation of safety standards for design, production and use of equipment.
- If you can't find effective treatment measures in the instructions, please contact your supplier to provide technical support. Any operation mode not provided by our company will not be recognized.

High pressure danger!

- High-voltage power supply provides power for the operation of equipment, and direct contact or indirect contact with high-voltage power supply or grid through wet objects will bring fatal danger.
- Special tools must be used during high voltage and alternating current operation, and ordinary or self-carried non-special tools must not be used.
- Before installing and removing the power cable, the power switch must be disconnected. It is dangerous to remove or install the power cable live!
- Voltage can be generated on the battery side or the grid side. Always use a standard voltmeter to ensure there is no voltage before touching.
- Before connecting the cable, it must be confirmed that the cable and cable identification are consistent with the actual installation before connecting.



Delayed operation!

- When the power of the energy storage system is disconnected, there may still be electricity in the battery. Wait 10 minutes before operation to ensure that the equipment is completely voltage-free.

Pay attention to operation safety!

- When hoisting heavy objects, it is strictly forbidden to walk under the boom and hoisting objects.
- Hoisting tools can only be used after passing the inspection.
- Hoisting operation personnel must undergo strict training, master correct operation methods, understand various safety precautions, and have hoisting operation qualifications.

Be careful of static electricity!

- Static electricity generated by human body will damage electrostatic sensitive components on circuit boards. Before touching equipment (such as flashboard, circuit board, IC chip, etc.), in order to prevent human static electricity from damaging sensitive components, an anti-static bracelet must be worn and the other end of the anti-static bracelet must be well grounded.

1.3 Operator Requirements

The hoisting, transportation, installation, wiring, operation and maintenance of the energy storage system must be carried out by professional qualified engineers according to local regulations. Qualified personnel should have suitable personal protective equipment on site to protect personal safety and meet the requirements of local engineering laws and regulations. Personal protective equipment is shown below.

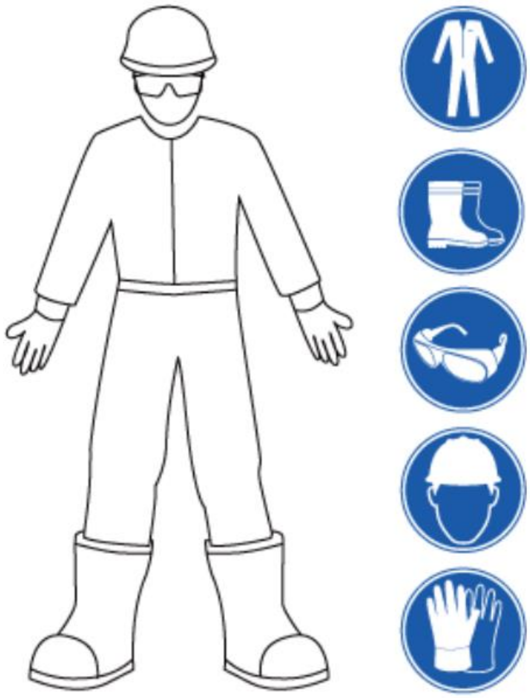


Figure 1-1 Schematic of personal protective equipment

Professional qualified engineers should have the skills and knowledge related to the construction, operation and installation of energy storage systems, have received safety training and are qualified, and can identify hazards and avoid related risks. The following requirements must be met:

- Familiar with the composition and working principle of the whole energy storage system, and operate according to the manual description.
- Understand the relevant knowledge of power, electronics, electrical wiring and mechanical expertise, and be familiar with mechanical and electrical schematics.
- Received professional training related to electrical equipment installation, commissioning and trial operation.
- Able to quickly respond to dangerous or emergency situations during installation and commissioning. Have the ability to respond urgently to dangers or emergencies.

- Familiar with the relevant standards and norms of the country/region where the project is located, and abide by the laws, regulations and relevant standards of the country/region when carrying out transportation, transit, installation, wiring and maintenance.
- Avoid using skills and techniques that pose an electric shock hazard.
- During installation, operation and maintenance, it is strictly forbidden to wear watches, bracelets, bracelets, rings, necklaces and other easily conductive objects to avoid being injured by electricity.

2. Overview

2.1 Scope of application

This manual is applicable to the liquid-cooled energy storage container system independently developed by SVOLT Energy Technology Co., Ltd. (hereinafter referred to as SVOLT). The manual is mainly prepared for the operators of battery energy storage system, and stipulates the transportation, installation, wiring, debugging and other operation contents of this product. The information contained in this manual should be fully familiar to plant personnel, whether trained in the operation of energy storage systems or not.

SVOLT has the final right to interpret the contents of this document.

2.2 Product Description

This energy storage system is suitable for intelligent buildings, industrial parks, wind farms, large and medium-sized energy storage power plants, etc. It can effectively complete the functions of peak shaving and valley filling, load smoothing, standby power supply, etc., so as to improve the reliability of power supply, reduce power consumption cost and improve the power quality of power grid.

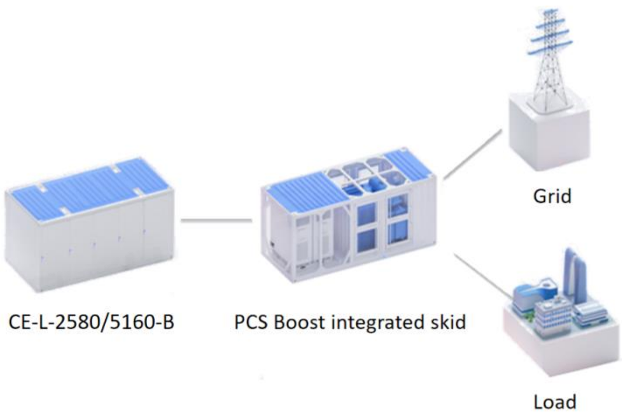


Figure 2-1 Typical application of energy storage system

This product is a 1500V high voltage DC side energy storage container. The battery adopts lithium iron phosphate 350Ah short-blade battery with high energy density and long cycle life, and is internally integrated with BMS control, fire fighting, lighting, thermal management and other auxiliary systems to realize the integration of energy storage system. The container adopts 20ft size(6058*2438*3100mm), which makes transportation more convenient. Through the modular design of PACK and external single-side door opening design, the installation, operation and maintenance of the system are better realized.

The system topology diagram is as follows:

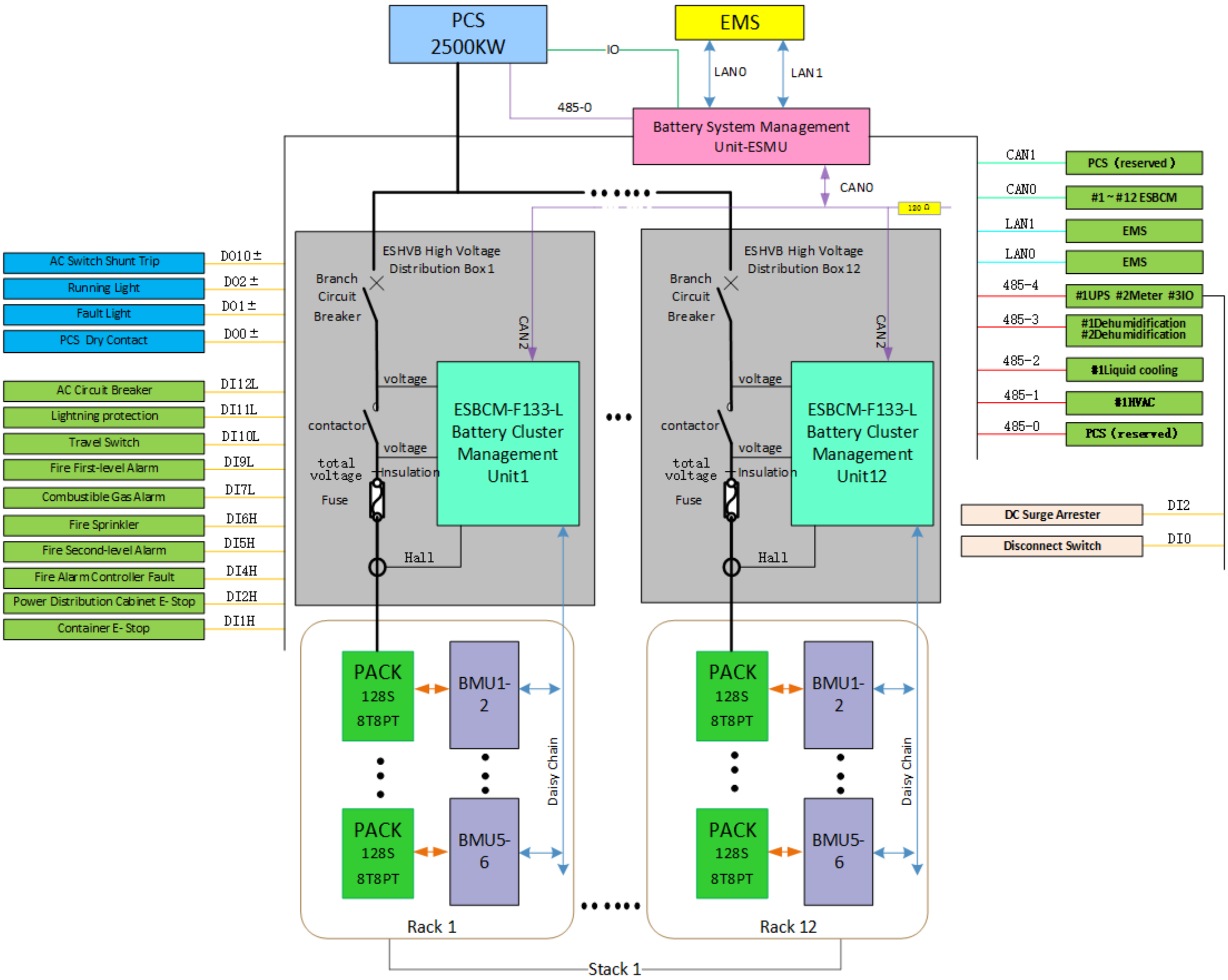


Figure 2-2 Topology of Energy Storage Container System

The main components of energy storage containers are as follows:

1. **Battery system:** The energy source of the energy storage system is used for the storage of electric energy, and the battery rack is output through the DC combiner cabinet;

2. **Distribution system:** Provide electric energy for electrical appliances in the energy storage system, including grid power and ups;

3. **Control system:** The control system composed of EMS, BMS and other control devices is used to control the normal operation and fault protection of the ESS;

4. **Thermal management system:** Used to ensure that the energy storage system works in a suitable temperature environment, including chiller and pipeline components;

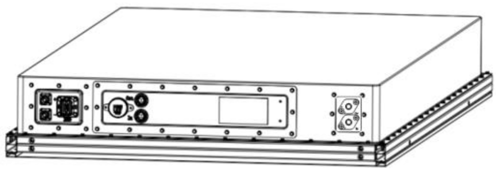
5. **Fire protection system:** Provide system fire warning and fire extinguishing function;

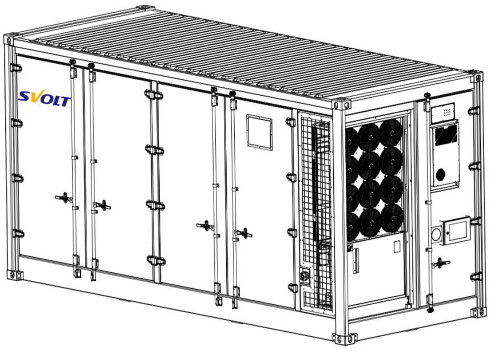
6. **Container:** Ensure the mechanical strength of ESS and provide protection.

This product control system carries out condition monitoring, logic control, data recording, etc., and cooperates with the thermal management system to provide the state in line with the best working condition of the battery, ensuring the efficient and reliable operation of the system; At the same time, early warning and rapid positioning of faults are carried out.

2.3 System Specification

Table 2-1 System Specification

Battery Cell	Parameter item	Specification parameter
	Cell type	LFP short-blade
	Nominal capacity	350Ah
	Nominal voltage	3.2V
	Rated rate	0.5P
	Voltage range	2.5~3.65V（＞0℃） 2.0~3.65V（≤0℃）
	Operating temperature	-20~60℃（discharge） 0~60℃（(charging）
	Dimensions (D*W*H)	26*500*215mm
	Weight	≤6.6kg
Battery PACK	**Parameter item**	**Specification parameter**
	Configuration	1P128S
	Nominal voltage	409.6V
	Voltage range	332.8 ~ 460.8V
	Nominal energy	143.36kWh
	Rated rate	0.5P
	Operating temperature	-30 ~ 55℃
	IP Grade	IP65
	Dimensions (W*D*H)	2149mm*1170mm*246mm
	Weight	950kg

Battery Container	Parameter item	Specification parameter
	Configuration	12P384S
	Number of racks	12
	Rated voltage	1228.8V
	Voltage range	1075.2V ~ 1363.2V
	Nominal energy	5160kWh
	Rated rate	0.5P
	Initial system RTE	93%（≤0.5P, DC side，Excluding auxiliary power supply）
	Auxiliary power	42kW
	Environment temperature	-10℃ ~ 50℃
	Humidity	5 ~ 95%
	Altitude	3000m
	IP Grade	IP55
	Average Noise	≤85dB
	Auxiliary power	380V 50Hz
	Dimensions (W*D*H)	60580*2438*3100mm
	System weight	≤42t
	Lifetime	15years, 1 cycle/day @25℃、DOD 95%
	BESS Certificate	IEC62619-2022、IEC-63056-2020 EN 61000-6-2、EN 61000-6-4 (The old standard has been abolished and replaced by EN/IEC 61000-6-2/4-2019. It will not be implemented) EN/IEC 61000-6-2、 EN/IEC 61000-6-4、 EN/IEC62477-1、 IEC62040、IEC60529-2013、 IEC60068-2-52 (Deemed equivalent to EN/IEC 62477-1, no additional certification is required）、 ISO4892-4(Only lable Certificate）

2.4 Product Appearance Description

The appearance of the energy storage container is shown in the following figure:

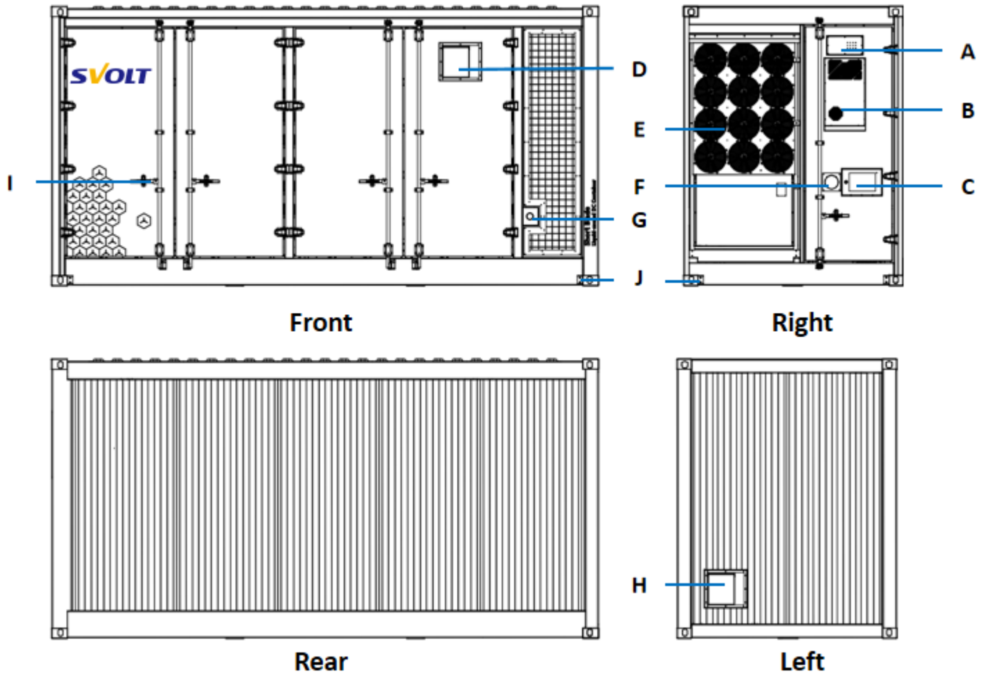


Fig. 2-3 Appearance of energy storage container

Table 2-3 External Component Description

S/N	Part name	S/N	Part name
A	Sound and light alarm/alarm bell	F	Emergency stop button
B	Air conditioning	G	Water fire inlet
C	Fire extinguishing control box (emergency stop button)	H	Air inlet
D	Air outlet	I	Lock lever
E	Chiller	J	Ground point

Emergency stop button!

- In case of emergency, press the emergency stop button on the container panel, the system stops and exits the charging and discharging operation.
- After pressing the emergency stop button, the internal FSS power will remain energized, so do not touch it!

2.5 Product Structure Description

2.5.1 External structure

The dimensions of energy storage container is shown in the following figure:

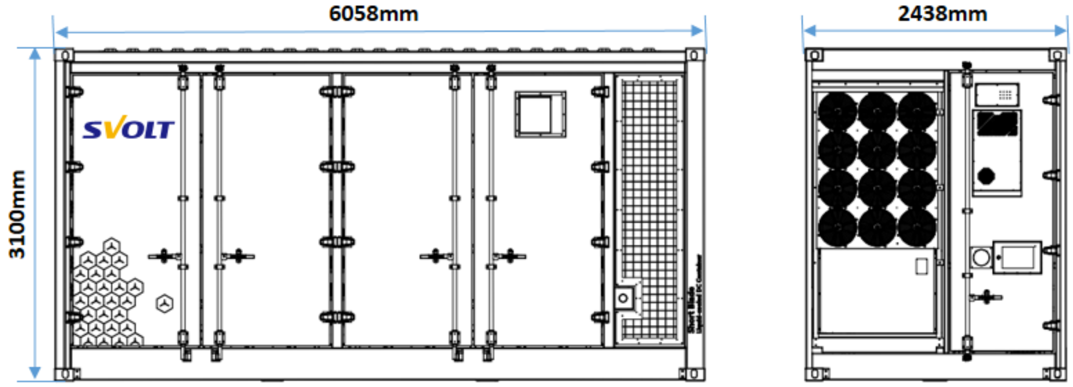


Fig. 2-4 Dimension drawing of energy storage container

The opening size and space of the energy storage container is shown in the following figure:

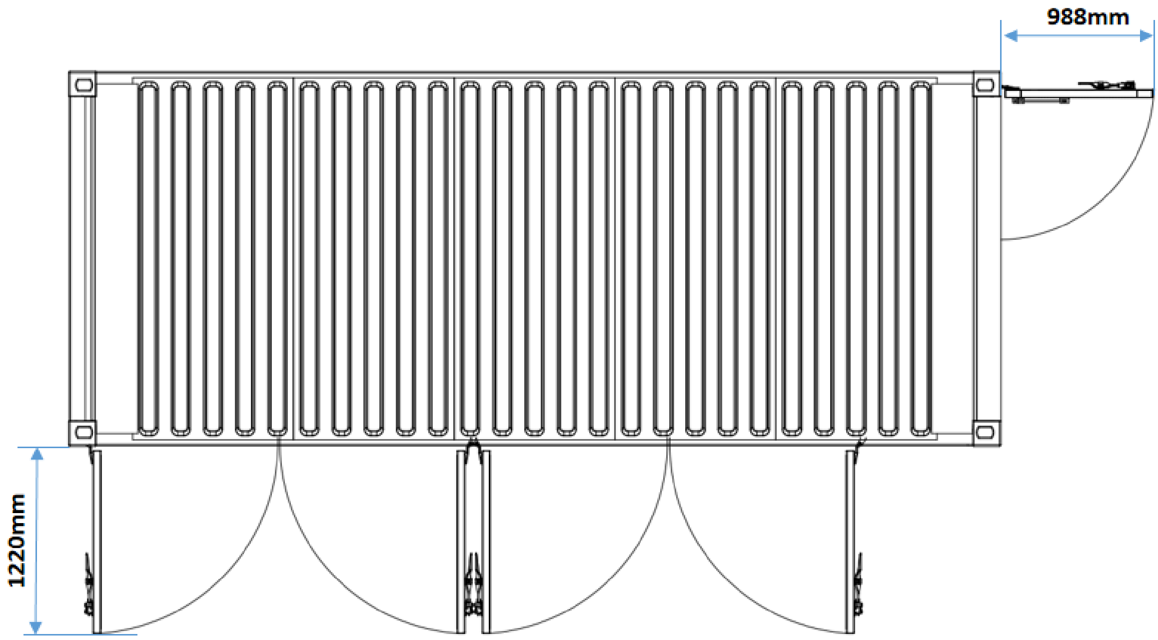


Fig. 2-5 Schematic diagram of opening space of energy storage container

2.5.2 Internal structure

The main electrical equipment such as battery PACK, high-voltage box, chiller, control cabinet and combiner cabinet are designed inside the energy storage container, among which battery racks are distributed in the battery compartment, with a total of 4-column of batteries. There are 9 battery PACKs in each column, which are divided into 3 racks, and every 3 battery PACKs are connected in series to form one rack. Each rack is connected to the bottom high voltage box. The internal arrangement of containers is shown in the following figure:

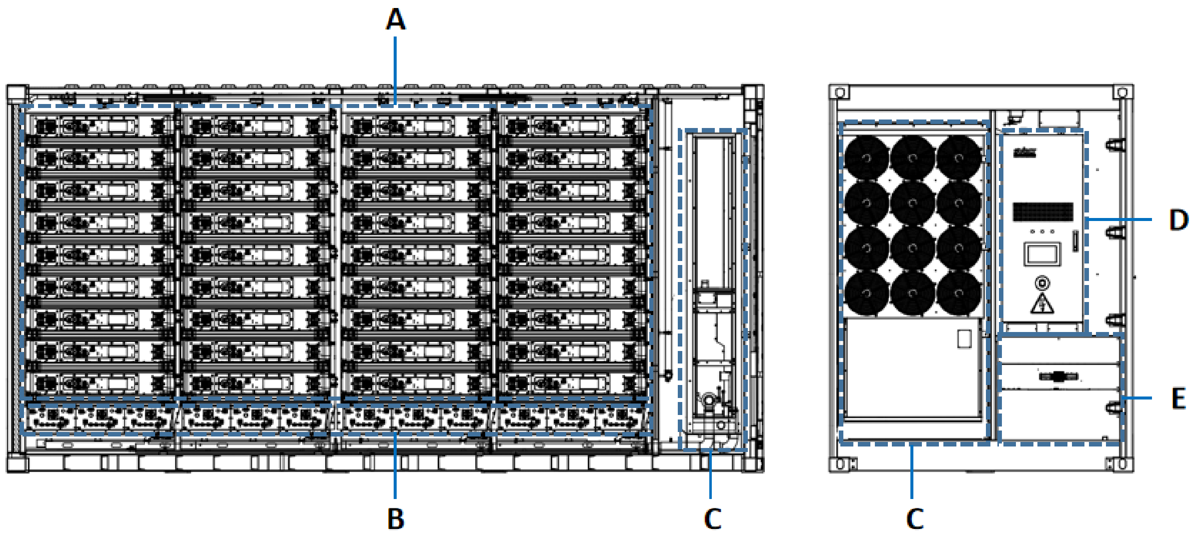


Fig. 2-6 Internal Equipment Components of Energy Storage Container

Table 2-4 Description of internal main equipment

S/N	Part name	S/N	Part name
A	Battery racks	D	Control cabinet
B	High voltage boxs	E	Combiner cabinet
C	Chiller		

2.6 Product Components Introduction

2.6.1 Battery PACK

The battery system in the energy storage container is composed of 36 battery PACKs, which is the basic energy unit of the battery system and is used for the energy interaction between charging and discharging of the energy storage container system.

Single battery pack is IP65 high protection design, Anti-corrosion grade, strong environmental adaptability. Battery PACK contains fuse design, which realizes single PACK level short circuit protection and makes single PACK maintenance more convenient. Built-in BMU collects battery voltage and temperature information and completes balance function.

The appearance of the battery PACK is shown in the following figure:

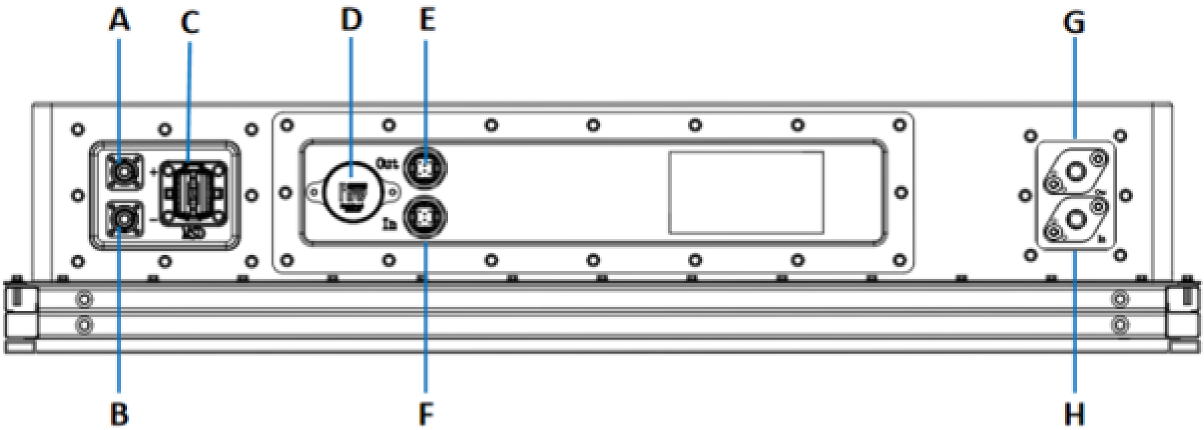


Figure 2-7 Battery PACK

Table 2-5 Interface Description of Battery PACK

S/N	Part name	Explanatory remarks
A	B+	Battery positive
B	B-	Battery negative
C	MSD	Manual maintenance switch
D	Explosion relief valve	Used for internal pressure relief when out of control
E	Communication interface	PACK communication (Out)
F	Communication interface	PACK communication (In)
G	Liquid outlet	Liquid outlet of the battery pack
H	Liquid inlet	Liquid inlet of the battery pack

2.6.2 High-voltage box

Every 3 battery PACKs in the energy storage container are connected in series to form a rack, with a total of 12 racks, and each rack is respectively connected to the high-voltage box. High-voltage box is the control unit of battery rack, which is used to monitor and control battery pack and complete signal interaction. The appearance of the high-voltage box is shown in the following figure:

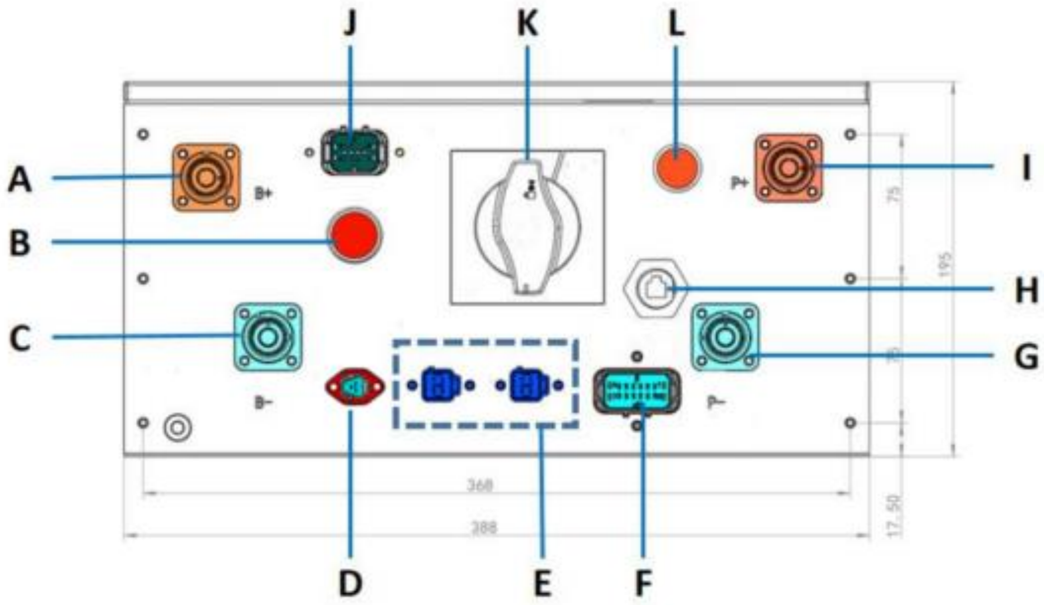


Fig. 2-8 Appearance of High-voltage Box

Table 2-6 Description of High-voltage Box Panel

S/N	Part name	Explanatory remarks
A	B+	DC input positive
B	Push button switch	Auxiliary power supply button switch
C	B-	DC input negative
D	Power supply interface	220VAC External Power Supply
E	CAN communication	External CAN communication
F	Dry contact	External dry contact output
G	P-	DC output negative
H	LAN port	External communication network port
I	P+	DC output positive
J	Chrysanthemum chain communication	Internal communication connection
K	Switch handle	Disconnector handle, for maintenance
L	Running indicator light	No trouble: Green LED lights up Two or three faults: Red LED lights up

2.6.3 Control cabinet

The control cabinet is the core control system of energy storage container, which includes BMS, signal and communication wiring and power distribution. The control cabinet panel is equipped with an operation display screen, which can display the system information and running status. The appearance of the control cabinet is shown in the following figure:

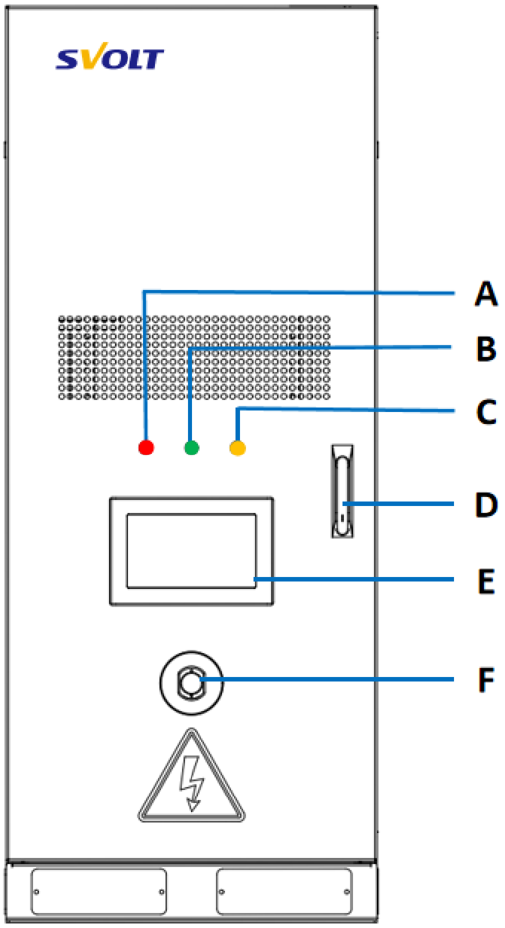


Fig. 2-9 Appearance of Control Cabinet

Table 2-7 Description of Control Cabinet Pane

S/N	Part name
A	Power indicator light (Red)
B	Running indicator light (Green)
C	Fault indicator light (Yellow)
D	Door lock
E	BMS Display Screen
F	Emergency stop button

2.6.4 UPS

The control cabinet contains UPS power supply, and the UPS operation panel is shown in the following figure:

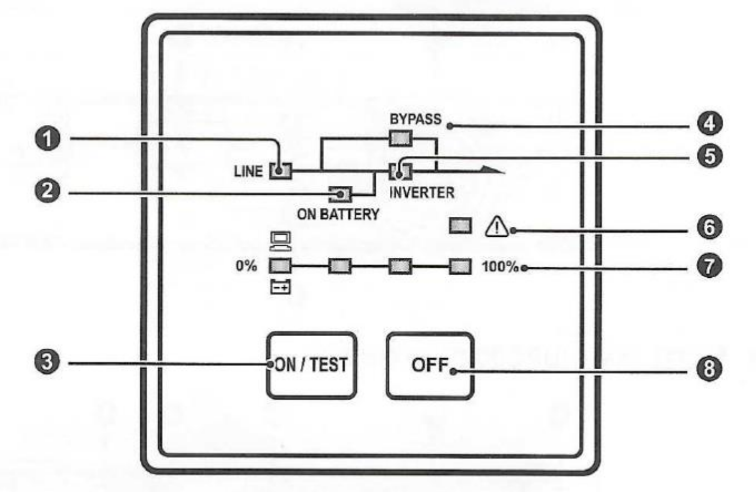


Figure 2-10 UPS Operation Panel

The UPS power operation panel is described in the following table

Table 2-8 Description of UPS Operation Paneln

S/N	Project	function
①	Input indicator light	The input indication is green: 1. Light is on: It means that UPS input voltage is normal. 2. Flicker: Indicates abnormal input.
②	Battery indicator light	Battery indicator light is two-color light: 1. Green light: It means the battery is normal. 2. Yellow light: indicates abnormal battery or low battery voltage alarm.
③	Boot button/ Test button/ Buzzer off button	1. Start button: Press this button for 3 ~ 5 seconds and release it after hearing the buzzer beep to start UPS. 2. Test button: In online mode, press this button for 3 seconds to do 10 seconds battery discharge test. 3. Buzzer off button: When the buzzer sounds, press this button to turn off the buzzer.
④	Bypass indicator	Bypass indicator light is yellow: the light on means that UPS is powered by bypass.
⑤	Inverter indicator light	Inverter indicator light is green: 1. Light is on: It means that UPS inverter is in normal state. 2. Flicker: Indicates that UPS is in startup detection, battery discharge test or DC BUS discharge state.
⑥	Abnormal indicator light	Abnormal indicator lamp is two-color lamp: 1. Red light: It represents the internal anomaly of UPS. 2. Yellow light: Represents other warnings.
⑦	Load capacity /battery remaining capacity indicator light	1. In online mode, the indicator light is on to represent the percentage capacity of connected load. 2. In battery mode, the indicator light is on to indicate the current battery

⑧	Shutdown button	In the UPS boot state. Press this button for 3 seconds and then release it after hearing the buzzer beep to turn off UPS.

2.6.5 Combiner cabinet

The combiner cabinet is the concentrated area of DC output of battery racks. The appearance of the combiner cabinet is shown in the following figure:

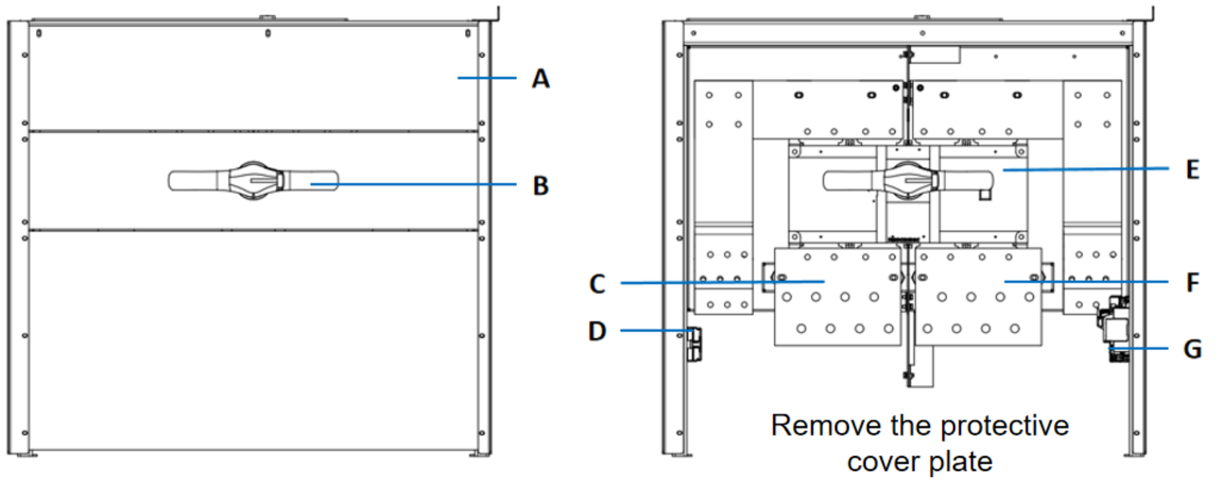


Fig. 2-11 Appearance drawing of combiner cabinet

Table 2-9 Description of the combiner cabinet

S/N	Part name	S/N	Part name
A	Protective cover plate	E	Isolating switch
B	Isolating switch handle	F	Negative DC busbar
C	Positive DC busbar	G	SPD
D	AC terminal block		

2.6.6 Thermal Management System

The main function of the thermal management system is to maintain the temperature of the battery system within the allowable operating temperature range. The thermal management system mainly includes chiller and three-level liquid cooling pipelines. The following is the layout of thermal management pipelines.

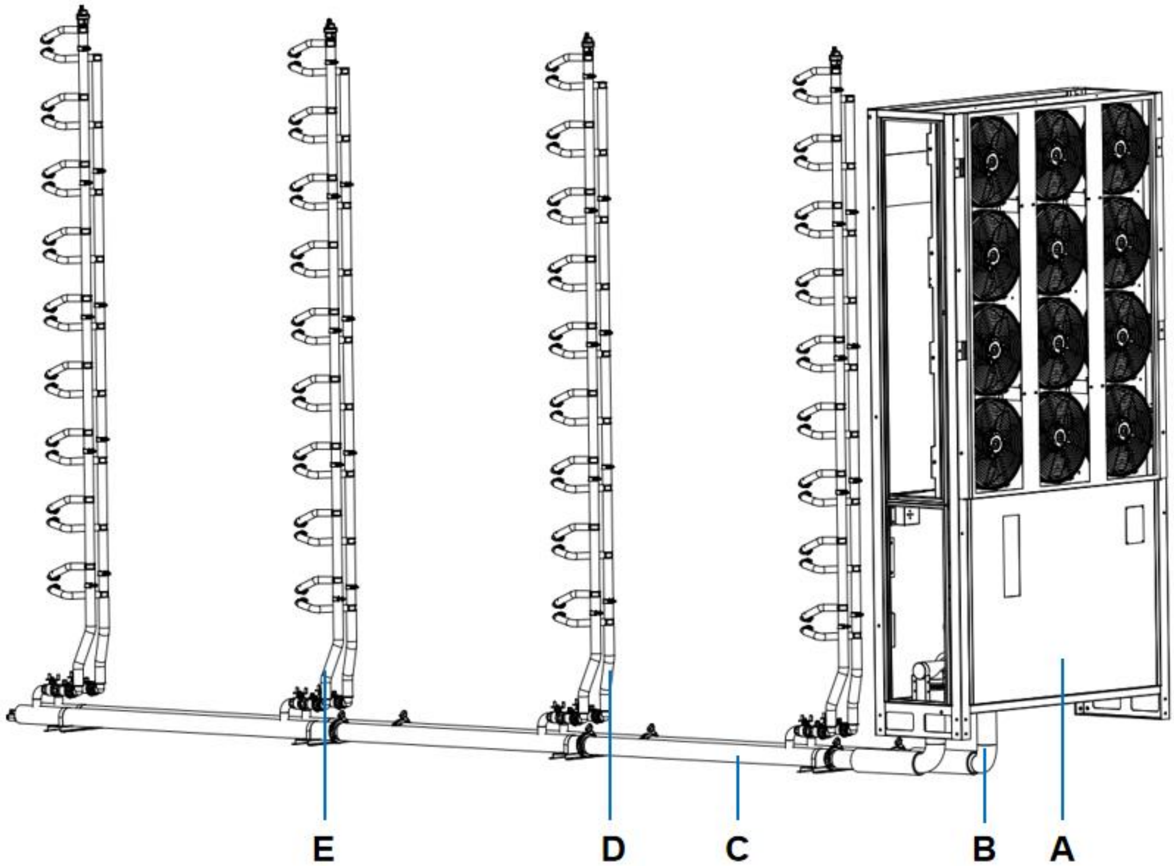


Fig. 2-12 Layout diagram of Liquid cooling system

Table 2-10 Description of the Liquid cooling system

S/N	Part name
A	Chiller
B	1-Level liquid supply pipe
C	1-Level return supply pipe
D	2-Level liquid supply pipe
E	2-Level return supply pipe

The chiller is arranged in the chiller compartment of the energy storage container, that is, the water engine room (side by side with the electrical room on the side of the container). The chiller exchanges heat with fluorine system through coolant, which provides cold and heat sources for the battery and ensures that the battery works in the best temperature range. The coolant after heat exchange in the chiller is transported to the battery pack through the cooling pipe network to ensure the uniform temperature distribution of the battery. The coolant of liquid cooling system is 50% water +50% ethylene glycol solution.

The chiller needs to exchange heat with the external space when working, and ventilation areas are designed on both sides of the container. The product layout should be considered not to be blocked, and regular maintenance is required to ensure that there are no foreign objects blocking. The ventilation schematic is shown in the following figure:

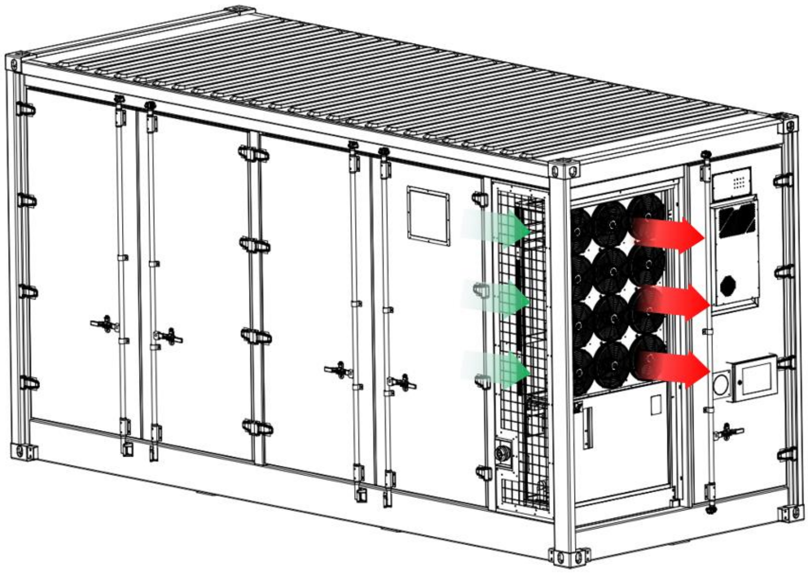


Fig. 2-13 Ventilation area of Energy Storage Container

2.6.7 Fire Protection System

As an outdoor non-step-in battery energy storage system, energy storage container provides a complete fire protection system solution with detection, explosion control and fire extinguishing functions. The fire protection system consists of three parts: fire detection system, fire extinguishing system and fire ventilation system. In addition, the fire protection system is linked with the control system. If any abnormality is detected, the fire protection controller will send the alarm signal to the EMS, it is used for alarm, shutdown control of energy storage containers, and judgment and execution of other corresponding logics.

The main components are listed below.

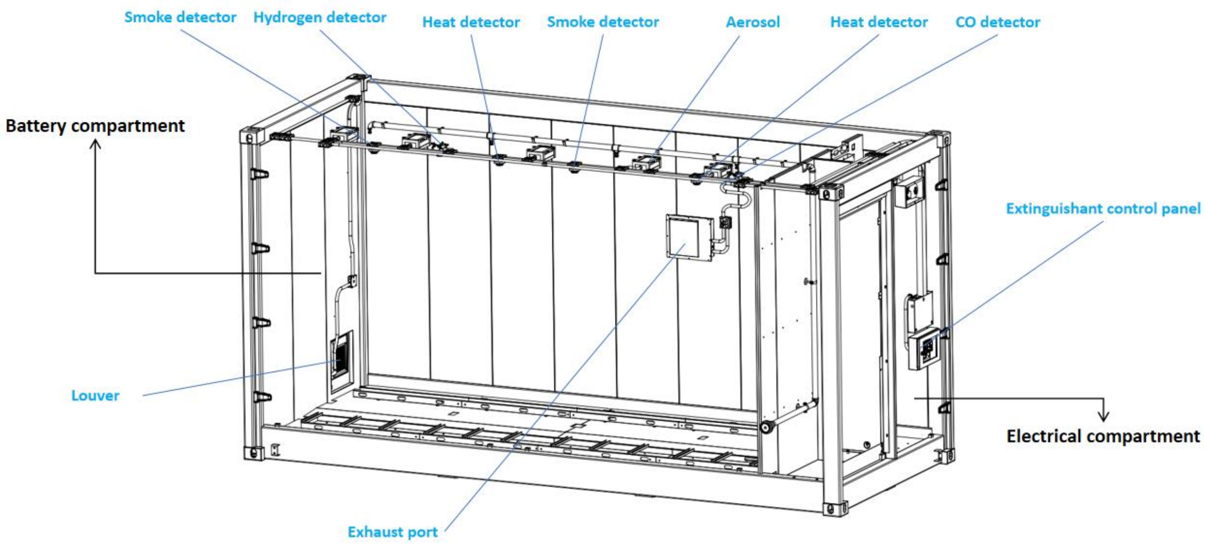


Fig. 2-14 Composition of fire protection system

The overall fire protection system diagram of container is as follows.

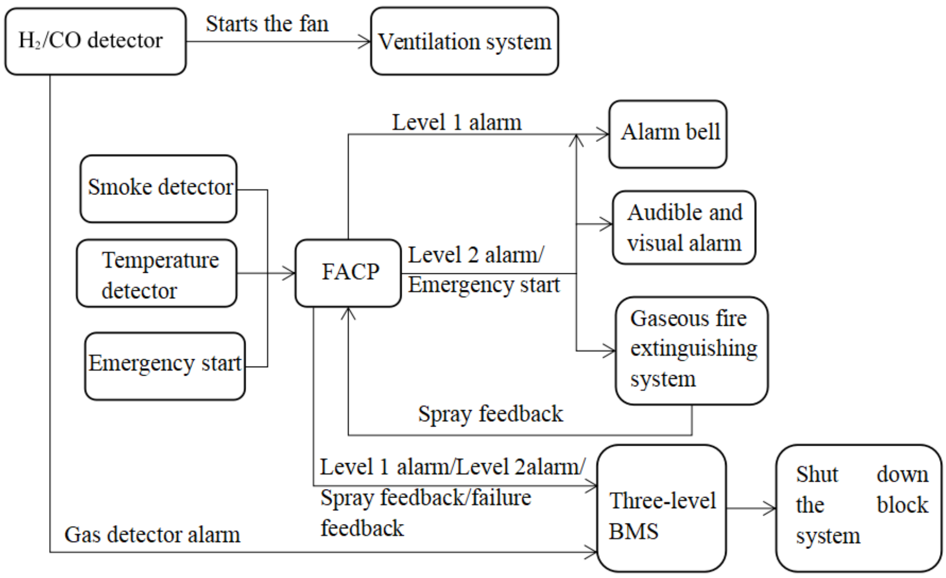


Figure 2-15 Fire Control Logic of Energy Storage Container

(1) Fire extinguishing control box

The fire extinguishing control box is designed inside the electrical compartment. There is a manual release button and a manual/automatic transfer switch in the control box.

When a fire occurs in the container, the fire detector installed in the compartment detects the fire and sends a signal to the fire extinguishing controller. The fire extinguishing controller then starts the aerosol fire extinguishing device in the compartment to extinguish the fire.

In order to ensure that the fire protection system can operate normally for at least 24 hours and maintain alarm function for 2 hours without external auxiliary power supply, standby power supply has been equipped in the control panel, and two 12VDC batteries have been connected in series.

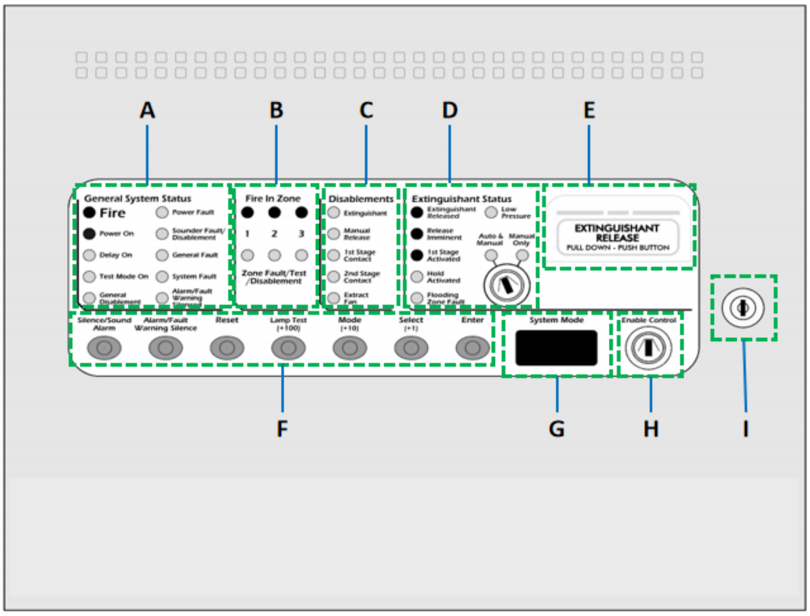


Fig. 2-16 Schematic diagram of fire extinguishing controller

Table 2-11 Component description of the Fire Extinguishing controller

S/N	Part name	Explanatory remarks
A	Overall system state	Indicate the current working status of the fire fighting main engine
B	Area status	Indicates the current working status of the protected area
C	Disabled state	Indicates the current disabled state of the subsystem
D	Automatic status/hand switch for fire extinguishing system	1. Indicate the current working state of the fire extinguishing system 2. Switch between manual and manual automatic state by rotating
E	Manual release	Open the yellow protective cover, press the red button, directly enter the second-class fire alarm state, and start the fire extinguishing device after the countdown of 30s
F	Control panel	Control the host after entering the second-level access authority
G	Display screen	30s countdown is displayed when the second alarm is given. Under secondary access, press the Control Panel Mode button to display the function selection
H	Enable switch	Turn secondary access on or off by rotating
I	Key switch of main engine cover plate	Open or close the fire main engine cover by rotating

The fire extinguishing system has the following two control modes:

① Automatic control: The system sends out an alarm when it detects a fire, then confirm the fire, and triggers the fire extinguishing operation after 30 seconds.

② Manual control: press the emergency stop button outside the energy storage container, so that the fire extinguishing system can be triggered after 30 seconds.

(2) Fire detection and gas extinguishing system

Aerosol is used to extinguish the fire in the container. When the detector in the battery compartment detects the initial fire, it sends out a fire alarm signal, and the fire extinguishing system automatically controls the release of aerosol, which can also be triggered manually.

When any one of the smoke and temperature detectors sends out an alarm, it will be judged as a level 1 alarm and start the alarm bell. The fire control controller sends out the alarm signal into the third-level BMS, and the third-level BMS receives the first-level alarm signal, and the system stops;

When the smoke and temperature detectors work at the same time, it will be judged as a level 2 alarm and start the sound and light alarm. After the 30-second countdown ends, the aerosol is sprayed, and the aerosol start-up signal is fed back to the fire control controller;

The fire control controller will send the alarm signal/start-up feedback signal into the three-level BMS, and after receiving the signal, the system will stop.

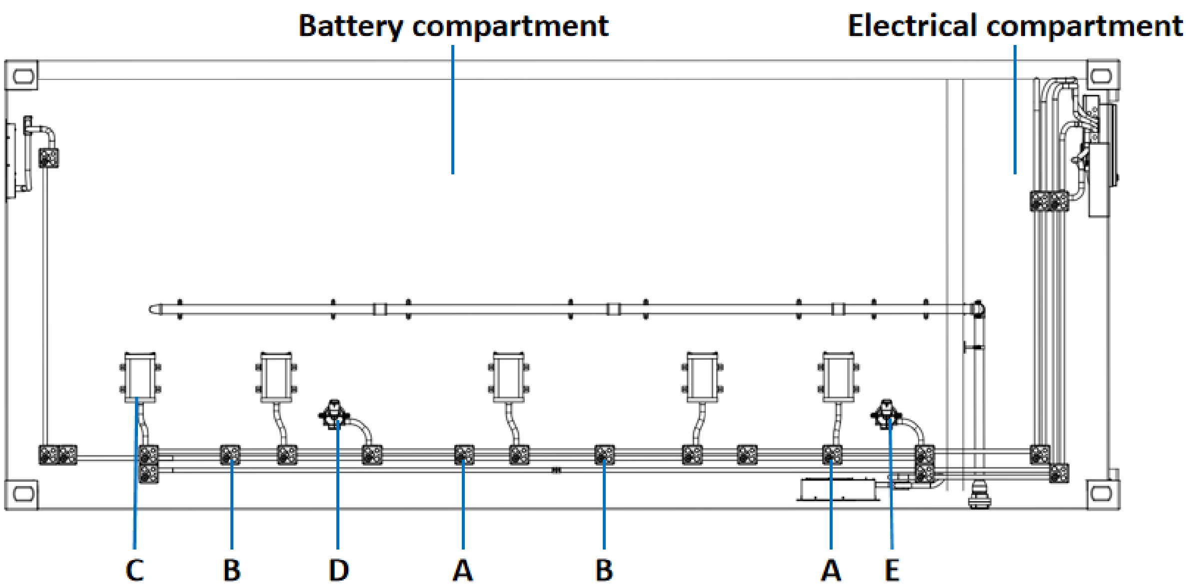


Fig. 2-17 Fire Detection and Fire Extinguishing System Layout (Top View)

Table 2-12 Description of Fire Protection Components

S/N	Part name
A	Temperature detector
B	Smoke detector
C	Aerosol
D	H2 detector
E	CO detector

No one is allowed to enter the energy storage system during the action of the fire fighting system.

(3) Ventilation system

The energy storage container is equipped with an emergency ventilation system. Once the H2 concentration in the protected area reaches 25% LEL or the CO concentration reaches 400ppm, the combustible gas detector will give an alarm and control the exhaust system to open (open the intake louver and exhaust fan), with a maximum displacement of 830 CFM 15%.

The combustible gas detector transmits the alarm signal to the three-level BMS, and the three-level BMS receives the signal, and the whole system stops.

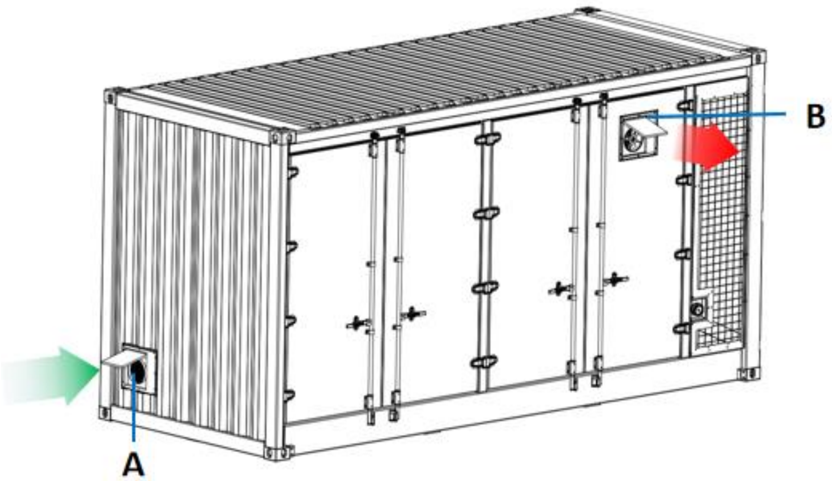


Figure 2-18 Exhaust System Layout

Table 2-13 Component description of the Exhaust System

S/N	Name
A	Intake louver

B	Exhaust fan

(4) Water fire extinguishing system

Container is designed with water fire extinguishing as backup protection, and water can be used to suppress fire when fire occurs. When the automatic fire control system fails or the fire spreads, it is necessary to connect the emergency water supply system for emergency treatment to prevent serious consequences such as deflagration.

The energy storage container is equipped with water spray prefabricated pipes, and the water supply pipes and equipment outside the energy storage container need to be connected to the nozzle of the energy storage container. The sprinkler system adopts drooping sprinkler to ensure that water can be sprayed to all areas in the cabinet.

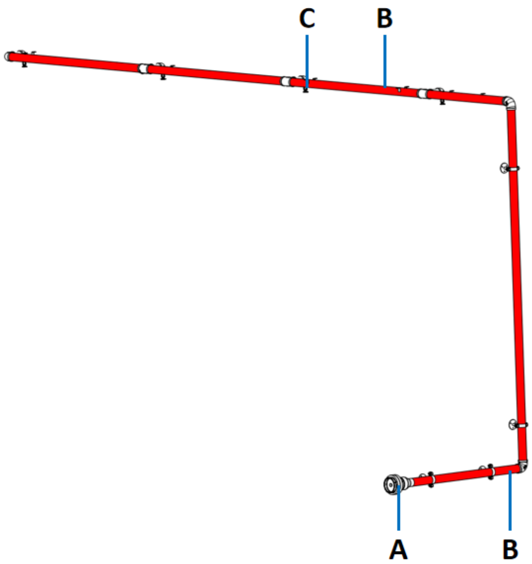


Fig. 2-19 Schematic diagram of water fire protection

Table 2-14 Component description of the water fire protection

S/N	Name
A	DN65 Water Fire Fighting Interface
B	Water fire-fighting steel pipe
C	Sprinkler head

The external interface specifications of water fire protection are as follows:

Table 2-15 Water Fire protection Interface

Parameter item	Specification	Parameter item	Specification
Model	DN65	Rank	PN 16

Standard	DIN14334，DIN14335	Thread type	EN ISO228-1
Material	EN AW-6082	Material of sealing ring	NBR

3. Transportation and storage

3.1 Mode of Transport

Energy storage containers can be transported by land and sea. At present, energy storage containers are not allowed to be used for air transportation, and there is no specific guidance on railway transportation.

3.2 Transportation Requirements

Energy storage containers are highly integrated and convenient for transportation. In the process of loading and unloading containers, they should be carefully operated to avoid dumping, rolling and bearing excessive pressure; During transportation, external mechanical impact should be avoided.

The following basic conditions should be checked and met during transportation:

- All cabinet doors are locked and fixed.
- Take appropriate protective measures to ensure that the system will not penetrate foreign liquids (such as water, oil, etc.).
- Select suitable crane or lifting tool according to site conditions. The lifting tools used shall have sufficient bearing capacity, boom length and rotating radius.
- Containers should be transported and moved in good weather conditions.
- Warning signs or warning areas shall be set up for transportation and loading and unloading to prevent non-workers from entering the loading and unloading area and avoid accidents.
- Do not drag the container when installing or removing the lifting device. Otherwise, the container may be damaged.
- Environmental requirements for transportation: temperature-30 ~ 60 ℃, moderately ≤ 95% (no condensation)

- State of Charge of Transported Products:It is recommended that the transportation SOC be maintained at a level of ≤30% and ensure that the system will not be short-circuited.

(1) Land transportation requirement:

- ➢ When transported by road, it is important to secure the bottom corner of the container to the transport vehicle to avoid excessive tilting during transportation.
- ➢ The maximum driving speed of vehicles (trucks) shall not exceed 100 km/h, and must comply with the relevant laws and regulations of the local transportation area;
- ➢ In the process of driving, it is strictly forbidden to brake suddenly and turn sharply;
- ➢ If containers need to be transported on slopes, extra traction and fixation need to be considered.
- ➢ Keep the vehicle in good condition, regularly check the loading situation of the vehicle, and find and solve the problems in time;
- ➢ Remove all obstacles existing or possibly existing on the road, such as branches, cables, etc., to avoid transportation difficulties or potential safety hazards caused by uneven road surface.
- ➢ In most cases, the total weight of trucks and goods exceeds the limit of general roads. In this case, it is necessary to obtain an overweight permit from the transportation country or region.

(2) Maritime transportation requirement

- ➢ All personnel engaged in port handling and mooring should have received relevant training, especially safety training.
- ➢ Under the condition of sea transportation, the traction force of products exists in multiple directions and dimensions, so it is necessary to consider additional strengthening and fixing.

3.3 Hoisting

Pay attention to operation safety!

It is recommended to use top hoisting for energy storage containers. In the process of hoisting, the operation must be carried out in strict accordance with the safety operation procedures of cranes, and attention should be paid to the safety of on-site operation. The hoisting work shall comply with the relevant standards and specifications of the country/region where it is located. No one is allowed to stay within the coverage area of the operation area, especially it is strictly forbidden to stand under the lifting arm and the lifted machine, and should be 5 ~ 10m away from the crane, so as to avoid casualties.

Lifting operations must be conducted by a professional on-site to guide and operate the entire hoisting process to ensure that at least the following requirements are met:

- All cabinet doors are locked and fixed.
- Lift from the top lifting hole and ensure on-site safety during lifting.
- Select suitable hoisting machinery according to site conditions. It is recommended that the load capacity of the selected hoisting machinery should be ≥ 100000 kg.
- The strength of the slings used shall be able to bear the weight of the energy storage container.
- Ensure that all slings are connected safely and reliably, and ensure that slings of equal length are connected to corner fittings.
- The length of sling can be adjusted according to the actual situation on site. It is suggested that the length is > 8m and the lifting angle is > 60.
- Take all necessary auxiliary measures to ensure safe and stable lifting of energy storage containers.
- Energy storage containers should be lifted vertically and slowly. Never drag containers on the ground or on top of lower containers, and never push or pull containers on any surface.
- In the process of lifting, it is theoretically necessary to ensure that the center of the hanger and the center of the top of the energy storage container are completely correct. In practice, minimize the deviation between the two centers.

- Ensure that the energy storage container remains stable and does not tilt during lifting, and the swing angle should be ≤ 10 and the tilt angle should be ≤ 5.
- When the energy storage container is in place, place it gently and smoothly. It is strictly forbidden to throw it beyond the vertical landing site.
- If the weather is bad, such as heavy rain, fog, gust, etc., the lifting work should be stopped and the lifting operation time should be reevaluated.

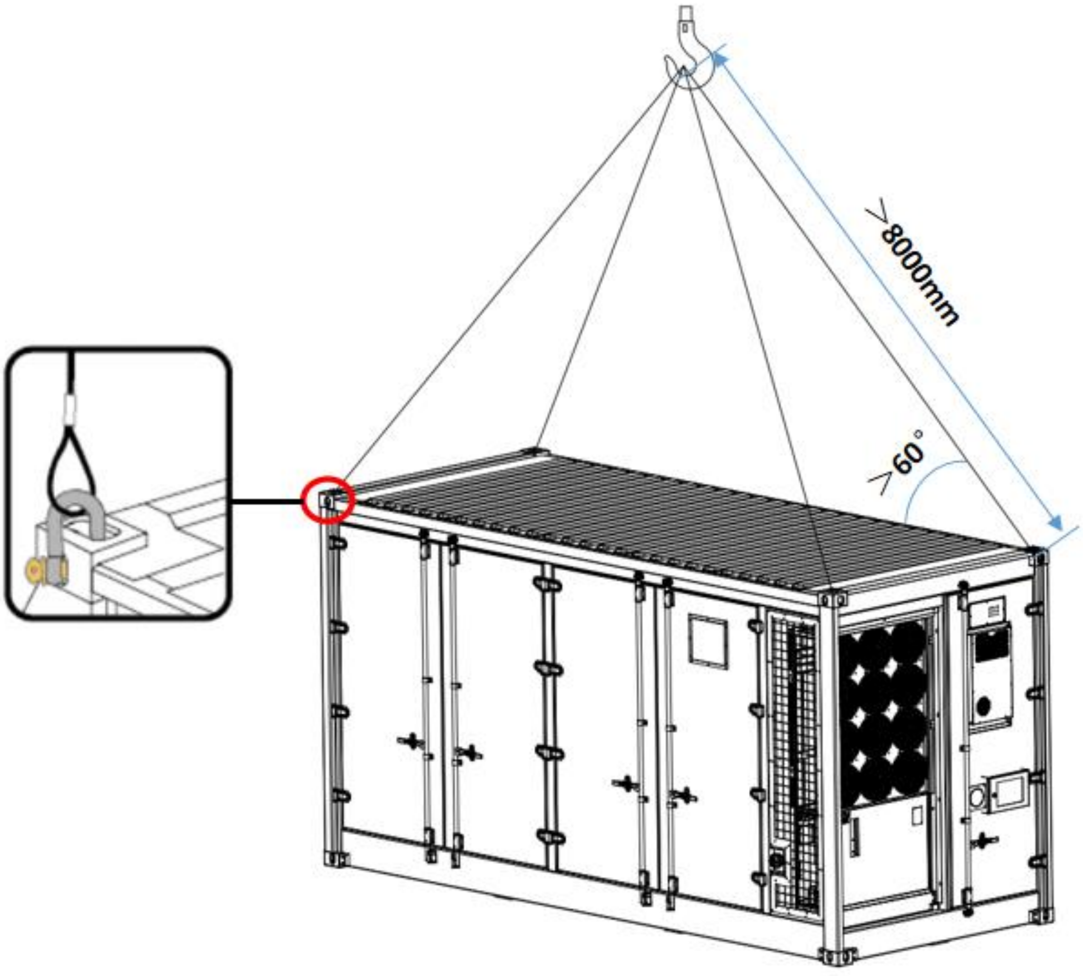


Fig. 3-1 Schematic diagram of hoisting

3.4 Storage Requirement

Energy storage containers should be stored in high-lying places. To prevent possible condensation or its bottom from being soaked by rain in rainy season. The storage floor shall be dry, flat, stable, with sufficient carrying capacity and free of any vegetation cover.

It is recommended that the storage environment be arranged indoors. If the energy storage container must be stored outdoors due to site conditions, please raise the container base. The specific height should be reasonably determined according to the geological and meteorological conditions of the site. Surface flatness shall not be greater than 5 mm.

Harmful gases, inflammable and explosive products or corrosive chemicals shall not be placed in storage places. Protect the battery from mechanical impact, heavy pressure, strong magnetic field and direct sunlight.

Before storage, ensure that the container door and all internal equipment are locked. And effectively protect the air inlet and air outlet of the container to prevent rainwater, sand and dust from infiltrating into the container.

System storage environment temperature:-30 ℃ ~ 55 ℃, recommended storage temperature: 0 ℃ ~ 35 ℃, storage relative humidity should be between 0% ~ 80%.

Check the equipment regularly during storage. Check at least once every half month to see if the equipment is damaged, ensure that the outer packaging or outer surface is not damaged in any way, and prevent any damage that may be caused by pests and animals.

Long-term storage of batteries is not recommended, as this may lead to decreased battery capacity and discrete battery consistency. Even if the battery is stored at the recommended storage temperature, irreversible capacity decay and dispersion will still occur during the storage period. The longer the time, the greater the capacity attenuation and dispersion.

Under the above conditions, the system with storage period exceeding 3 months should be charged and discharged once, and the longest period should not exceed 6 months, so that the SOC of the system can reach 20% ~ 40%.

4. System Installation

4.1 Pre-installation inspection

Before system installation, please conduct a product inspection to ensure there are no issues or damage to the system. If any abnormalities are found or the energy storage model does not match, do not unpack and contact your supplier immediately.

(1) Delivery inspection

- Check if the delivered items and accessories are complete..

(2) Product inspection

- Check if the model and type of the delivered energy storage container are correct..
- Check for any damage to the exterior of the energy storage container and its internal equipment.

4.2 Preparation before Installation

It is recommended to engage professional construction teams or design institutes for container installation and engineering construction to address complex on-site installation conditions and environmental factors. Before installation, conduct an environmental survey to accurately assess the installation location and construction plan.

4.2.1 Installation location

- When choosing the installation site, the climatic environment and geological conditions should be fully considered, and stress wave emission, groundwater level and low-lying areas should be avoided.
- The environment around the installation site should be dry and well ventilated.
- There should be no trees around the installation site to prevent branches and leaves blown away by strong winds from blocking the doors or air intakes of the energy storage system.
- The installation site should be far away from the concentrated areas of toxic and harmful gases, dust and smoke, and free of flammable, explosive and corrosive substances.
- The installation site should be far away from residential areas to avoid noise.
- Avoid equipment installation in vibration and strong electromagnetic field environment. The magnetic field strength is not more than 30 A/m.

- Determine the installation position and orientation of the energy storage container according to the site conditions.

The reference steps are as follows:

Step 1: Determine the ESS installation reference point on the concrete platform. Mark the reference point with a marker.

Step 2: Based on the reference point, mark the installation position of ESS quadrangle accessories with ink fountain and soft tape measure.

Note that when marking the position of corner parts, ensure that 4 lines form a rectangle.

4.2.2 Infrastructure

The foundation of container must be designed and constructed according to certain standards to meet the requirements of mechanical support, cable wiring and later maintenance and overhaul.

Foundation construction shall at least meet the following requirements:

- The soil on the installation site should be tight. It is recommended that the relative density of soil on the installation site should not be less than 98%. In the case of loose soil, take relevant measures or necessary engineering construction to compact and fill the foundation pit to ensure the stability of the foundation and provide sufficient and effective support for the container.
- Container foundation shall be made with reference to the foundation plan provided by the manufacturer or confirmed by both parties, and the flatness of the top surface of the foundation shall be less than 5mm. The construction requirements should not be lower than the recommended drawing provided by the manufacturer.
- The height of container from the ground shall be implemented according to the requirements of local design specifications. It is suggested that the foundation height should be higher than the highest water level in history to prevent the bottom and interior of the container from being eroded by rain. The specific height shall be determined by the construction party according to the site conditions.
- The height of each foundation base was measured by laser level meter, and the height difference of each foundation base was less than 5mm.

- Construct cement foundations or other non-combustible surfaces with sufficient cross-sectional area. The installation surface must be level, stable, smooth, dry, and steady, with adequate load-bearing capacity to support the weight of the energy storage container without risk of collapse or sliding. Depressions or tilting are prohibited, and water accumulation is strictly forbidden
- Construct appropriate drainage systems based on local geological conditions. The drainage system should meet the requirements for handling the local historical maximum rainfall, preventing the container bottom and internal equipment from being submerged during rainy seasons or heavy rainfall periods.
- If welding fixation is used, No. 30 channel steel must be embedded. Anti-corrosion treatment must also be properly applied.

This section does not apply to foundation construction, and the following foundation drawings cannot be used as final construction drawings, but only as reference for foundation design. Final engineering drawings must be prepared by professionals.

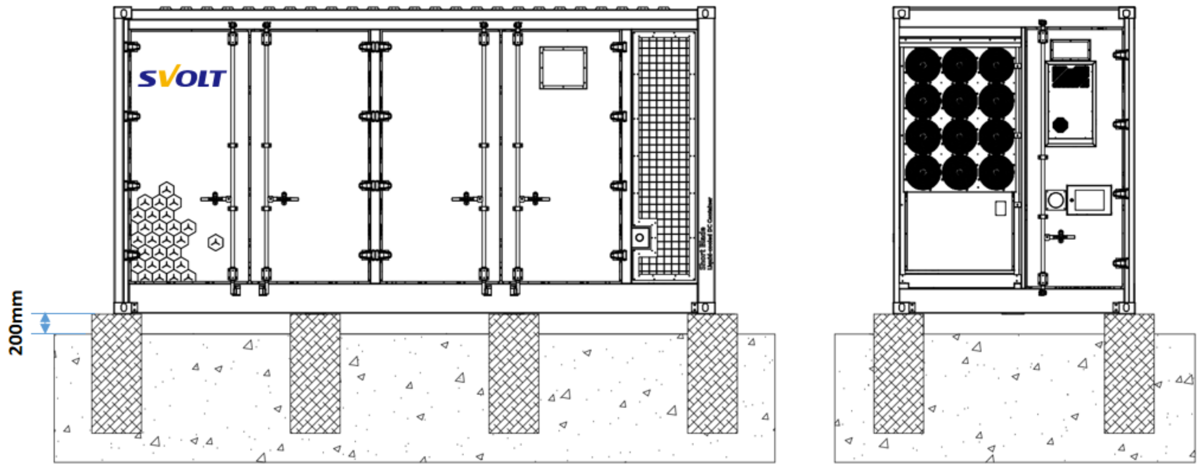


Figure 4-1 Foundation Reference Scheme

The overall foundation construction effect is referred to in the above figure.

The foundation construction for energy storage containers requires at least 8 load-bearing points, with their relative positions arranged as shown in the reference diagram. Site planning and construction conditions must meet or exceed this standard. The height of the foundation platforms should be adjusted according to site conditions and actual requirements, ensuring that the bottom of the equipment is above the local historical highest water level, with a recommended minimum height of no less than 200mm. The recommended diameter of the foundation platform is 500mm. When the steel shims of the foundation platform structure are under uniform load, ensure that the minimum load-bearing capacity of each foundation platform is not less than 50,000kg.

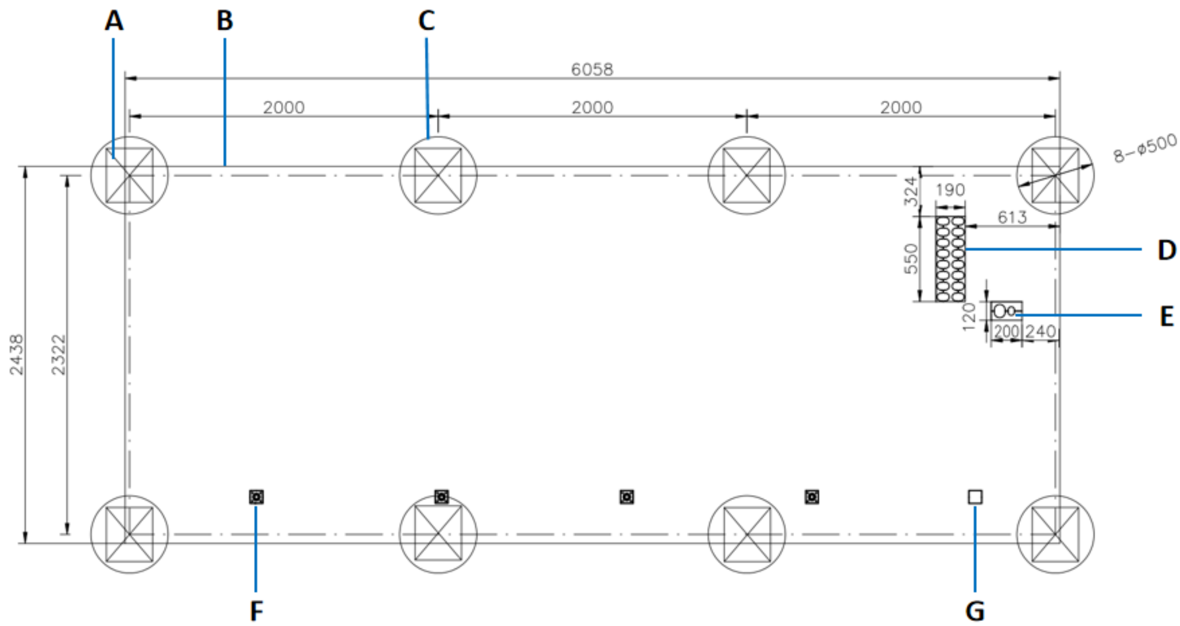


Fig. 4-2 Foundation Reference Scheme (Top View of Single Container)

Table 4-1 Foundation Drawing Description

Serial number	Name
A	Embedded No.30-C Channel Steel for Welding Fixation
B	Container Outline
C	Column foundation
D	DC cable outlet
E	AC power supply inlet
F	Drainage floor drain
G	Drainage floor drain

4.2.3 Trench construction

The energy storage container uses bottom cable entry. To prevent foreign objects from entering, there are no cable entry holes on the sides of the container, and cables must be routed through ground trenches. Therefore, ground trenches need to be pre-installed at the site.

- Cable routing must be considered during foundation construction. To facilitate subsequent electrical wiring, it is recommended to pre-install cable trenches, cable trays, entry/exit holes, and maintenance passages in the foundation according to the positions and sizes of the container's cable entry and exit points. Conduits should also be pre-embedded.
- The construction of cable trenches, cable trays, and maintenance passages must include necessary dust-proof and rodent-proof designs to prevent foreign objects from entering.
- Necessary waterproof and moisture-proof designs are required to prevent cable aging and short circuits, which could affect the normal operation of the energy storage container.
- Drainage construction must be considered to prevent water accumulation during rain and snow, which could lead to cable soaking and corrosion.
- If pre-embedded conduits are used, both ends of all buried conduits should be temporarily sealed to prevent debris from entering and affecting subsequent wiring.
- Due to the high-power rating of the energy storage container, thicker cables are required. The cable cross-sectional area must be fully considered when designing the ground trench. The inner diameter of the protective conduit should not be less than 1.5 times the outer diameter of the cable (including protective layer).

4.2.4 Pre-embedded Grounding

The energy storage container body requires reliable grounding on site, and a reliable grounding grid should be planned for the site.

The recommended solution is as follows: Bury the grounding grid underground and reserve a grounding wire or grounding bus at the corresponding ground position of the container's external grounding points. Connect one end of the grounding wire or grounding bus to the buried grounding grid, and after the container installation is completed, connect the other end to the container's grounding points. When embedding the grounding wire or grounding bus, estimate and reserve sufficient and appropriate length to ensure the grounding wire can reach the grounding points on the container.

The grounding resistance of the energy storage container shall not be greater than 4 Ω.

4.2.5 Installation spacing

To ensure normal operation of the energy storage container and equipment door access, meet the required spatial distances for air inlets and outlets, and avoid hot air interference between adjacent systems, sufficient clearance should be maintained around the equipment. The equipment layout spacing is shown in the figure below, where the marked distances represent the minimum requirements considering installation, operation, heat dissipation, and maintenance. Customers may increase these distances according to local fire regulations, vehicle traffic main road requirements, and overall site construction needs. Fire walls can be added when necessary.

Note: The following distance requirements are the distance between containers, not the distance between foundations.

(1) Single system distance requirement

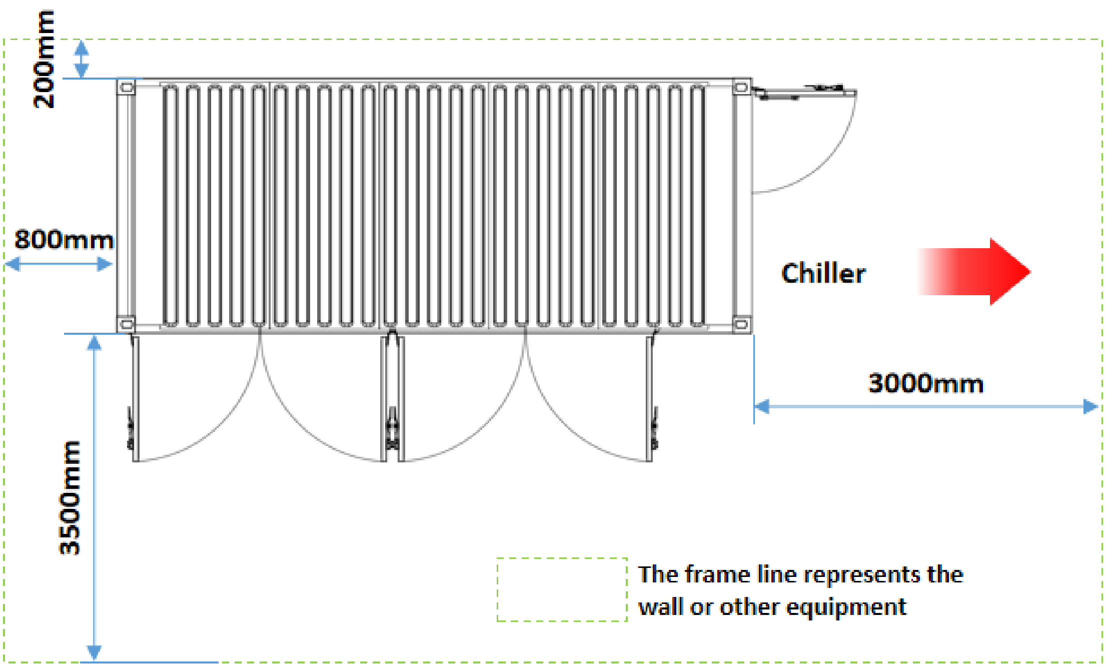


Figure 4-3 Installing a single container

(2) Multi-system distance requirement

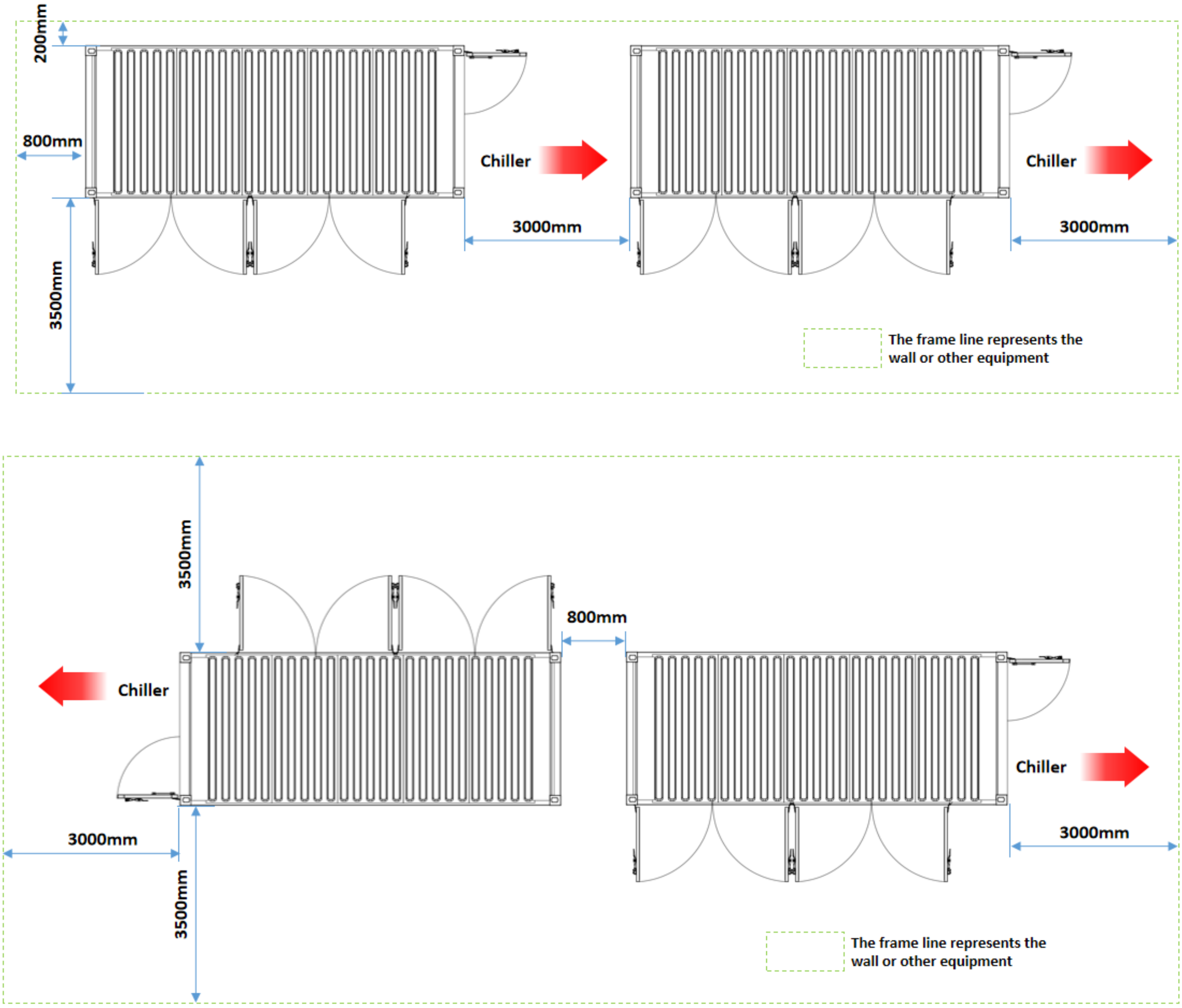


Figure 4-4 Installing multiple containers (side by side)

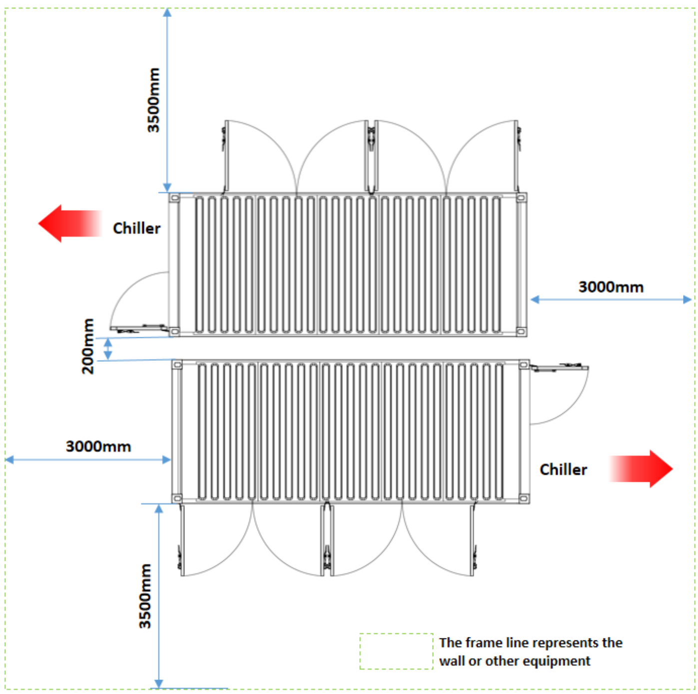


Figure 4-5 Installing multiple containers (back to back)

4.2.6 Installation and fixing

After the energy storage container is transported and installed at the designated location, it should be securely anchored.

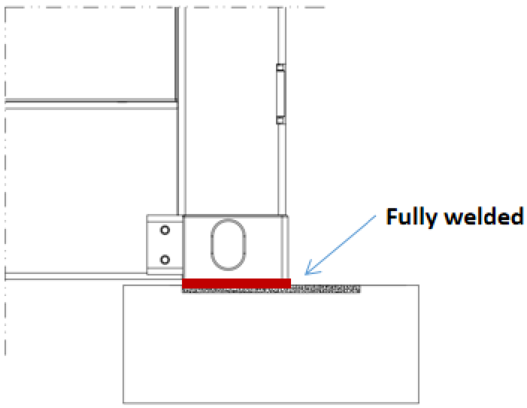


Fig. 4-6 Schematic diagram of full welding

The recommended anchoring method is bottom welding fixation: Embed No. 30 channel steel and weld the container bottom to the foundation. After completion, the welded areas should be painted for corrosion protection. All welded components must be fully welded, and after welding is completed, the weld seams should be painted for corrosion protection.

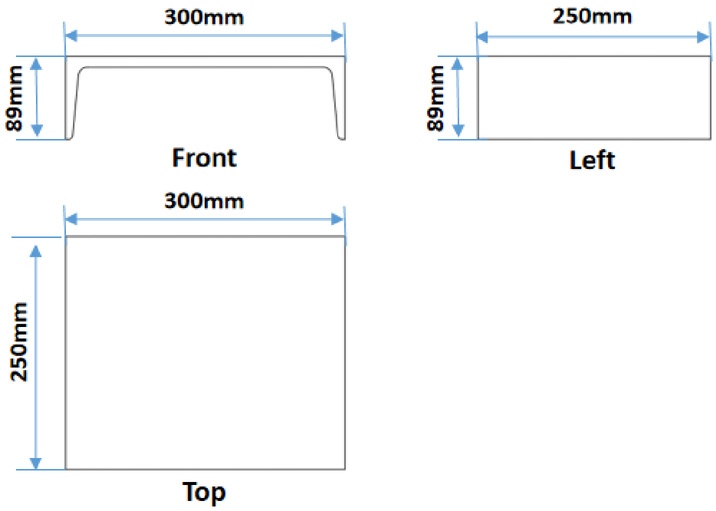


Figure 4-7 Dimensions of 30 #-c Channel Steel

4.2.7 Post-installation inspection

After the installation is completed, review and check the installation work to ensure that the work is qualified and meets the product requirements.

(1) Verify if the installation is at the designated or pre-designed location; make adjustments if there are any deviations.

(2) Check if the flatness of the container's bottom installation surface meets requirements; if not, perform leveling adjustments or use shims to correct.

(3) Verify if the surrounding space of the container meets requirements and if maintenance vehicles can properly navigate and turn; if not, adjust the position of obstacles or modify the access routes

(4) Check if the reserved grounding locations align with the product's grounding points; if not, adjust the grounding positions or modify the layout arrangement.

(5) Check if there are foreign objects or blockages in the reserved cable trenches/ducts that might affect subsequent cable laying; if so, clean them promptly.

5. Electrical wiring

5.1 Wiring Precautions

Pay attention to the following matters before wiring, pay attention to operation safety and avoid potential safety hazards.

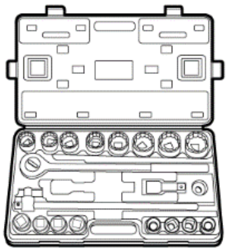
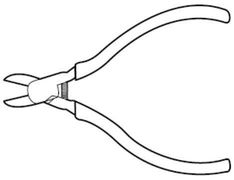
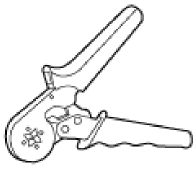
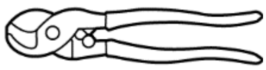
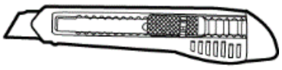
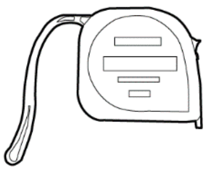
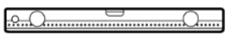
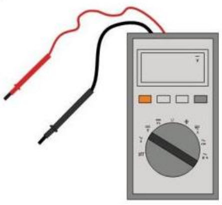
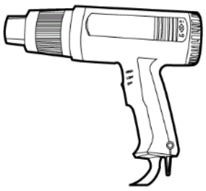
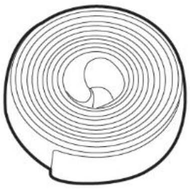
- Personal protective equipment should be worn as required for electrical wiring operation.
- Before wiring, check and ensure that all wiring harnesses and cables are polarized correctly.
- In the process of electrical wiring, if you encounter drag resistance, do not pull any wires or cables forcibly, otherwise it may damage the insulation skin and damage the insulation performance.
- Make sure all cables and wires have enough space for bending. Forcibly squeezing and bending cables in extremely small areas is prohibited. And when conditions permit, necessary auxiliary measures should be taken to reduce the stress applied to cables and wires.
- After each electrical connection operation, carefully check and ensure that the connection is correct and safe.
- Wiring operation should be carried out in sunny and dry weather. Avoid electrical connection in case of severe weather such as rain, snow, wind and sand or when the relative humidity of the surrounding environment is greater than 95%. Wind, sand, dust and moisture penetration may damage the electrical equipment in the energy storage container or affect its operation performance!

5.2 Preparation before wiring

5.2.1 Prepare Tools

Wiring personnel must be professionals who are familiar with the contents of the manual. In addition to wearing the necessary personal protective equipment, at least the following operating tools should be prepared to assist the wiring work.

Table 5-1 List of wiring tools

			
Torque screwdriver	Torque wrench	Stripping wire pliers	Hydraulic tongs
			
Torque socket wrench	Oblique pliers	Wire crimping pliers	Thread cutter
			
Rubber hammer	Tool knife	Tape measure	Horizontal ruler
			
Multimeter	Hot air gun	Heat shrinkable sleeve	cable tie

5.2.2 Wire selection

➢ Select the cable with appropriate diameter according to the maximum load or maximum current to ensure that the current carrying capacity of the cable meets the requirements.

➢ Select the cable with appropriate length according to the connection position.

➢ All DC input cables must have the same specifications and materials.

➢ AC cables must have the same specifications and materials.

➢ Communication harness must be shielded harness. If network cable is involved, it must be above CAT 5e.

➢ Use flame retardant cables, and the cables used should meet the requirements of local laws, regulations and standards.

The recommended cable specifications are shown in the following table:

Table 5-2 Specifications of Connecting Cable for Energy Storage System

Cable name	Cable specification (mm²)	Connecting terminal
HV+(PCS+)	Monopole 6*240mm²	DT240-16
HV-(PCS-)		DT240-16
Phase A of power grid	YJV0.6/1kV-4*25+1*16	SC25-6
Phase B of power grid		SC25-6
Phase C of power grid		SC25-6
N phase of power grid		SC25-6
Communication cable	RVVSP 2*1.5	E0508
Network cable	CAT5e	RJ45

5.2.3 Wiring area

The wiring areas of energy storage containers are uniformly concentrated in the electrical cabin, which is located in the right half of the container side, that is, the right side of the water engine room. The door can be opened by surface door lock for wiring.

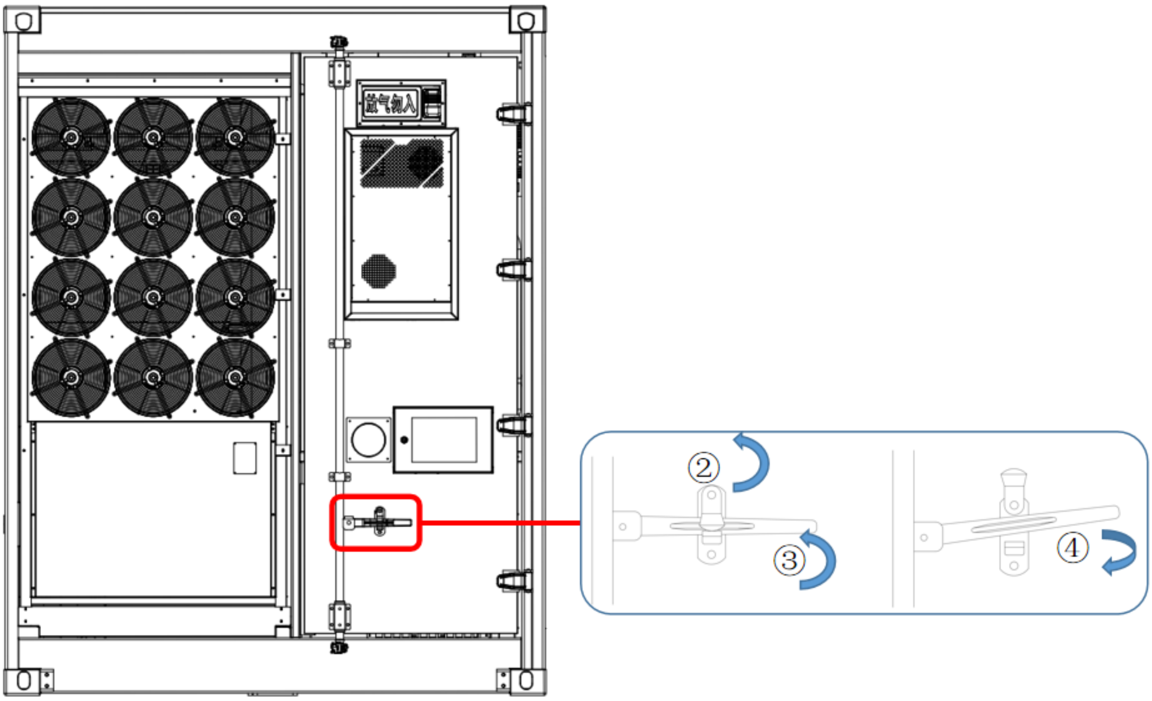


Fig. 5-1 Illustration of electrical cabin door unlocking

Table 5-3 Door Opening Steps

S/N	Description
1	Use the key to open the padlock and remove it.
2	Pull the upper cover on the handle counterclockwise to expose the long handle
3	Rotate the long handle up to the position on the figure
4	Turn the long handle outward to open the electric cabin door

5.2.4 Cable Inlet

Cables between the energy storage container and external equipment are wired from the bottom of the container. Take measures to protect all cables of energy storage containers, such as laying cable protection pipes to prevent rodents from damaging cables. The cable inlet and outlet holes at the bottom of the energy storage container are shown in the following figure.

After the wiring is completed, the gap between the cable and the opening must be sealed with fireproof/waterproof materials such as fireproof mud to prevent foreign matter or moisture from entering, which will affect the normal operation of the product for a long time!

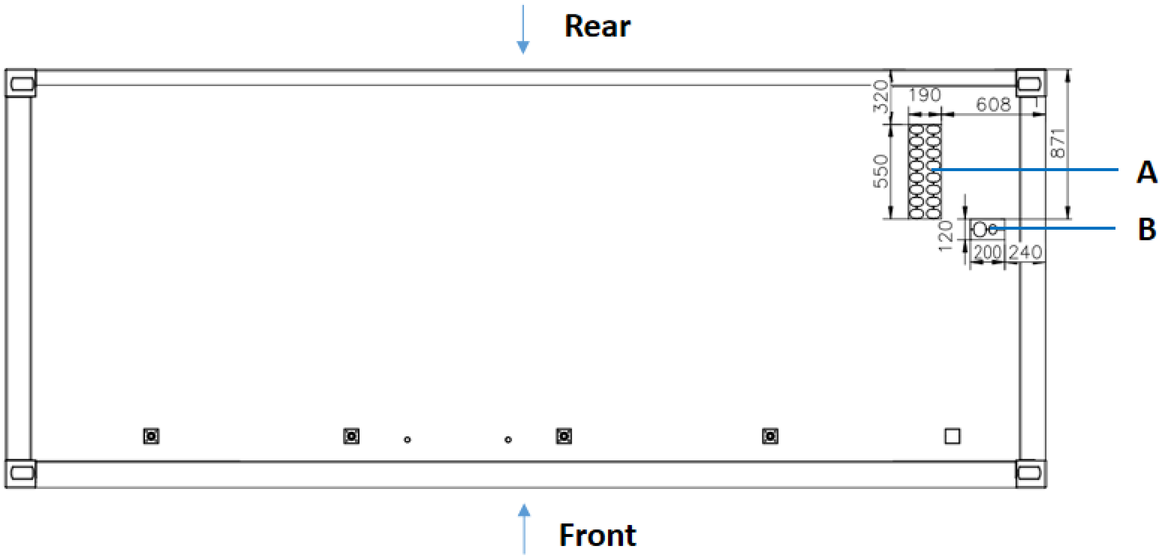


Fig. 5-2 Schematic diagram of cable entrance and exit at the bottom of container (top view)

Table 5-4 Container Bottom Cable Entrance and Exit

S/N	Part name	Explanatory remarks
A	DC cable outlet	550mm*190mm
B	AC/communication outlet	200mm*120mm

5.3 DC Cable Connection

5.3.1 Wiring location

The DC cable connection position of the energy storage container is located in the bus bar of the bus cabinet in the middle below the electrical cabin. DC bus bars are divided into positive and negative copper bus bars, and each bus bar provides 8*M16 single-hole connections respectively. As shown in the following figure.

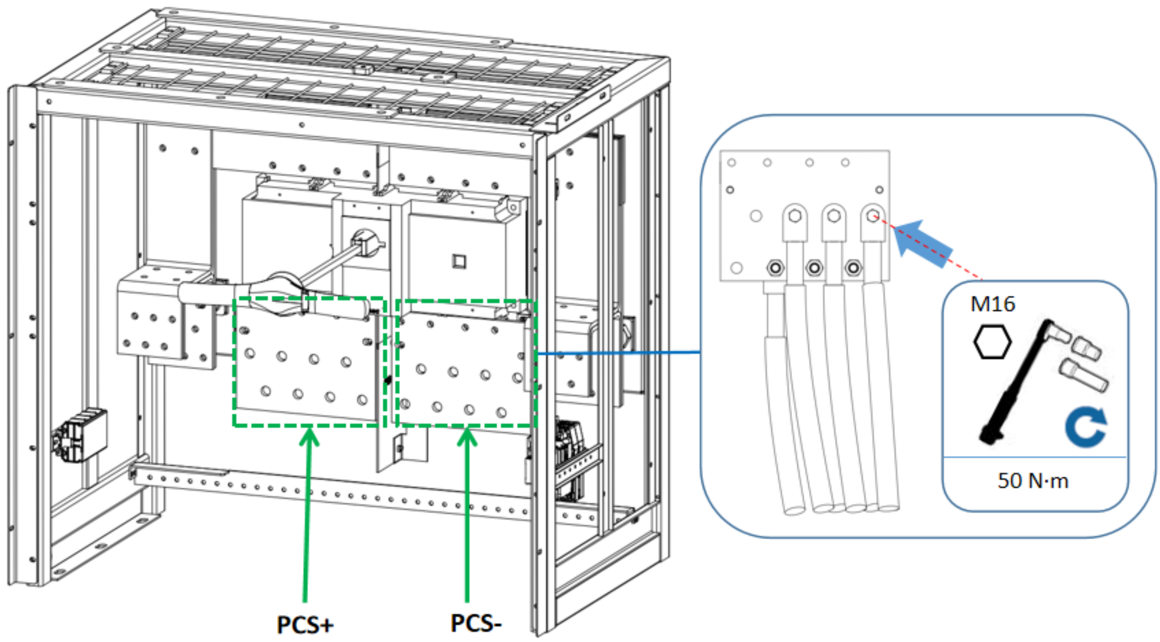


Fig. 5-3 Connection Position of DC Cable

Table 5-5 Description of DC Cable Connection Port

Port name	Parameter item	Description
HV + (PCS +)	Copper bar position	Copper bar on left side
	Recommended cable	Copper cable
	Wiring hole	M16
	Number of wiring holes	8
	Recommended wiring	6 * 240mm^2
	Torque	50 N.m
HV-(PCS-)	Copper bar position	Right copper bar
	Recommended cable	Copper cable
	Wiring aperture	M16
	Number of wiring holes	8
	Recommended wiring	6 * 240mm^2
	Torque	50 N.m

5.3.2 Cable wiring

When connecting the cable, the cable must be put on and out from below through the lower outlet hole, pulled to the corresponding position of positive and negative copper bus, and marked with positive and negative electrodes to avoid polarity error. Ensure that the communication cable connector or RJ45 is securely locked and not loosened after wiring.

Step 1: Confirm that the output switches of the front and back stages of the confluence area and control area are in open state.

Step 2: Peel off the insulating skin at the end of the cable and expose the copper core. The length of peeling off the insulating skin at the end of the cable should be the depth of the wire pressing hole of the connecting copper nose plus about 5mm. At this time, heat shrinkable sleeve should be sleeved in advance.

Step 3: Sleeve heat shrinkable sleeve.

- Choose the heat shrinkable sleeve which is more consistent with the cable size, and the length should exceed the connecting copper nose pressure pipe by about 2cm.
- Sleeve the heat shrinkable sleeve on the cable in advance.

Step 4: Crimp the wiring copper nose. According to the selected cable specifications, equipped with appropriate wiring copper nose.

- Put the exposed copper core part of the peeled thread end into the wire pressing hole of the connecting copper nose. Be careful not to expose copper.
- According to the cable or harness to be crimped and its matching terminals, select the appropriate crimping tool and press the copper nose of the wiring tightly, and the number of crimping should be more than two.

Step 5: Tighten the heat shrinkable sleeve.

- Sleeve the heat shrinkable sleeve on the connecting copper nose to completely cover the wire pressing hole of the connecting copper nose.
- Tighten the heat shrinkable sleeve with a hot blower.

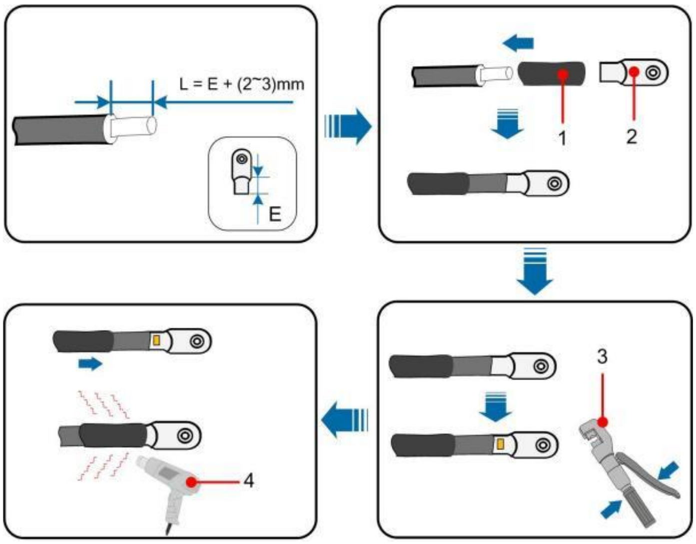


Fig. 5-4 Cable crimping schematic

Table 5-6 Crimping Description

S/N	Description
1	Heat shrinkable tube
2	OT/DT terminal
3	Crimping tool
4	Hot air gun

Step 6: Wiring.

- Choose the terminal matching with the bus hole, and recommend using single hole terminal.
- The connection copper nose is crimped at the designated connection position. When contacting, it should be in accordance with the maximum contact area, and it should be flat without tilting after contacting.
- Choose bolts that match the connection terminals and bus connection holes. Pay attention to bolt fastening should be firm and reliable.

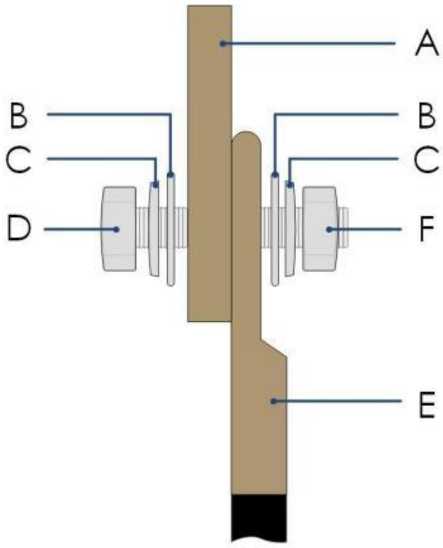


Fig. 5-5 Cable Connection Reference Schema

Table 5-7 Wiring Locking Instructions

S/N	Description
A	Copper bus bar
B	Flat gasket
C	Spring washer
D	Bolt
E	Copper terminal (OT)
F	Nut

5.4 AC Cable Connection

5.4.1 Wiring location

The AC cable connection position of the energy storage container is located at the terminal row J0 on the left side below the electrical cabin, which is mainly used as the auxiliary power supply of the energy storage container system. The power supply adopts three-phase alternating current input, and its access position is J0: 1 ~ J0: 4 of the terminal block. As shown in the following figure.

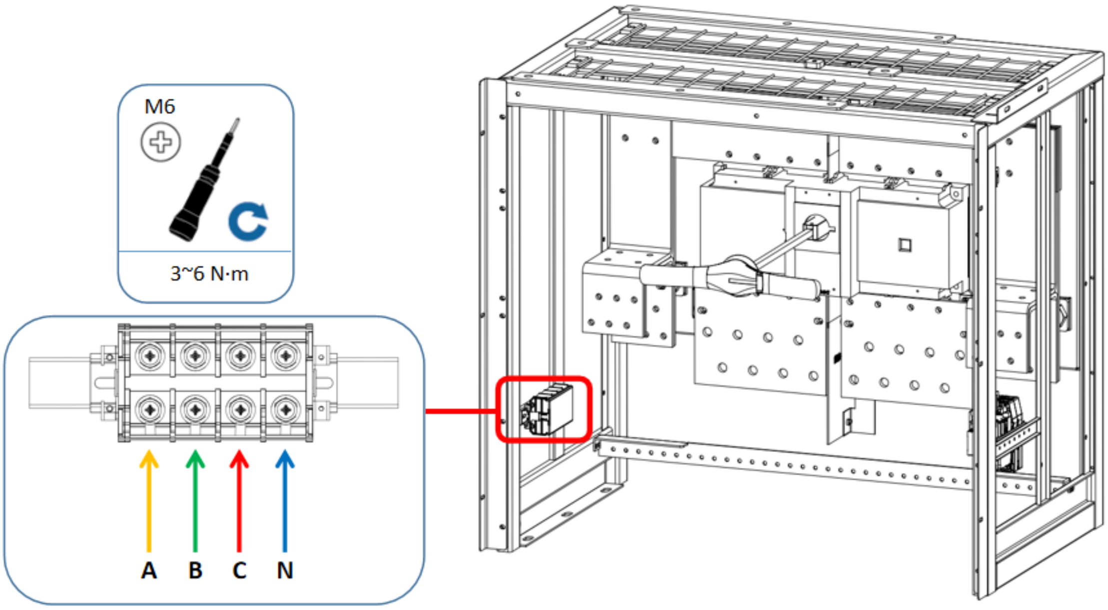


Fig. 5-6 AC Cable Connection Position

Table 5-8 Description of AC Cable Connection Port

Port	PIN definition	Cable specification	Recommended cable	Wiring aperture	Torque
J0: 1	A	1 * 25mm^2	Copper cable	M6	3~6 N m
J0: 2	B	1 * 25mm^2	Copper cable	M6	3~6 N m
J0: 3	C	1 * 25mm^2	Copper cable	M6	3~6 N m
J0: 4	N	1 * 16mm^2	Copper cable	M6	3~6 N m

5.4.2 Cable wiring

When connecting the cable, the cable must be put on and out from below through the lower outlet hole, pulled to the corresponding AC wiring row position, and made polarity marks to avoid polarity errors. Make sure to lock the OT terminal after wiring. Refer to the previous chapter for wiring steps.

5.5 Communication cable wiring

5.5.1 Wiring location

The wiring position of the communication wiring harness of the energy storage container is located on the central control panel on the control cabinet panel above the electrical cabin, and the specific position is at the LAN communication wiring jack at the bottom of the back of the central control display screen. As shown in the following figure.

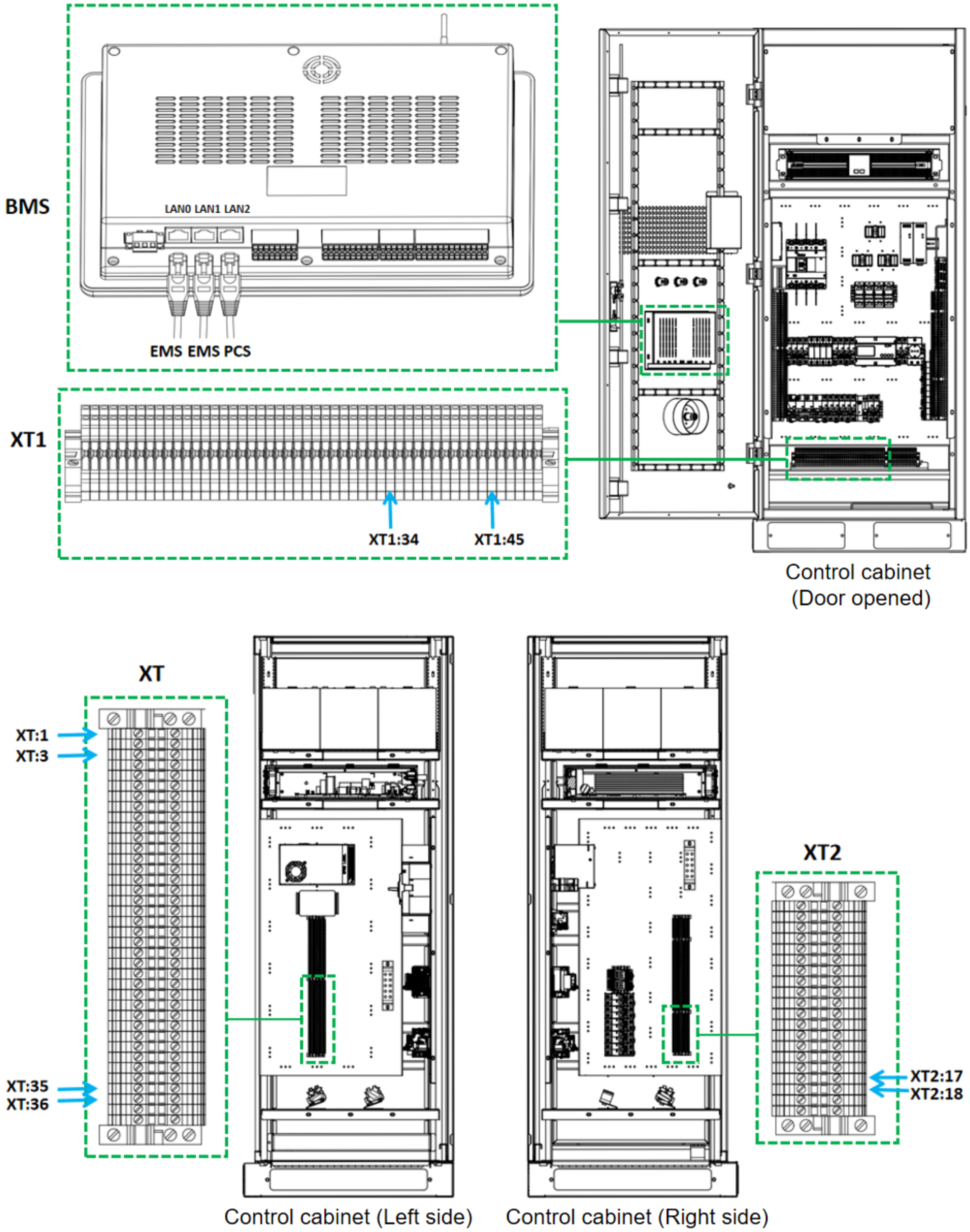


Fig. 5-7 Wiring Position of Communication Harness

Table 5-9 Communication Harness Connection Port Description

Port	Interface definition	Recommended cable	PIN definition	Interface form
LAN0	EMS-LAN	CAT-5e	/	RJ45
LAN1	PCS-LAN	CAT-5e	/	RJ45
XT	PCS-RS485	RVVSP 2*1.5	XT:1：485-A XT:3：485-B	E1508
	PCS-CAN	RVVSP 2*1.5	XT:35：CAN-H XT:36：CAN-L	E1508
XT1	PCS-COM(TO BMS)	RVVSP 2*1.5	XT1:34 XT1:45	E1508
XT2	PCS-COM(TO PCS)	RVVSP 2*1.5	XT2:17 XT2:18	E1508

5.5.2 Harness wiring

When wiring harness, the network cable must be put on and out from below through the lower outlet hole, pulled to the wiring area of the control cabinet, and led to the communication wiring position of the corresponding display screen, and marked with wiring harness to avoid wiring errors. Ensure that RJ45 is locked and not loosened after wiring.

When piercing wiring, a fixing point of binding belt is designed on the left side of the combiner cabinet, which can be used for fixing the wiring harness when it is led up. The top of the combiner cabinet is provided with threading holes, and the four corners of the bottom of the control cabinet are also provided with threading holes for wiring harness. After the network cable and other wires are connected, they can be fixed to the original wiring harness through tie ties. After entering the control cabinet, it can be wired in the original trunking.

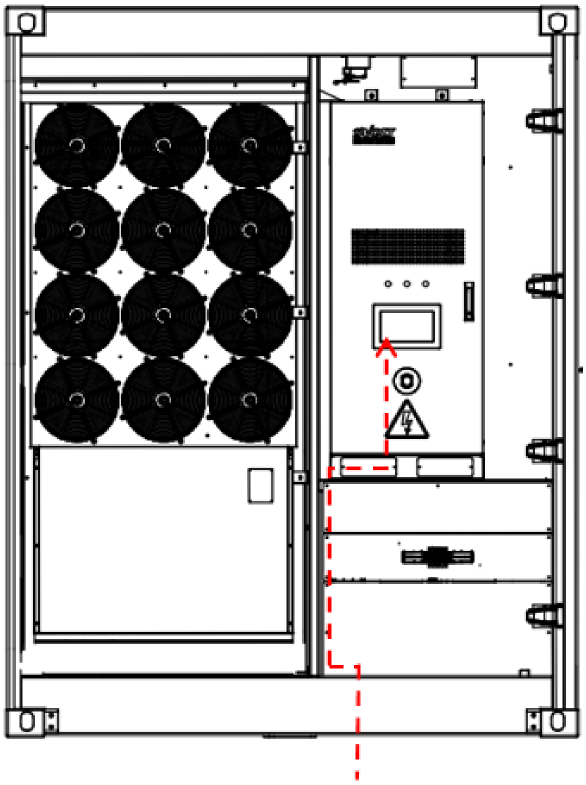


Fig. 5-8 Schematic diagram of communication line wiring

5.6 Grounding Cable Connection

5.6.1 Grounding position

The internal grounding of the energy storage container has been completed before delivery, and the external grounding of the container needs to be completed on the project site.

The energy storage container is grounded through an external grounding point. In order to facilitate on-site cable connection, two grounding points are designed at the bottom diagonal position of the container, which are located at the lower left corner of the front of the battery compartment and the lower right corner of the front of the electrical compartment. Please refer to the following figure for positions. It is used for connecting and grounding with the grounding grid pre-arranged in the case site. Refer to "4.2. 4 Embedded grounding" for the construction of grounding grid.

The product must be double grounded, and both grounding points should be reliably grounded. And grounding the products in strict accordance with local standards and regulations.

5.6.2 Grounding mode

There are two kinds of grounding methods: welding and fixing with grounding flat steel or locking and fixing with grounding cable. No matter the method of grounding flat steel or grounding cable is adopted, the exposed metal at the connection should be well protected against corrosion after grounding operation.

The recommended size of the grounding cable is half the cross-sectional area of the phase conductor, and the grounding cable must be identical to the phase conductor.

The grounding resistance shall be measured after grounding connection, and the resistance value shall not be more than 4 Ω.

（1）Grounding flat steel

The product is a protective grounding point, which will be covered with protective tape on the surface. When grounding the site, remove the protective band from the ground point and weld the hot-dip galvanized flat steel to the ground point (flat steel to cabinet lap area 50mm * 80mm). Spray the entire fixing surface after grounding.

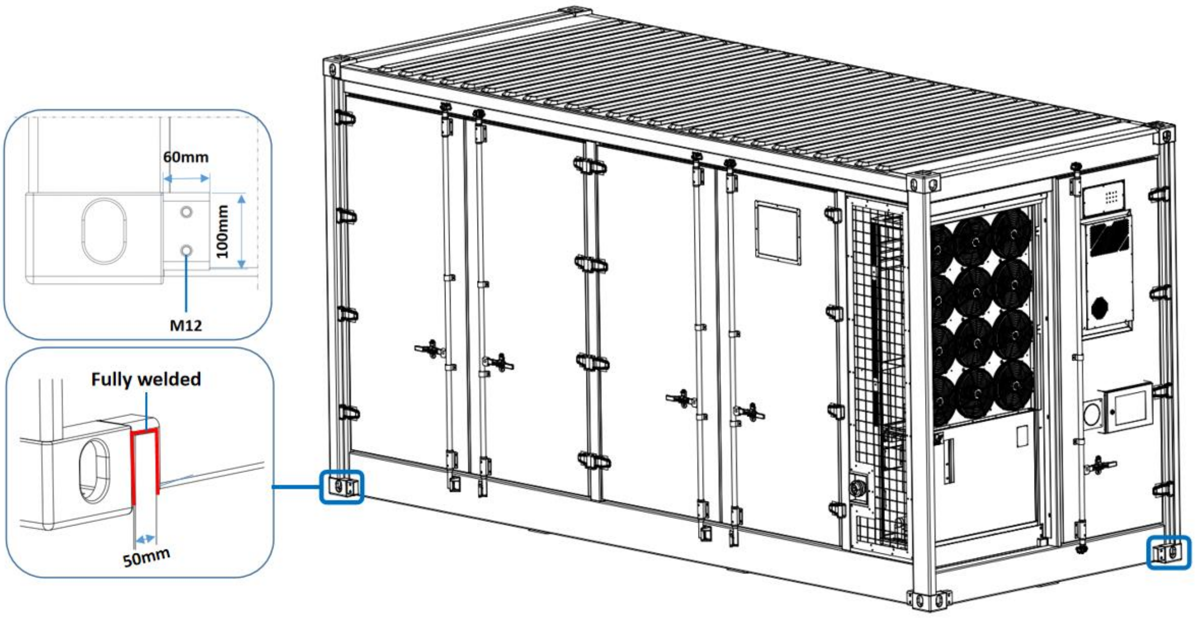


Fig. 5-9 Welding of Container Grounding Flat Steel

Table 5-10 Grounding Specifications for Flat Steel

Port	Grounding quantity	Specification (W*D)	Grounding form	Overlapping area
Earthing	2	2 * 50mm*6mm	Grounding flat steel	50mm*80mm

（2）Grounding cable

The product is a protective grounding point, which will be covered with protective tape on the surface. When on-site grounding, remove the protective band from the ground point and secure it using standard ground cable, using OT locking, torque 47 N. m to ensure reliable connection between the ground point and the system ground point.

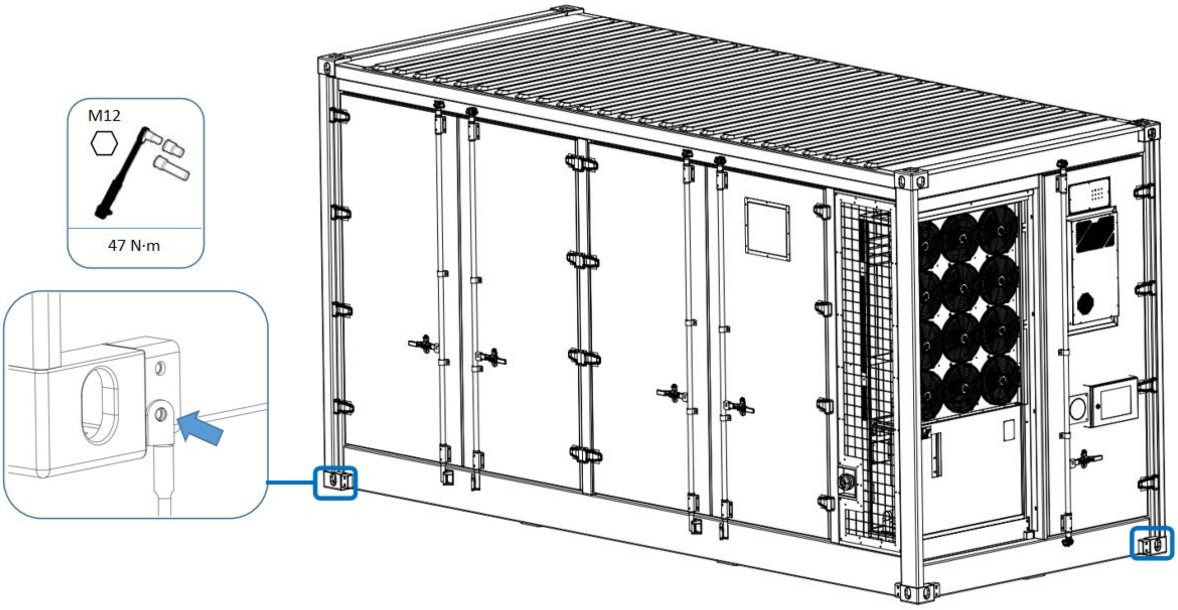


Fig. 5-11 Container grounding cable fastening

Table 5-11 Cable Grounding Specifications

Port	Grounding quantity	Specification	Grounding form	Wiring aperture	Torque
Earthing	2	2 * 180mm^2	Grounding cable	M12	47N m

5.7 Post-wiring inspection

After all electrical connections are completed, the system and wiring conditions need to be thoroughly and carefully inspected to prevent product or personnel damage caused by wrong wiring or improper operation.

➢ Check whether the polarity of DC cable wiring is correct, whether the nut is installed tightly, and whether the cable identification is correct. Check whether there is short circuit between positive and negative electrodes on DC side.

- Check whether the phase sequence of AC cable is correct, whether the terminal is fastened and whether the cable identification is correct. Check whether there is short circuit between phases and ground of AC cable terminals.

- Check whether the grounding cable is fastened, whether the cable identification is correct, and whether the grounding conduction is good.

- Confirm that the cable connection is reliable, without aging, fracture, insulation damage and abnormal bending.

After confirming the correct wiring and completing the inspection, seal the wiring hole at the bottom of the container and seal the gap around the cable inlet hole. After completion, close the cupboard door and lock it tightly to ensure that the closing and plugging do not interfere.

If the container is improperly sealed, foreign bodies such as moisture, rats and insects may enter.

6. Power up and down operation

6.1 Check before power-on

Before the energy storage container is powered on, it is necessary to recheck and check the equipment status and wiring to confirm that the system is in a safe and suitable power-on state. At the same time, confirm that the external equipment is in normal condition, and the system or project has grid connection or startup conditions.

Check at least the following:

- Check whether the wiring is correct.
- Check whether the protective cover inside the equipment is firmly installed.
- Check whether the emergency stop button is released.
- Check and make sure there is no ground fault.
- Check whether the AC and DC voltages meet the start-up conditions and ensure that the multimeter is not overvoltage.
- Check and ensure that no tools or parts are left in the equipment.
- Check all air inlets and outlets for blockage.

6.2 Power-on Operation

The control cabinet is equipped with various switches, which are used for on-off and protection control of each circuit. The switch should be closed step by step for power-on operation. The electrical switch in the control cabinet is shown in the following figure.

After each switch is operated according to the steps, the DC side of the energy storage container is closed and powered on, and then PCS is powered on. For PCS power-on steps, refer to the operation manual of corresponding models of each machine.

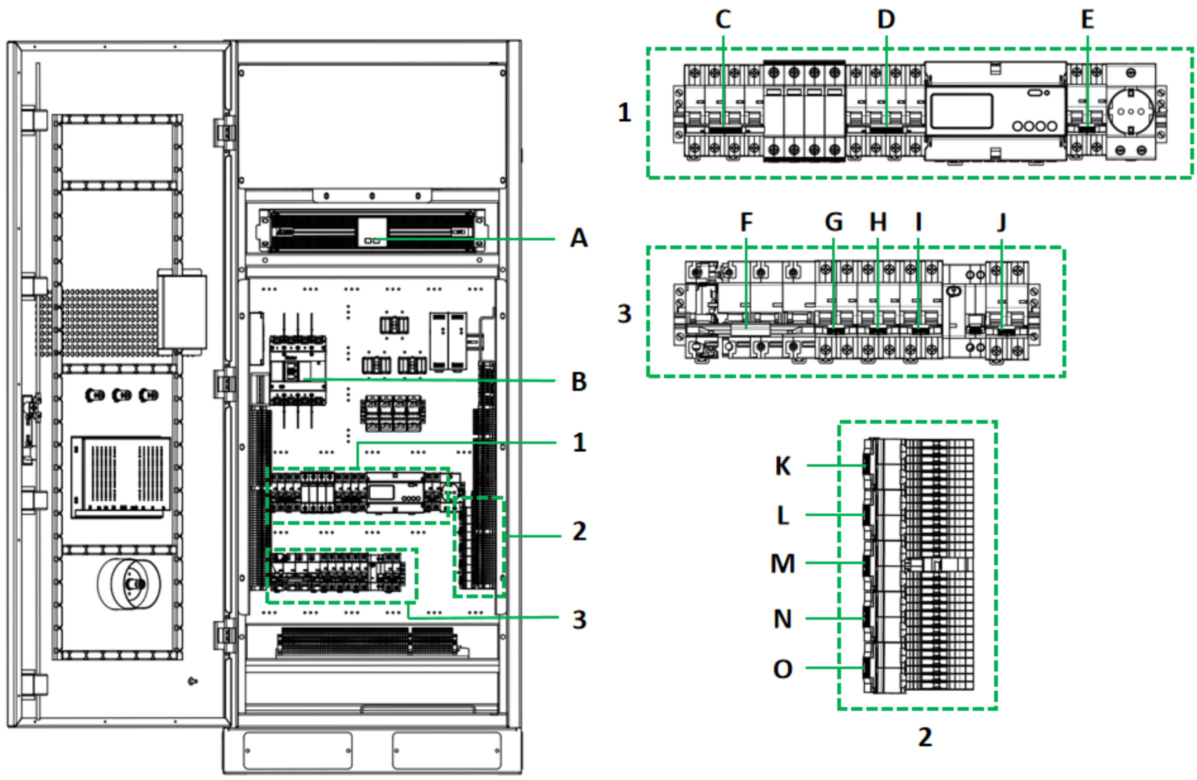


Fig. 6-1 Internal layout of control cabinet

Table 6-1 Control Cabinet Electrical Switch Description

S/N	Part name	S/N	Part name
A	UPS Start Button	I	QF5 UPS Micro circuit Break Switch
B	QF1 AC Master Switch	J	QF7 Lighting Switch
C	QF10 Surge Backup Protection Switch	K	QFA1 High Voltage Box 1~6 Power Supply Switch
D	QF11 Meter Switch	L	QFA2 High Voltage Box 7~12 Power Supply Switch
E	QFZ UPS Backup Switch	M	QFA3 Control Power Supply Switch
F	QF2 Cooling Machine Switch	N	QFA4 Fire Fan Switch
G	QF3 Air Conditioning Switch	O	QFA5 Fire Engine Switch
H	QF4 Dehumidifier Switch		

Please follow the following steps for power-on operation.

① Determine that the emergency stop switch is in an open state;

② Close the AC main incoming line switch QF1;

③ Close the lightning protector switch QF10 and the meter switch QF11, and then carry out the next operation after the voltage and other data are displayed on the display screen of the AC meter;

④ Close the UPS power supply switch QF5, then close the UPS standby switch QFZ, then press the UPS startup key for 3s for a long time, and carry out the next operation after the UPS is started;

⑤ Closing the Cooling machine power supply switch QF2, the air conditioner power supply switch QF3, the dehumidifier power supply switch QF4 and the lighting power supply switch QF7 in turn;

⑥ Closing of firefighting power supply switches QFA4 and QFA5;

⑦ Close the power button switch of the high-voltage box, then close the power supply switches QFA1 and QFA2 of the high-voltage box, and carry out the next operation after the power indicator lamp of the high-voltage box is red and the status indicator lamp is green;

⑧ Close the control power switch QFA3 in the cabinet. After the running indicator lamp on the control cabinet is always bright green, the power-on is completed;

Note: When the level 3 BMS is powered on in this step, it may give an alarm because the isolating switch in step 9 is not closed;

⑨ Close the general isolating switch in the electrical combining cabinet (before closing, it must be ensured that there is no insulation problem in the circuit to PCS end), and the power-on is completed.

Note: (1) If the isolating switch in step 9 is closed, there is an alarm , then need to check the fault; (2) All equipment has been powered on, and the DC side of the container is in a conductive state;

6.3 Power down operation

6.3.1 Conventional power down

Planned routine power-down operation is generally carried out during commissioning, repair, testing and maintenance of facilities during operation and maintenance, which belongs to planned power-off.

Please follow the following steps for power-on operation.

Ensure that the energy storage converter has been turned off before power down, and there is no power fluctuation in the energy storage container system

① After disconnecting QS (isolating switch) on the high voltage box of each battery cluster. The system can be powered down according to the operation;

② Disconnect the control power switch QFA3 in the cabinet;

③ Disconnect the high-voltage box power supply switches QFA1 and QFA2, and then disconnect the high-voltage box power button switch;

④ Disconnect the UPS power supply switch QF5, and then disconnect the UPS standby switch QFZ;

⑤ Disconnect cooling machine power supply switch QF2, air conditioner power supply switch QF3, dehumidifier power supply switch QF4, lighting power supply switch QF7 and fire fighting power supply switch QFA4、QFA5 in turn;

⑥ Disconnect the main incoming line switch QF1;

⑦ Disconnect the isolating switch. Power down is completed.

6.3.2 Emergency power down

In case of emergency, when emergency power-down is needed, press the emergency stop button on the container door or control cabinet door (the two emergency stop buttons have the same function and operate according to the actual situation) to close, and the system will stop urgently. The operation indicator light is extinguished, and the fault indicator light is always bright yellow. See the following figure for the position description of container emergency stop button.

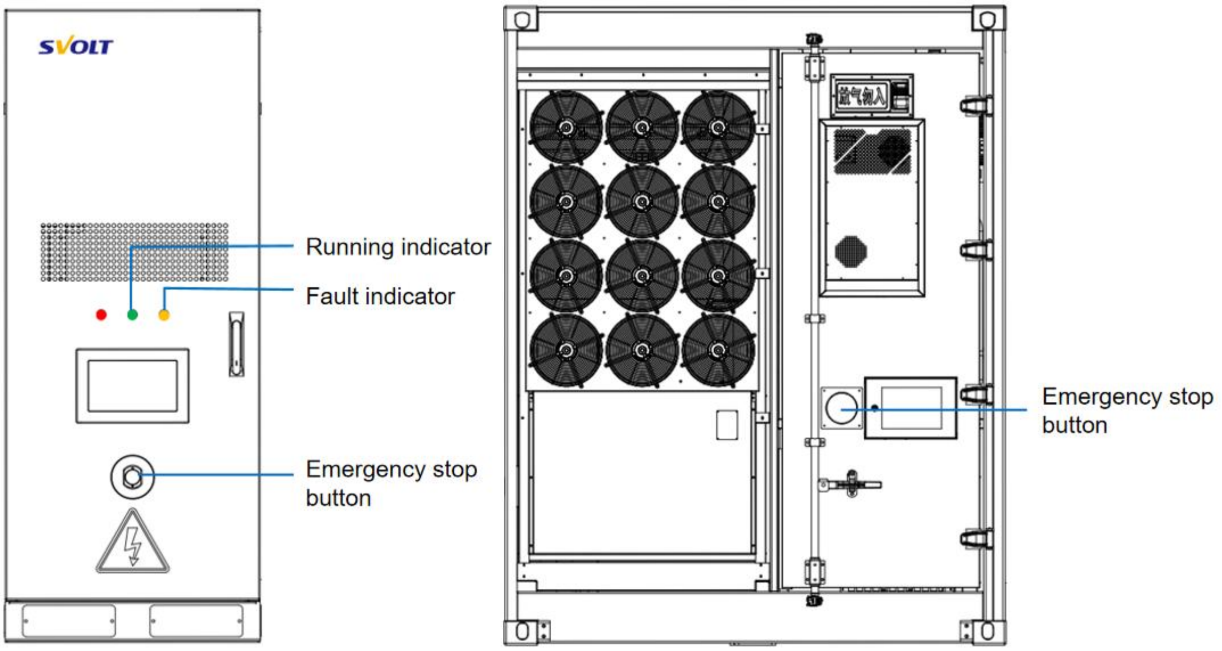


Figure 6-2 Position of Control Cabinet and Container Emergency Stop Button

7. Interface operation instructions

7.1 System Start up

When the ESMU is powered on, the system login interface is shown in Figure 7-1:

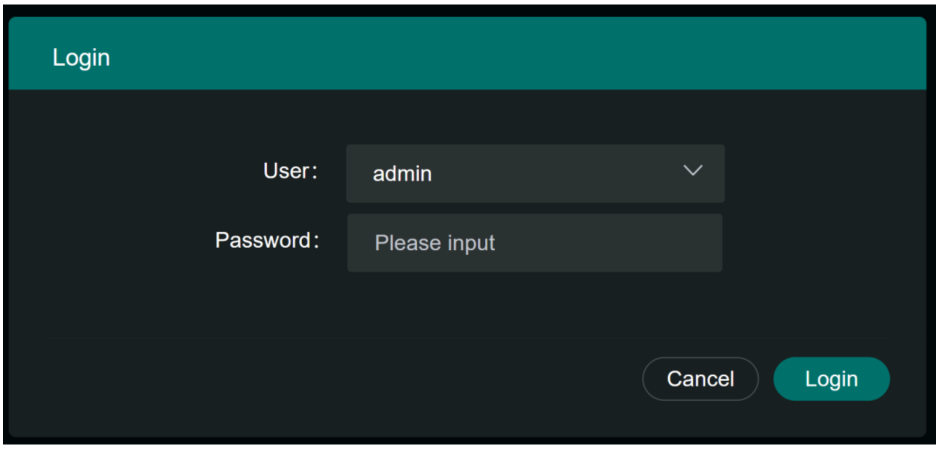


Figure 7-1 System Login Interface (Automatic Login Page)

Note: Login account is divided into administrator account (admin) and general user account. admin login password is 123456, and general user password can be set according to needs (default password: 123456). Administrator account is used to check battery running status, alarm information, basic parameter setting, data export, etc., and has the highest authority, while general user account does not have parameter setting authority.

7.2 Running Detection

- Basic information of cluster: View the basic information of stack connected to ESMU, including stack SOC, SOH, cumulative charge and discharge, insulation resistance, etc.;
- Auxiliary control information: Click to view the inquiry page of auxiliary control equipment (air conditioning, fire protection, DI/DO, temperature and humidity, electricity meter, etc.) connected to ESMU;
- View more: View the remote alarm and telemetry information of the battery stack;

- Cluster balance index: Consistency analysis of remaining electricity and current of battery stack;
- Current time: ESMU current time display;
- More operations: Drop-down menu of ESMU more operation functions;
- Alarm display: BMS communication status, battery stack, battery rack, battery cell real-time alarm display;
- Detail information for each cluster: Display the total voltage, current and power information of the battery stack, the voltage, current and power of each battery rack and the status of circuit breaker and contactor;
- Rack information summary: Display the summary information of each battery rack, including SOC/SOH, and extreme value information of voltage and temperature; The outer frame color of the battery rack indicates normal and primary/secondary/tertiary alarm states respectively;
- Rack information carousel: When you need to query the summary information of different racks, click the "carousel button" to scroll left and right to display the information of different racks;
- Cluster status: Display the current rack running status, including charging, discharging, stopping and other states;

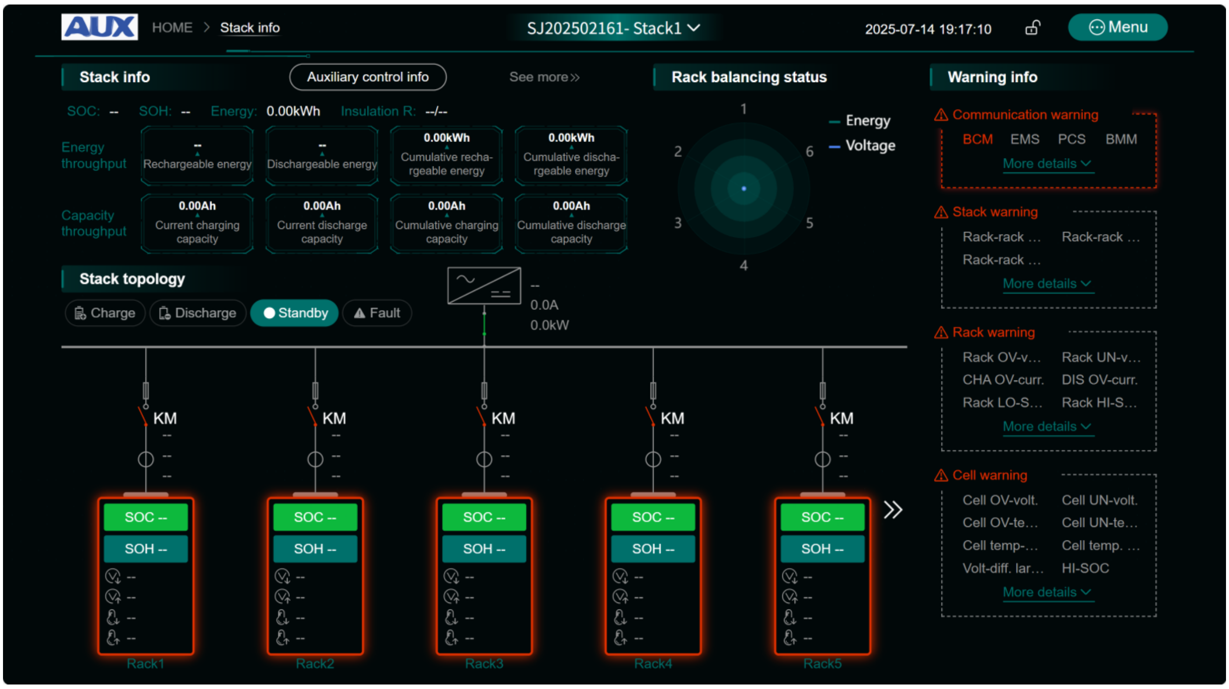
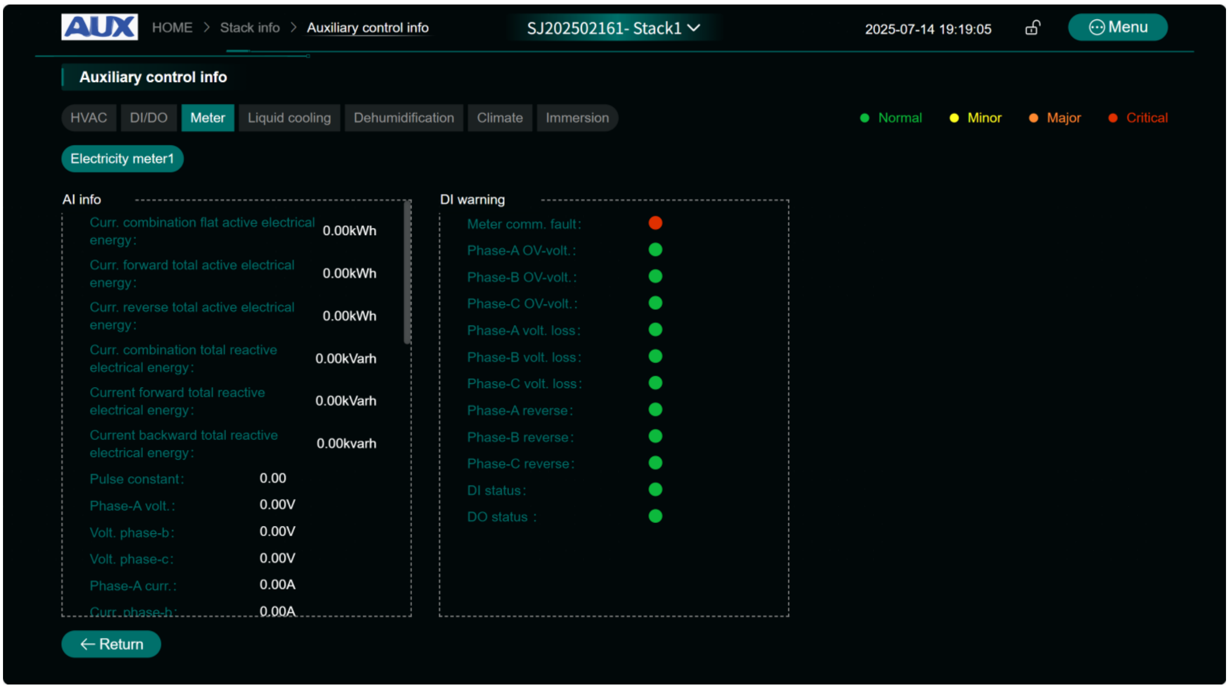


Fig. 7-2 ESMU Operation Chart

7.2.1 Auxiliary control information

Click [Auxiliary control info] on the auxiliary control information to inquire about the auxiliary control information accessed to ESMU, such as the detailed information of air conditioning, fire protection and other DIDO; If there is more information about fire fighting and air conditioning, it will be displayed in the form of a separate menu. Click [HVAC], [Meter], [Liquid cooling], etc. to inquire about detailed equipment information.



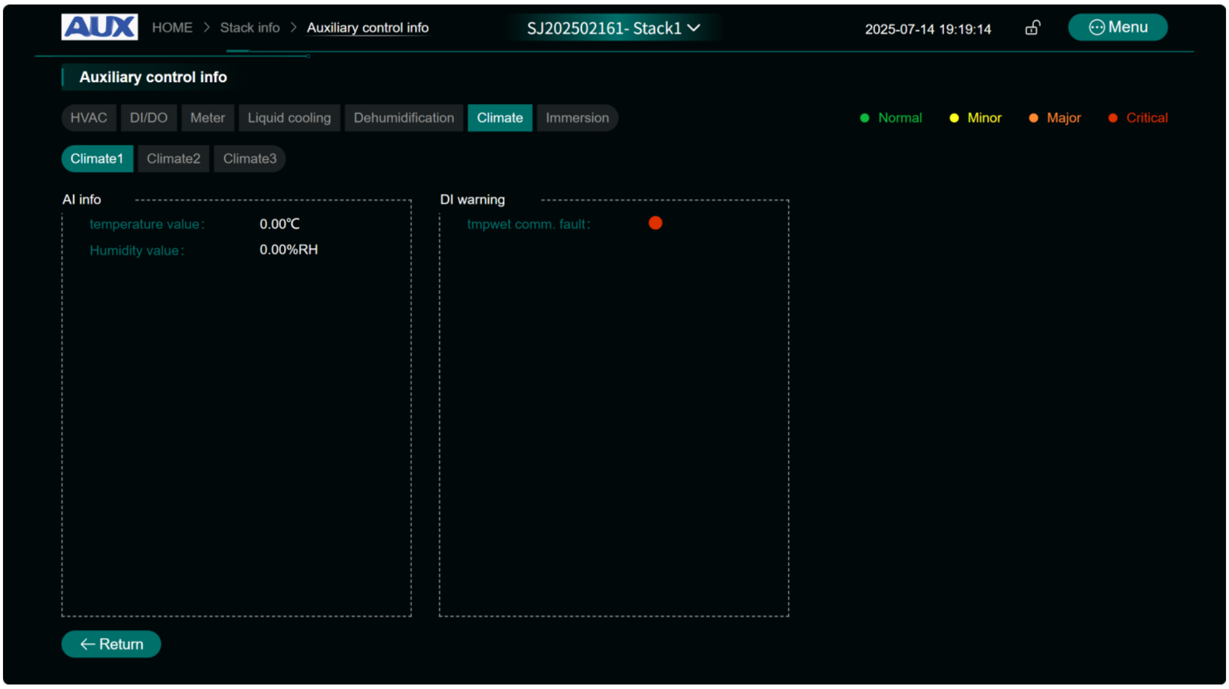


Fig. 7-3 ESMU Auxiliary control information

Click [See more] to view more for detail information and status of battery rack.

Fig. 7-4 Real-time AI information of Stack

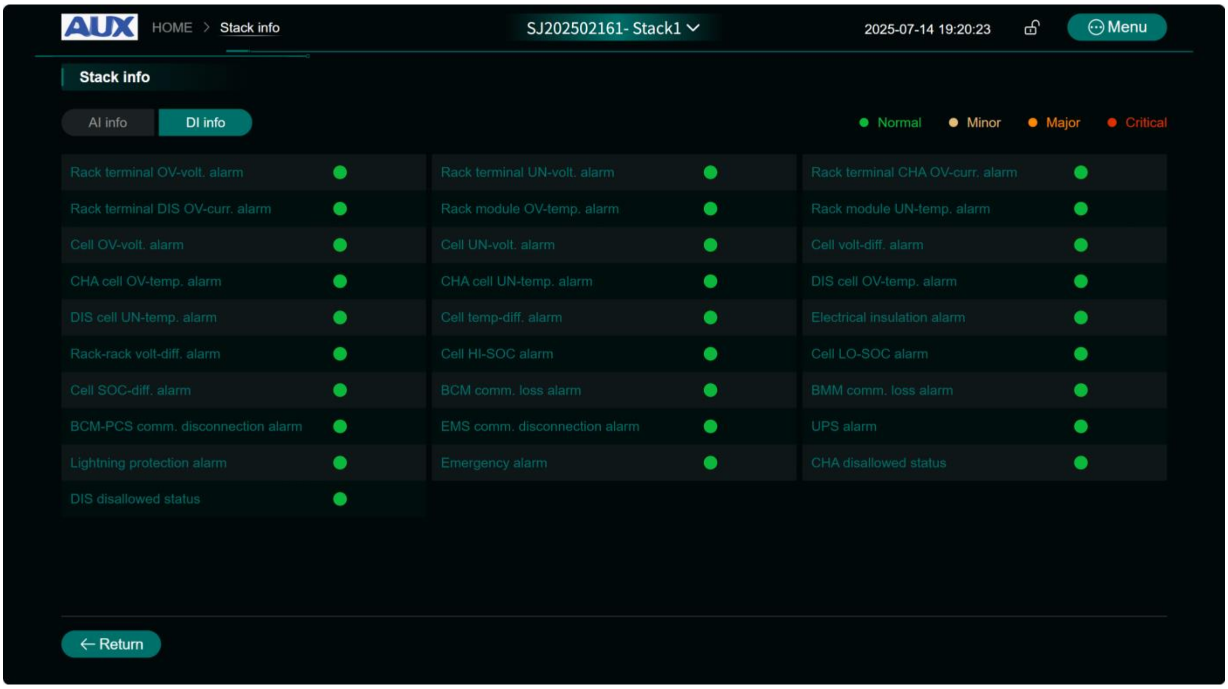


Fig. 7-5 Real-time DI information of Stack

7.2.2 Cluster details

Click on the summary information box of each rack in Figure 7-2 to query the detailed information of each rack respectively, as shown in the following picture.

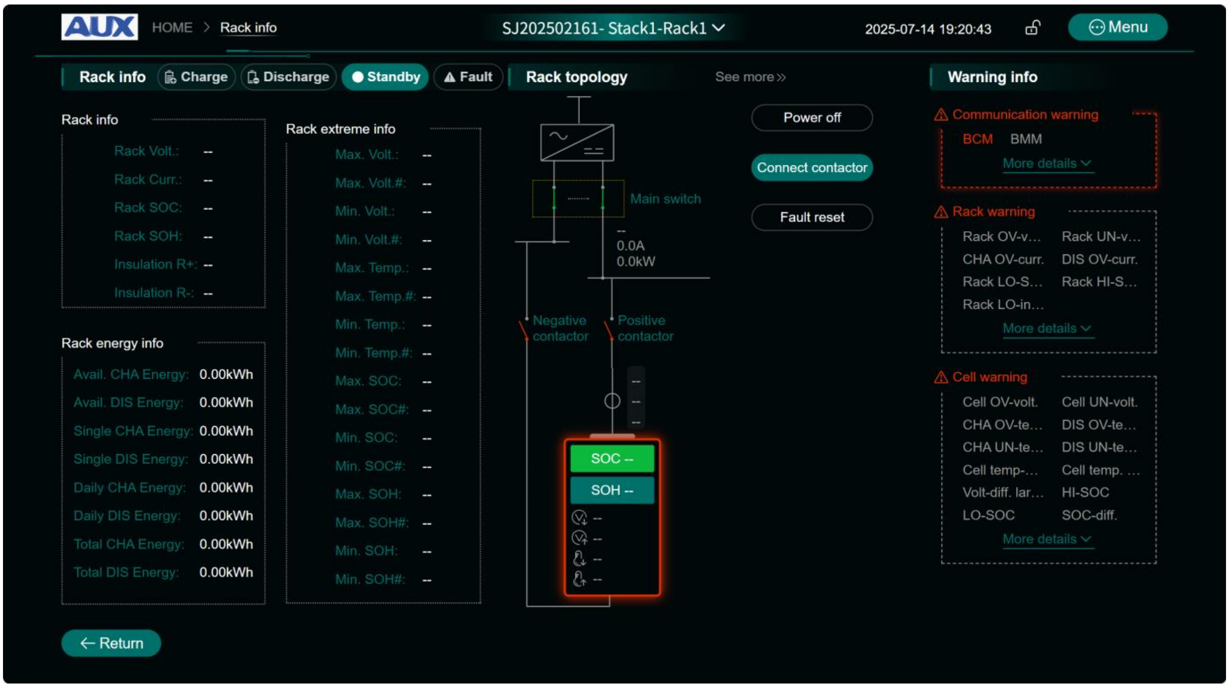


Figure 7-6 Cluster Information Interface

In the above figure, the rack status (charging, discharging or standby status, fault, etc.), rack summary information (rack voltage, rack current, rack SOC, rack SOH, rack insulation resistance), rack power information (single charge and discharge power, charging and discharging power, cumulative charge and discharge power), rack extreme value information, etc. are displayed in real time. In the middle of the page, the contactor status and current alarm status of the battery rack are displayed in real time, and realize power off, contactor control and fault reset, display all kinds of alarm information of this rack on the right side. Click on the rack Summary Information to query the details of cluster monomer, as shown in the following figure:

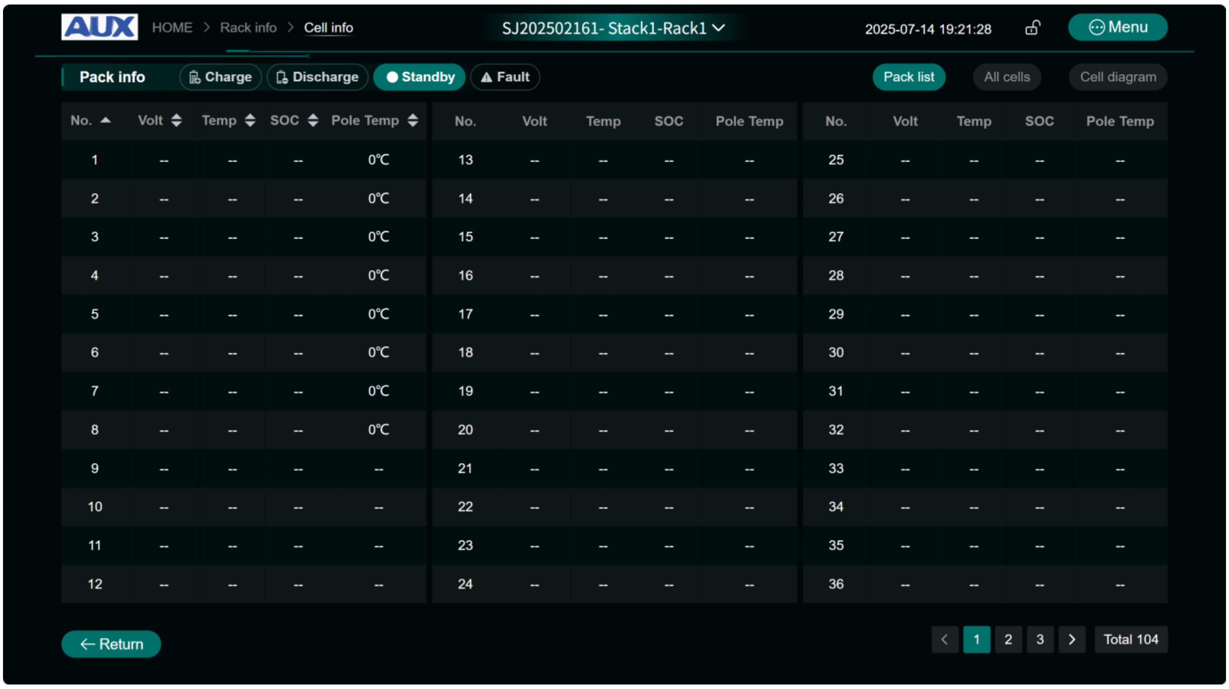


Figure 7-7 ESMU Monomer Information Interface

On this page, you can sort the cell voltage, cell temperature, cell SOC and cell SOH in the rack to query the real-time differences of different monitoring information according to different needs; Select a single cell, you can view the real-time curve of voltage, current, temperature and SOC of a single cell, and the time period of displaying data curve can be adjusted; You can also click on the [Cell diagram] or specific Cell row on the upper right side to query the real-time histogram and real-time curve of the whole cluster of cells, and display that the data period can be adjusted.

Fig. 7-8 Information histogram of whole cluster single cell

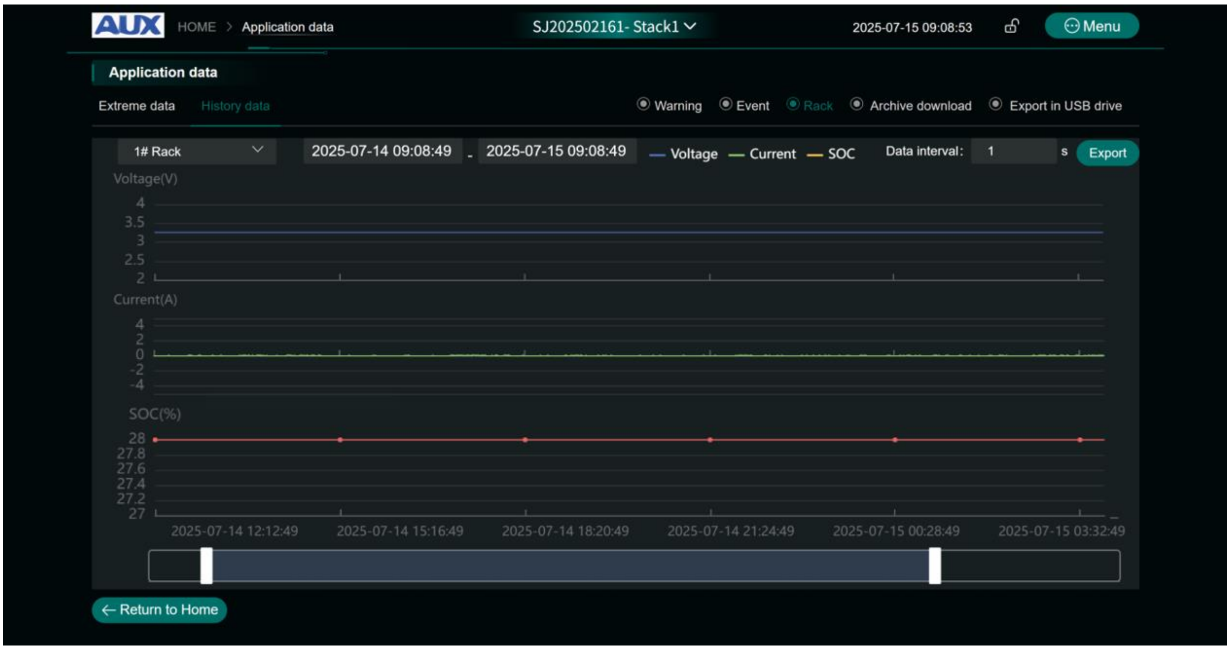


Fig. 7-9 ESMU Cell Information Curve

7.3 Real Time Alarm

In the alarm information column on the right side of Fig. 7-2, BMS communication status, real-time alarm information of battery stack, battery rack and battery cell can be displayed respectively; Click [See more] to inquire about the specific alarm type, alarm description and alarm start time of alarm items respectively; Under the [Warning info] column, you can click or select the contents of communication status, stack alarm, rack alarm, single cell alarm and Aux alarm respectively. By selecting the rack option drop-down menu on the right side, you can select the rack to be queried respectively.

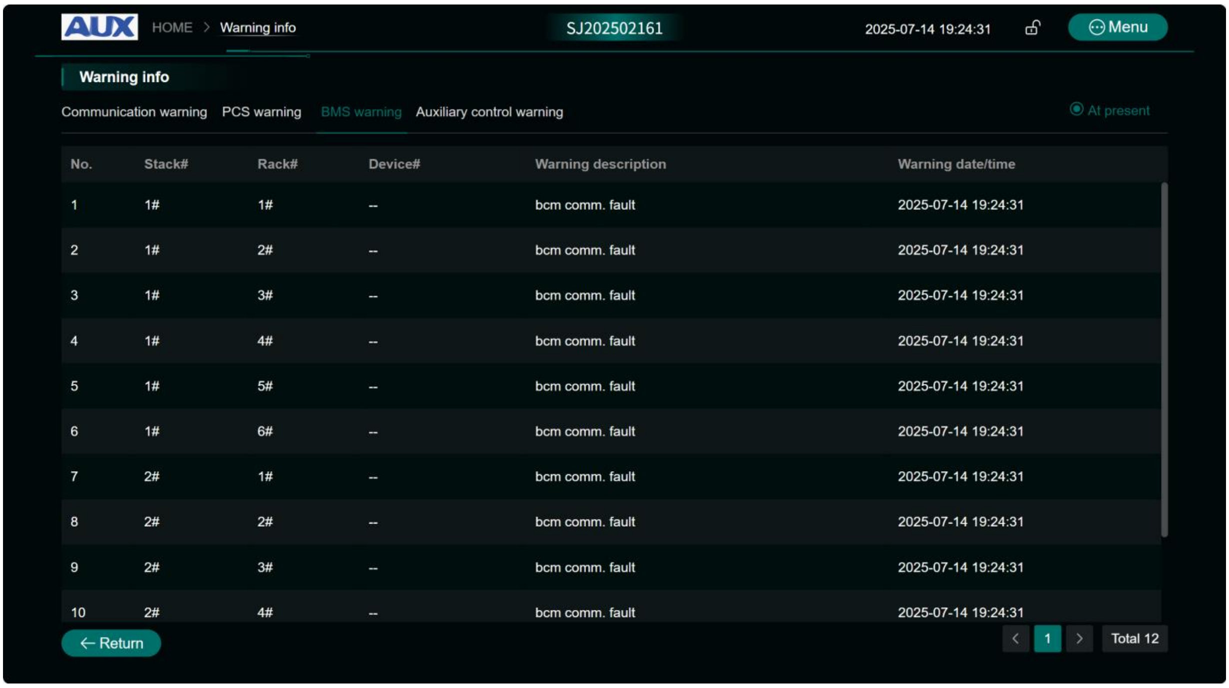


Figure 7-10 Real Time Alarm Information

On the right side of the real-time alarm interface, the alarm information range can be screened, and the real-time alarm information can be queried in single rack or multiple racks.

7.4 More Operations

Click [See more] on the right side of Figure 7-2, and select menus such as [Manipulation], [Parameter settings], [Device settings] and [About System] to make corresponding inquiries and operations.

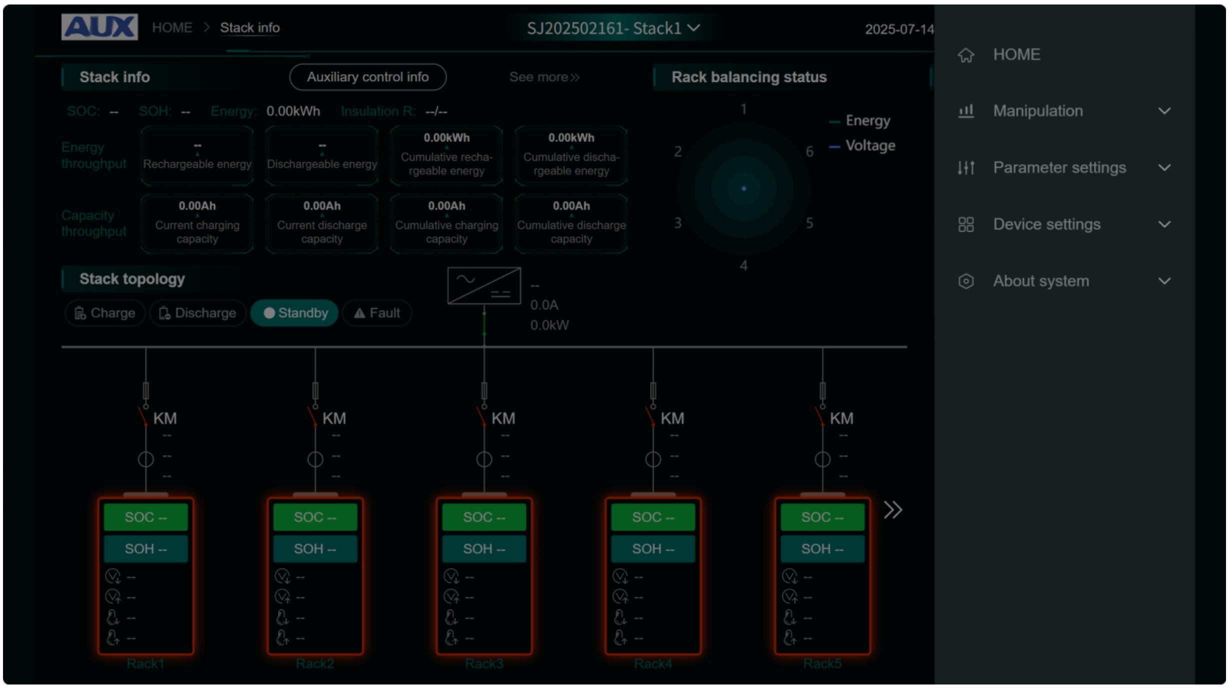


Figure 7-11 System Menu Bar

7.5 Working condition data

Working condition data include "container real-time information", "rack real-time information", "cell information", "extreme value information", "historical data" and "real-time alarm".

Select the "Container Real-time Information" menu to enter the Container Real-time Information interface.

Select the "Cell Information" menu to enter the Cell Information interface.

Select the "Extreme Value Data" menu, select this menu, and the system will display the extreme value information of each rack, including the max./min. Voltage value and serial number of cell, the max./min. Temperature value and serial number of cell, the max./min. SOC/SOH value and serial number, etc.

Fig. 7-12 Extreme Data Graph

Click on the [Extreme data] of each rack to view the extreme value change curve of voltage, temperature, SOC and SOH information, and select the time range, as shown in the following figure:

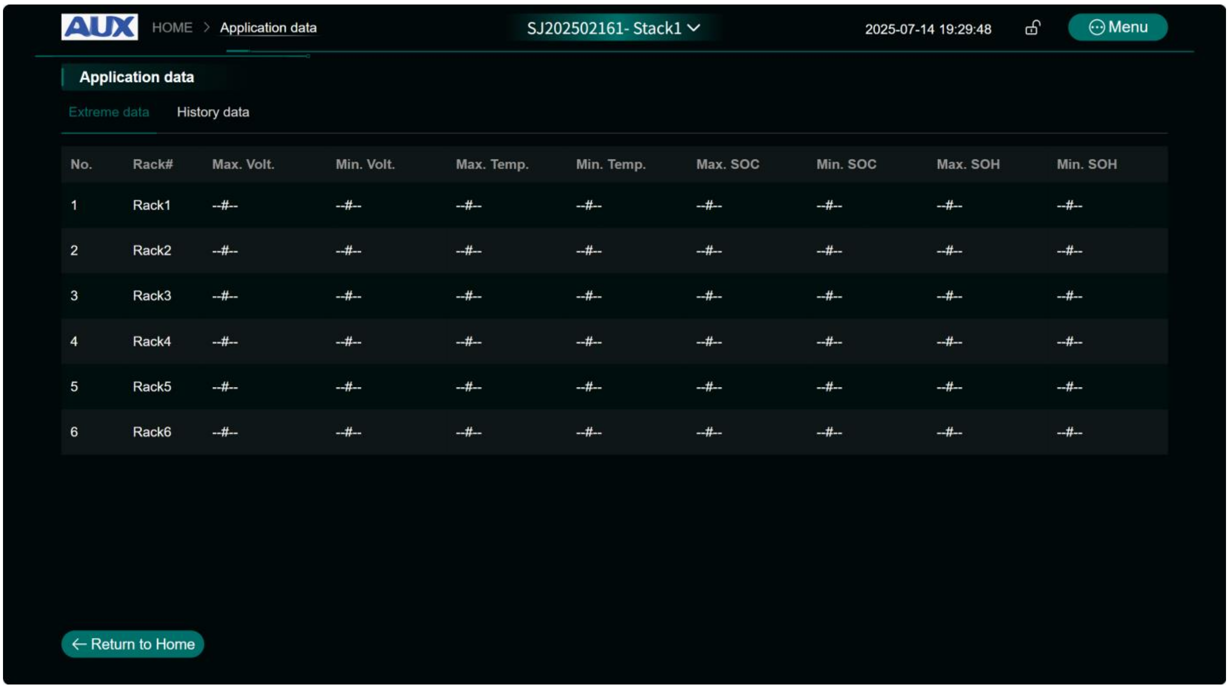


Figure 7-13 Extreme Data Interface

Select the [History Data] menu to query and export history data such as alarms and events, and support query by events and time periods according to equipment classification.

Select "Warning", "Event",etc. on the right to query the corresponding historical data respectively, and support the export of alarm and event historical data and the download of data.

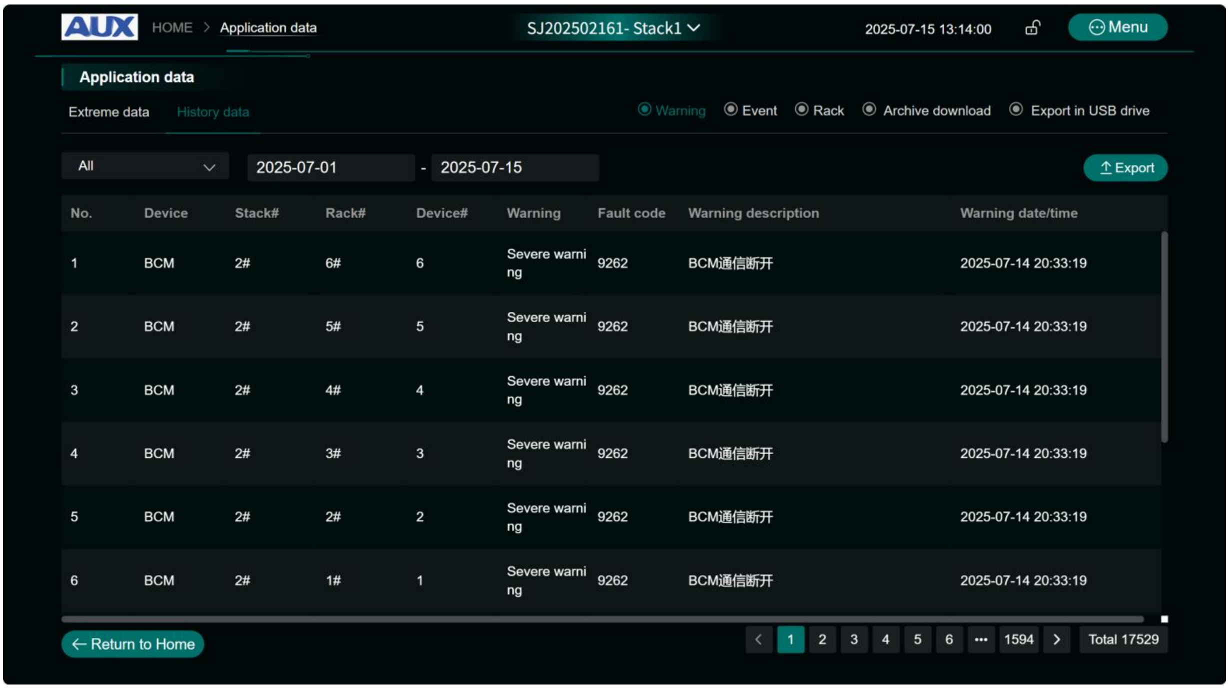


Figure 7-14 Historical Data Query Interface

7.6 Parameter Settings

Select the [Parameter settings] menu on the right to set the communication interface and system parameters.

Fig. 7-15 Parameter Setting Interface

The above figure shows the interface for Parameter settings, which can set the basic parameters of each rack and the basic protection parameters of single cell respectively.

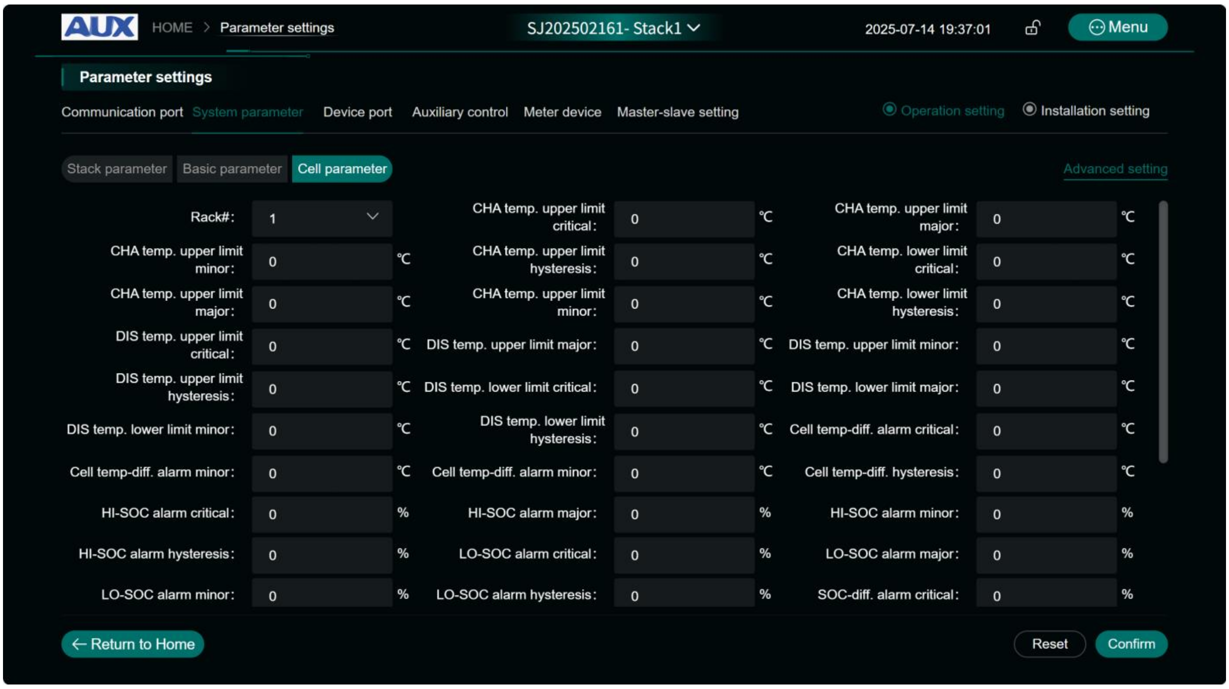


Fig. 7-16 Cluster Protection Parameter Setting Interface

The "Advanced Settings" menu carries out manual equalization control for each battery rack and initializes SOC, SOH and accumulated power of the rack.

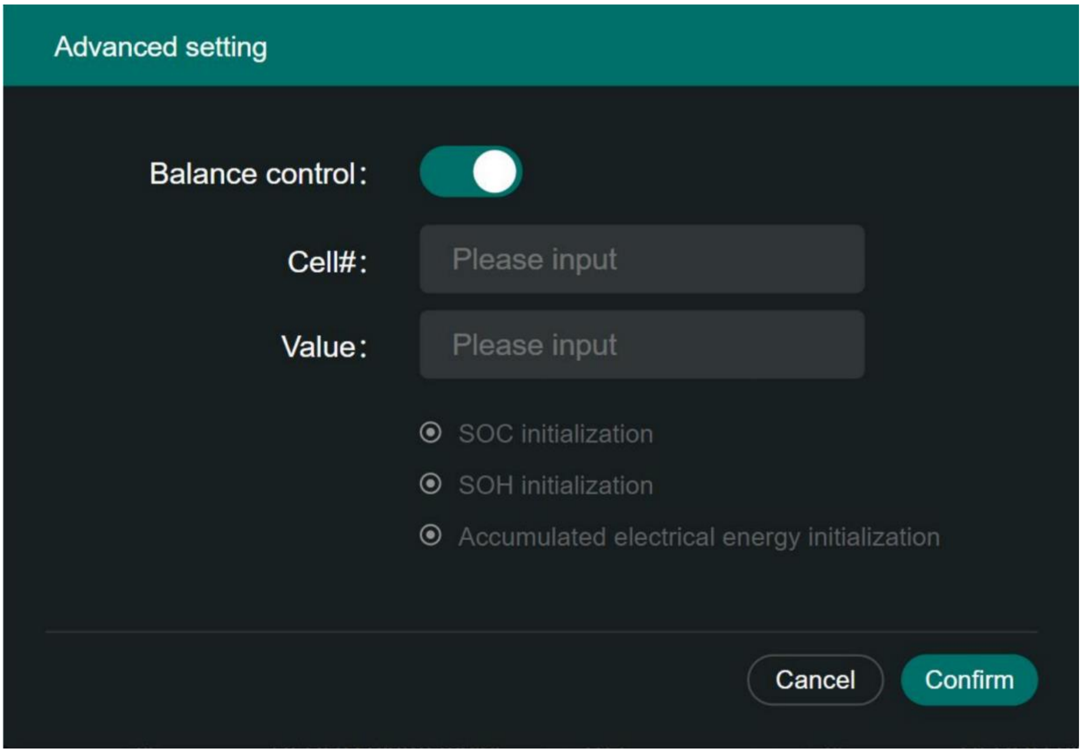


Figure 7-17 Advanced Settings

"Stack Parameters" can set the starting address of battery rack, the number of battery racks and the number of battery racks in a single battery rack.

"Rack Parameters" can be set with parameters such as the number of BMUs, battery type, nominal battery capacity, current range and acquisition deviation and system operation time.

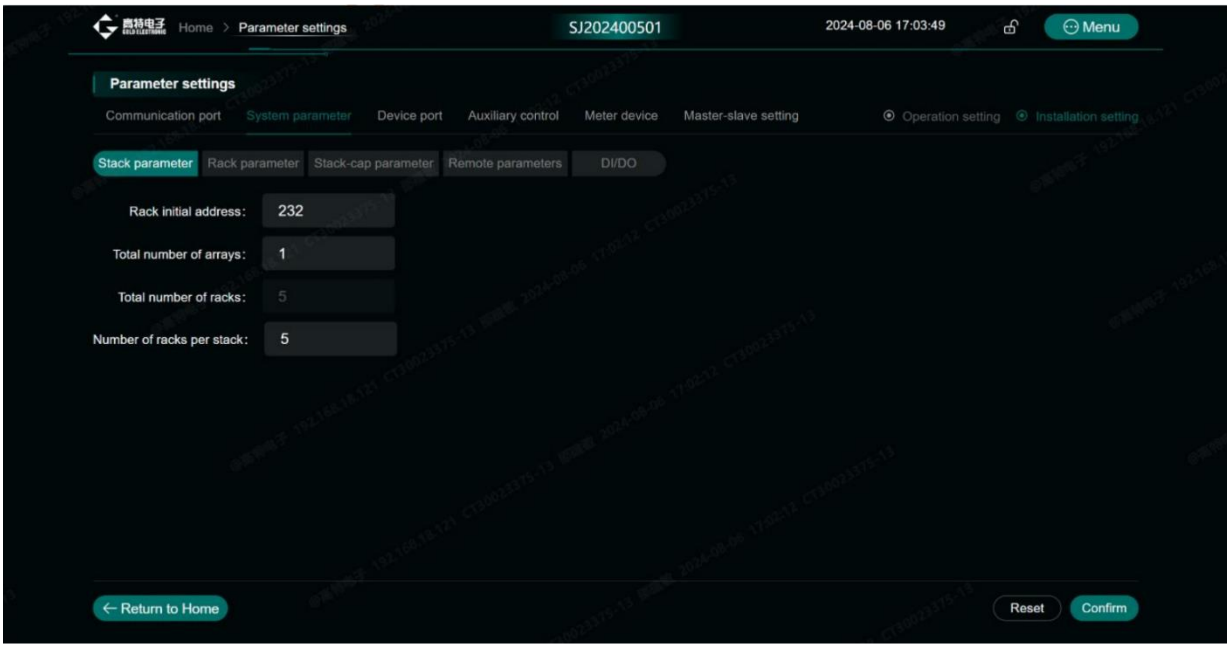


Figure 7-18 Heap Parameter Settings

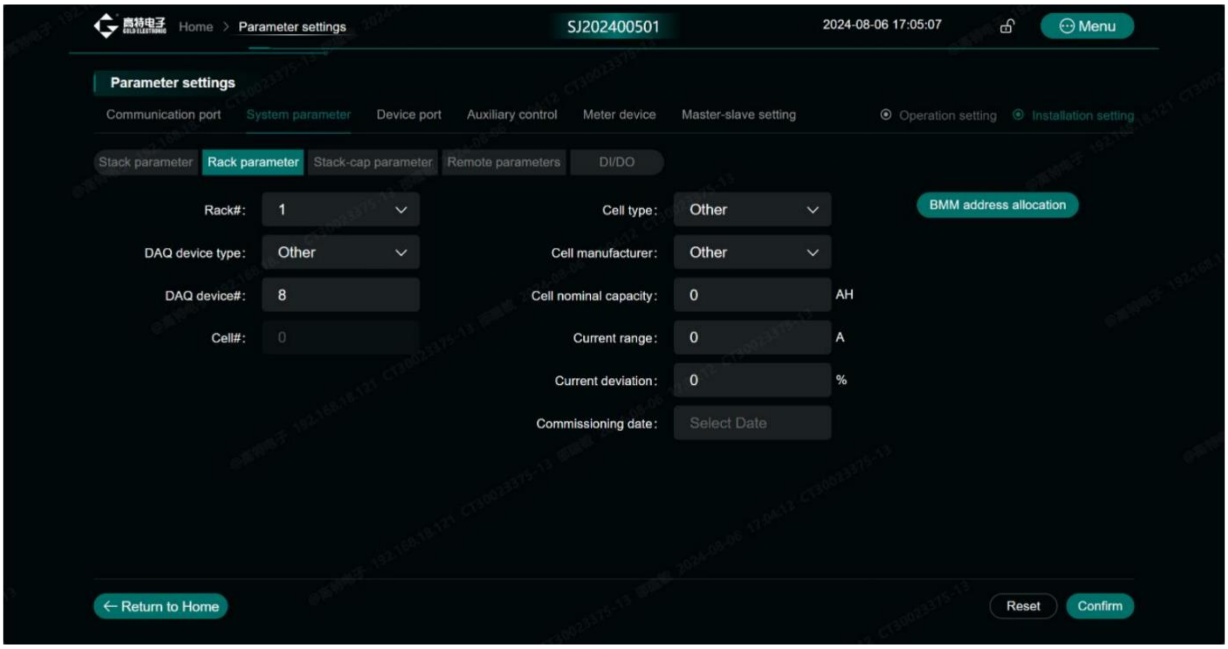


Figure 7-19 Cluster Parameter Settings

Parameters in "System Parameters" are read from BMS master device by default, or can be set separately as administrator. It is recommended to keep the parameters consistent with BMS master device.

8. Troubleshooting and handling

8.1 Troubleshooting of Container System

When the container energy storage system is abnormal or faulty, the fault information can be read from the current alarm interface on the display screen (the icon in the upper right corner of the main interface explains as follows: the alarm state is when the system is normal and becomes when the system is faulty, and clicking this icon can enter the current alarm interface). Please check and troubleshoot according to the treatment method in Table 8-1 first. If the problem persists, please contact the supplier for technical support. If you need to report the failure to our company or distributor, please record and inform the equipment code (refer to the equipment nameplate for details).

Table 8-1 System Failure and Treatment

Fault name	Treatment method
AC over-voltage	Check whether the mains voltage is within the range
AC under-voltage	Check whether the mains voltage is within the range
AC over-frequency	Check whether the mains frequency is within the range
AC under-frequency	Check whether the mains frequency is within the range
Reverse phase of grid	Check whether the phase of the grid is reversed
Power grid phase locking failure	Please contact your supplier
Auxiliary power failure	Please contact your supplier
High voltage of bus	Please contact your supplier
Low voltage of bus	Please contact your supplier
Voltage of bus is unbalance	Please contact your supplier
Output overload timeout	Check whether the load is within the specified range
Inverter overload	Please contact your supplier
DC input over voltage	Check whether the DC voltage is higher than the upper limit protection value of battery voltage

DC input low voltage	Check whether the DC voltage is lower than the lower limit protection value of battery voltage
AC soft start fault	Please contact your supplier
DC soft start failure	Please contact your supplier
DC over current fault	Check whether there is a short circuit at the battery port
Communication failure	Check communication lines and parameter settings
Slight alarm when the temperature of discharge is too low	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Slight alarm when discharge temperature is too high	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Slight alarm when low insulation	Insulation detection failure or system leakage
Slight alarm when voltage difference is too large	Detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Slight alarm when temperature difference is too large	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Slight alarm when SOC is too low	The alarm of SOC value lower than the protection value is a normal phenomenon, and the alarm is eliminated according to the control strategy. If the voltage does not match the SOC, the SOC is not calibrated or the SOC error accumulates, which requires a complete charge-discharge cycle of the system.
Slight alarm when charging temperature is too low	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Slight alarm when charging temperature is too high	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Slight alarm when discharge over current	Whether the charging and discharging current of the system is higher than the protection value, check whether the PCS is operating normally
Slight alarm when cell block under voltage	It is normal to alarm when the voltage is lower than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.

slight alarm when cell under voltage	It is normal to alarm when the voltage is lower than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Slight alarm when charging over current	If the charging and discharging current of the system is higher than the protection value, check whether the PCS is running normally
Slight alarm when over voltage at cell block	It is normal to alarm when the voltage is higher than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Slight alarm when cell over voltage	It is normal to alarm when the voltage is higher than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Moderate alarm of low discharge temperature	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Moderate alarm of high discharge temperature	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Moderate alarm of low insulation	Insulation detection failure or system **electric** leakage application remote background turn off PCS, BMS relay and contact your supplier.
Moderate alarm of low charging temperature	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Moderate alarm of high charging temperature	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Moderate alarm of discharge over current	If the charging and discharging current of the system is higher than the protection value, check whether the PCS is running normally
Moderate alarm when cell bolck terminal under voltage	It is normal to alarm when the voltage is lower than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Moderate alarm when cell under voltage	It is normal to alarm when the voltage is lower than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.

Moderate alarm when charging over current	If the charging and discharging current of the system is higher than the protection value, check whether the PCS is running normally
Moderate alarm of over voltage at block terminal	It is normal to alarm when the voltage is higher than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Moderate alarm of cell over voltage	It is normal to alarm when the voltage is higher than the protection value, and the alarm is eliminated according to the control strategy. If it cannot be eliminated, detect whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Severe alarm when discharge temperature is lower	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Severe alarm when higher discharge temperature is higher	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Severe alarm when low insulation	Insulation detection failure or system leakage application remote background turn off PCS, BMS relay and contact your supplier.
Severe alarm when temperature of pole is higher	Whether the connector connection of high voltage box is reliable; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Severe alarm when charging temperature is lower	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Severe alarm when Charging Temperature is higher	Whether the temperature sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check whether the chiller is working properly. If it is not working properly, please contact your supplier.
Severe alarm when discharge overcurrent	If the charging and discharging current of the system is higher than the protection value, check whether the PCS is running normally
Severe alarm when block terminal under voltage	If the voltage is lower than the protection value, check whether PCS stops when the alarm is moderate, and check whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Severe alarm when cell undervoltage	If the voltage is lower than the protection value, check whether PCS stops when the alarm is moderate, and check whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Severe alarm when charging over current	If the charging and discharging current of the system is higher than the protection value, check whether the PCS is running normally
Severe alarm when block over voltage	If the voltage is higher than the protection value, check whether PCS stops when the alarm is moderate, and check whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.

Severe alarm when cell over voltage	If the voltage is higher than the protection value, check whether PCS stops when the alarm is moderate, and check whether the battery sampling harness is broken or the sampling plug-in connection is reliable; Whether there is any abnormality in BMU operation; Check the system battery voltage. If the battery cell is damaged, please contact your supplier to replace the damaged battery cell.
Communication with master control is failure	Check communication lines and power supply lines
Communication with slave control is failure	Check communication lines and power supply lines
Slave control equipment failure	Please contact your supplier
Main positive disconnection fault	Please contact your supplier
Main positive closing fault	Please contact your supplier
Main negative disconnection fault	Please contact your supplier
Main negative closing fault	Please contact your supplier
Sensor failure	Please contact your supplier
High and low voltage alarm	Check AC power supply or voltage; Call or power up again or please contact your supplier.
High and low temperature alarm	Check whether the system air duct is unobstructed. Whether the ambient temperature is within the allowable range; Clean the condenser regularly; Please ask professionals to check and repair or contact your supplier;
High and low voltage alarm	Please contact your supplier
Compressor alarm	Clean the condenser regularly; Users are also required to check whether the voltage is stable and meets the design requirements; Or please contact your supplier;

For system detection fault problems,, the abnormal state is divided into three levels to distinguish the severity of the fault. The actions and severity of the system for each level of fault judgment are shown in the following table.

Table 8-2 Treatment Measures for Different Levels of Failure

Alarm level	Severity	Output of fault
Level 1	Slight	alarm only
Level 2	Moderate	No charging and no discharging
Level 3	Severe	Container system shutdown

Refer to "Handling Methods" for common faults and troubleshooting. If the problem cannot be solved according to this manual, please contact your supplier. Providing information requires at least the following to help you better.

1) Manufacturer, model and configuration information of equipment, product nameplate;

2) Project information corresponding to products, project operation location and environment;

3) The operation situation and operation method, operation method and steps of the product site;

4) The system serial number read by the display screen and the software version of the equipment;

5) Description of fault situation and phenomenon of fault display;

6) Live pictures.

8.2 Thermal Management System Troubleshooting

Refer to the following table for common faults and troubleshooting of thermal management system. If the problem cannot be solved, please contact your supplier.

Table 8-3 Treatment measures for fan faults

Phenomenon	Possible causes	Check the item or processing method
External circulation fan does not operate	1. The chiller is not powered on; 2. The circuit breaker tripped after being struck by lightning;	1. Check whether there is electricity at the power input of the chiller; 2. Check whether the internal circuit breaker of the chiller is closed;
	Abnormal power input of chillers, such as over voltage and under voltage of power supply	Determine whether there is a corresponding alarm in the chiller
	Fan jammed	Check whether there is any foreign matter stuck in the fan
	Loose terminal	Check whether the fan butt terminal is loose
	Compressor not started	The external fan starts after the compressor starts
	Control panel failure	Replace the control panel
	Fan failure	Replace the fan
Abnormal noise from the external circulation fan	Wear of fan bearings	Replace the fan
	Fan blades scratch other objects	Check whether there are cables and other interference with fan blades

Table 8-4 Treatment Measures of Refrigeration System

Phenomenon	Possible causes	Check the item or processing method
Compressor Do not start	Power is not turned on (standby)	Check the main power switch and check the operation display interface whether it has been turned on
	Loose circuit connection	Tighten circuit connector
	Open circuit or short circuit	Check the open circuit or short circuit of the circuit and repair the main power supply
	Frequency converter fault	Replace frequency converter
	Control panel damage	Replace the control panel
	Compressor motor fault	Replace the compressor
Compressor Not working	No cooling requirement	Check the outlet water temperature on the display screen interface and the output state of the compressor, and check whether it is in refrigeration state in the operation interface
	Downtime delay	The compressor has the shortest downtime under normal conditions. If the temperature rises to the open point again during this period, the compressor will still be delayed to start
High exhaust pressure	Dirty blockage of condenser	Clean condenser with compressed air or vacuum cleaner equipped with brush head
	External circulation fan does not operate	Refer to the fan fault table above
	The ambient temperature is too high	Check whether the ambient temperature exceeds the allowable operating range of the unit

Table 8-5 Troubleshooting Measures for Cooling Medium Circulation System

Phenomenon	Possible causes	Check the item or processing method
The internal circulating water pump does not start	Power is not turned on (standby)	Check the main power switch and check the operation display interface whether it has been turned on
	Loose circuit connection	Tighten circuit connector
	Fault of water pump frequency converter	Replace the pump frequency converter
	Self-protection of pump due to no coolant	Check whether there is coolant in the circulation system. If there is no coolant, it needs to be replenished
	Pump body failure	Replace ther pump
The electric heating tube does	No heating requirement	Check whether the outlet water temperature and heating set point are set reasonably

not work	Loose circuit connection	Tighten circuit connector
	Overheat protection of electric heater	After waiting for a period of time, restart the electric heating and observe whether the electric heater works normally
	Electric heater failure	Replace the electric heater
Automatic make-up pump is not running	Failure of make-up water pump	Replace automatic make-up water pump
	The make-up water pump is stuck	Check whether there are foreign bodies in the water supply tank, and remove them if there are; If the make-up water pump is damaged, please replace it;
	Loose terminal	Check whether the butt terminal of make-up water pump is loose
Abnormal sound of automatic make-up pump operation	Axis wear of make-up water pump	Replace automatic make-up water pump
	Loose fixing screw of make-up water pump	Fastening screw

8.3 Fire Protection System Troubleshooting

Refer to the following table for common faults and troubleshooting Fire Protection System. If the problem cannot be solved, please contact your supplier.

Table 8-6 Troubleshooting Measures for Fire Protection System

Phenomenon	Possible causes	Check the item or processing method
Mains fail LED is flashing	Power failure	The indicator light at this location is flashing, indicating that the 230V power supply does not exist and the system is running on the backup battery. If the power is not cut off, check the main fuse of the panel.
Batt fail LED is flashing	Battery failure	The indicator light at this location is flashing, indicating that the backup battery is disconnected or there is a fault in the charging circuit of the control panel. Check if the two batteries are connected together, disconnect the batteries, and ensure that 24V voltage can be measured.
LED is flashing	CPU failure	The indicator light at this location is flashing, indicating that the CPU failed to execute the code correctly and has been restarted by the system watchdog. The watchdog reset switch must be pressed to clear the CPU fault status. If the watchdog reset switch is pressed

		and the system does not return to normal, the panel may be damaged and the circuit board needs to be replaced.
24V fault LED is flashing	24V battery failure	The auxiliary 24V and R0V terminals provide 500mA, 24VDC power supply for the fire alarm auxiliary equipment. The indicator light at this location is flashing, indicating that the fuse protecting the R0V output has tripped and has exceeded the rated value of this output. The fuse is self-resetting type. Power supply will be restored after the fault is eliminated.
LED is flashing	Low battery power	When the system is running with a battery, the indicator light flashes when the battery voltage is between 21.5V and 20.5V (the minimum battery voltage).
LED is flashing	Communication failure	The indicator light at this location is flashing, indicating a loss of communication with the intermediate relay panel or auxiliary board. Check all the intermediate relays and auxiliary boards for communication faults to determine the root cause of the problem.
LED is flashing	Grounding fault	The indicator light at this location is flashing, indicating that some system lines are connected to the ground. Remove all system wiring and then reconnect them one by one until the ground fault is restored.
LED is flashing	Fuse failure	The indicator light at this location is flashing, indicating that the total rated power of the power supply has been exceeded and the system fuse has started to work. Remove and inspect all loads, reconnect one load at a time, until the circuit fuse with an excessively high rated power trips to determine a circuit fault.
LED is flashing	S1, S2, S3 failure	The indicator light at this location is flashing, indicating that the sounder output is short-circuited or open-circuited. Remove the wiring and reinstall the resistor at the end of the line. Check the wiring of the detection circuit.
LED is flashing	Release fault	The indicator light at this location is flashing, indicating that the release output is short-circuited or open-circuited. Remove the wiring and reinstall the terminal resistor to check for faults in the release circuit.
LED is flashing	Manual release fault	The indicator light at this location is flashing, indicating that the input end of the manual release switch is short-circuited or open-circuited. Remove the wiring and reinstall the terminal resistor, and check for faults in the manual release circuit.
Detectors does not alarm	The detector is damaged	Replace the detector

False alarm from detectors	The detector is contaminated.	Contact the service provider to clean the detector.
	The detector is damaged	Replace the detector.
	Water has entered the detector or the circuit	Water ingress into the detector or circuit causes the fire extinguishing control panel to fail to work properly. Replace the detector.
LED of the gas detector flashes yellow	The cable line fault has been contaminated	Check the circuit of the combustible gas detector.
	Calibration is required.	If it shows that calibration is needed, then calibrate it.
	Others.	Replace the detector.
Fan fault feedback alarm	The fan is working abnormally.	The fan did not rotate when the equipment was started, and it did not rotate when the fan was separately connected to a 220V power supply for testing. Replace the fan.
		The fan did not rotate when the equipment started, and the fan was tested normally when it was separately connected to a 200V power supply. Replace the electronic control.
	Abnormal louvers	The start-up louver of the equipment does not open, and the motor does not open when it is connected to a 24V power supply alone. Replace the fan.
		The start-up louver of the equipment can only be opened halfway. Replace the fan.
		The start-up louver of the equipment does not open, and the motor is tested normally when connected to a 24V power supply alone. Replace the electronic control.
	Wiring error	The incorrect connection of the circuit led to internal damage of the electronic control system. Replace the electronic control.
The appearance of the fan is damaged.	Fall, collision	Replace the entire machine

9. Operation and maintenance

9.1 System Operation

9.1.1 Fire Operation

The panel is divided into two parts. One is the control and indicating equipment part of EN54-2 standard, including three areas; Another is EN12094-1 extinguishing agent system, which separates extinguishing agent state from disabled state for clarity.

The button control and programming facilities of the two parts are shared.

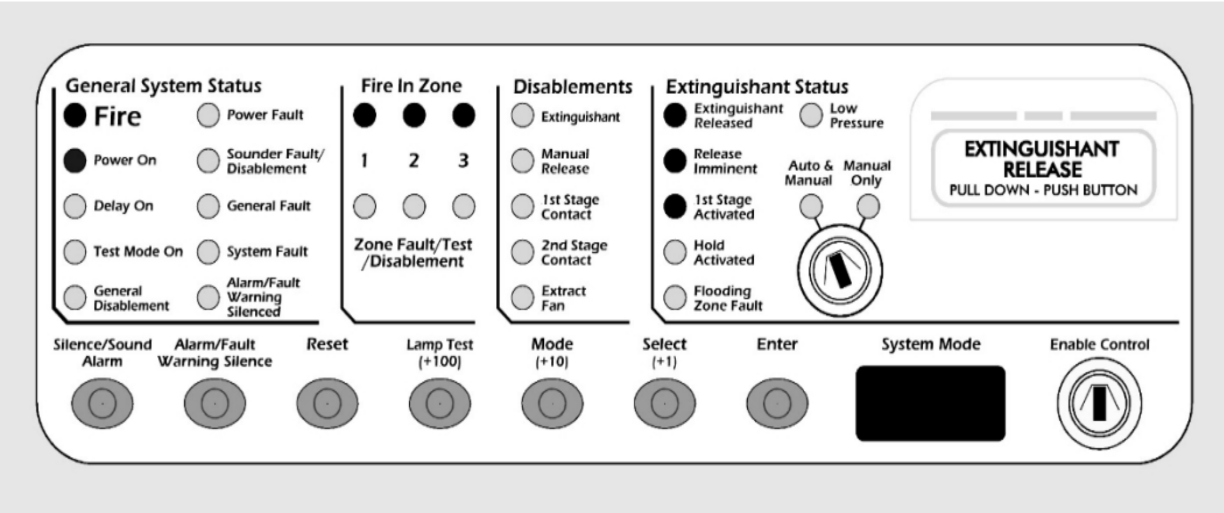


Fig. 9-1 Panel of Fire Fighting Main Engine

(1) Level 1 alarm

When the fire extinguishing controller is in normal monitoring mode, it is set to automatic mode. Use smoke temperature gun to trigger any smoke detector or temperature detector, and observe the system state: the fire extinguishing controller enters the first-class alarm state, the first-class alarm indicator flashes, the buzzer sounds, and the alarm bell rings.

Fig. 9-2 Detector Induction Test

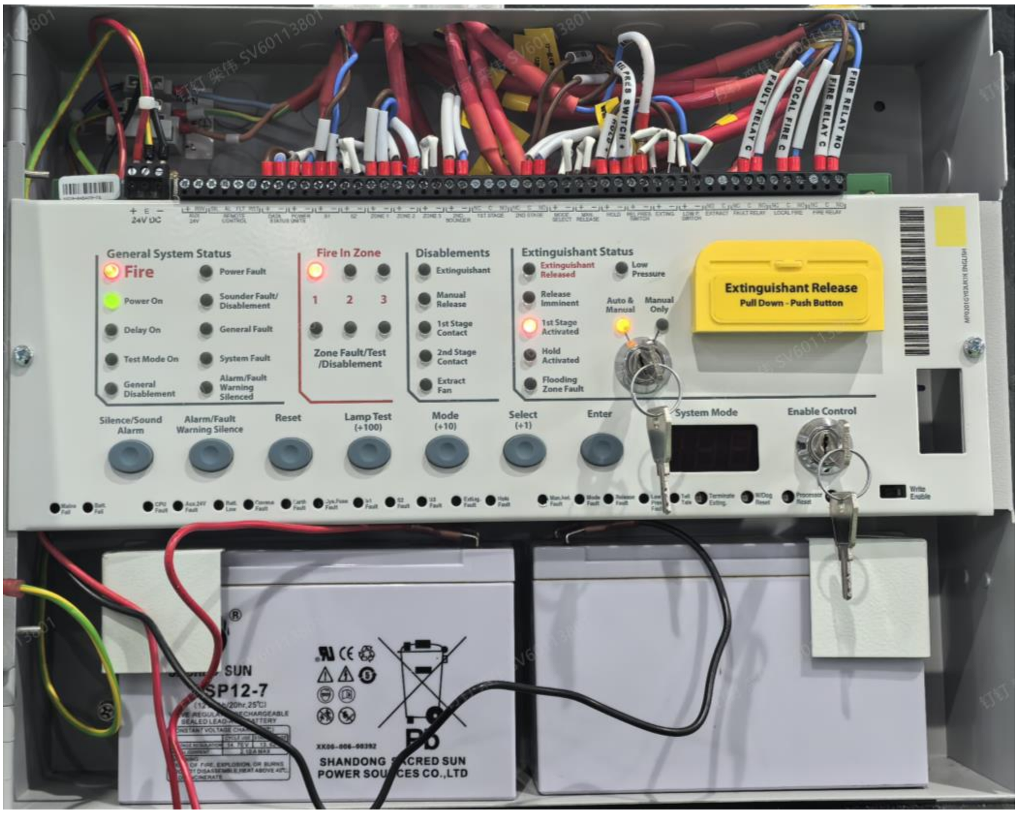


Fig. 9-3 Schema of Level 1 Alarm of Fire-fighting Main Engine

In the first-level alarm state, press the reset button on the panel or press the Processor Reset button on the panel to restore the fire extinguishing controller to the normal monitoring state.

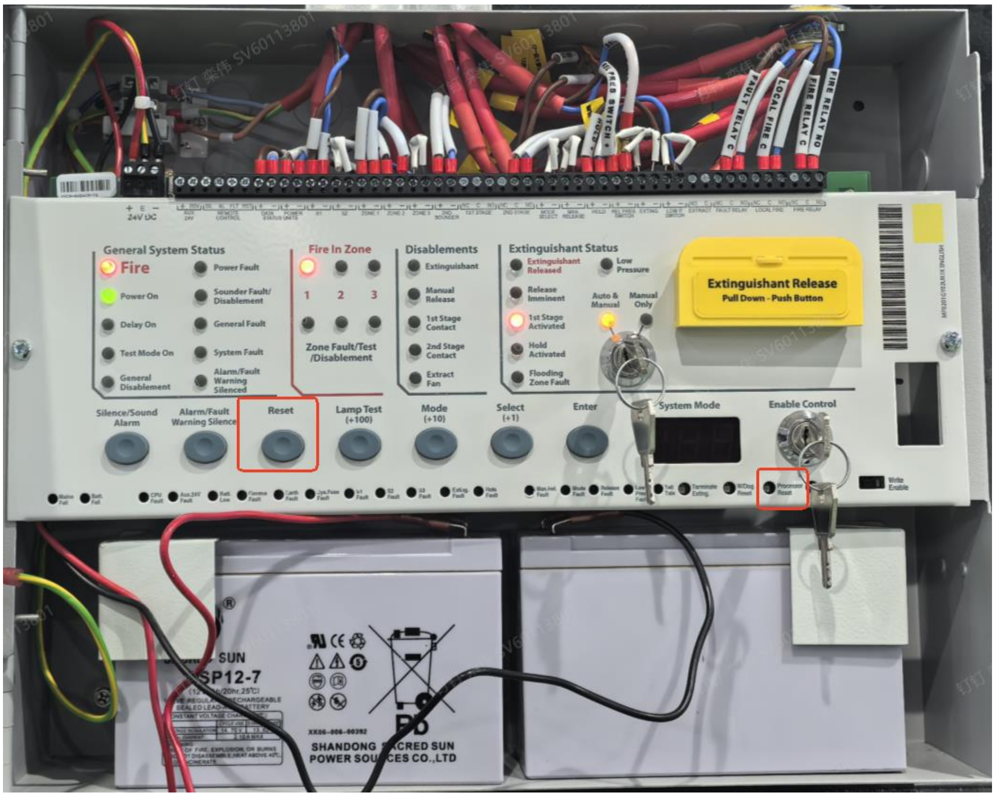


Fig. 9-4 Reset button of fire fighting main engine

（2）Secondary alarm

When the fire extinguishing controller is in normal monitoring mode, it is set to automatic mode. Use a smoke temperature gun to randomly trigger a set of smoke detectors and temperature detectors (Trigger the smoke and temperature detectors both). Observe the system status: the fire extinguishing controller enters the secondary alarm state, the secondary fire alarm indicator flashes, the buzzer of the fire extinguishing controller sounds, the alarm bell rings, the sound of acousto-optic fire alarm flashes, and the display screen of the fire extinguishing controller displays the countdown of 30 seconds gas release.

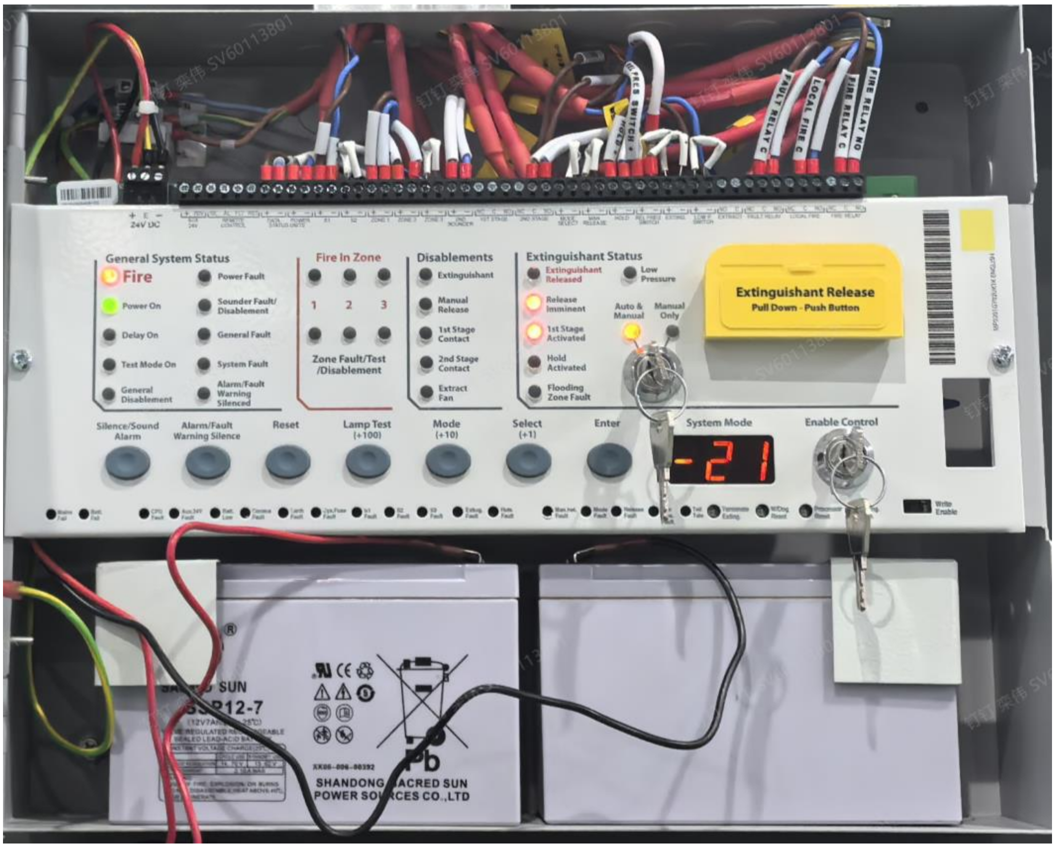


Fig. 9-5 Schema of Level 2 Alarm of Fire-fighting Main Engine

In the secondary alarm state, the "Reset" button cannot be used for Reset. Only the manual reset button (Processor reset) can restore the fire extinguishing controller to the normal monitoring state.

(3) Manual release

When the fire extinguishing controller is in normal monitoring mode, press the manual release button to observe the state of the fire extinguishing controller: the fire extinguishing controller enters the secondary alarm state, and the display screen enters the 30-second gas release countdown.

Figure 9-6　Manual Release Button

Figure 9-7　Manual Release Status

If fire simulation test is carried out for this operation, it must be disconnected from aerosol fire extinguishing agent before starting.

(4) Emergency stop

When the fire extinguishing controller is in the secondary alarm state, the display screen of the fire extinguishing controller will displays the countdown of 30 seconds gas release. Press the emergency stop button for a long time, and the countdown will resume and remain at 30s. After releasing the emergency stop button, the countdown will resume for 30s. The emergency stop button of this European standard system can be used many times.

Figure 9-8 Emergency Stop State

9.1.2 Operation of liquid chiller

(1) Home interface

The unit is equipped with a controller. The control layout is displayed by touch panel, and users can conveniently query, set up and monitor through the touch screen to ensure the normal operation of the unit.

The standby main interface of water cooling unit is shown in Figure 9-9.

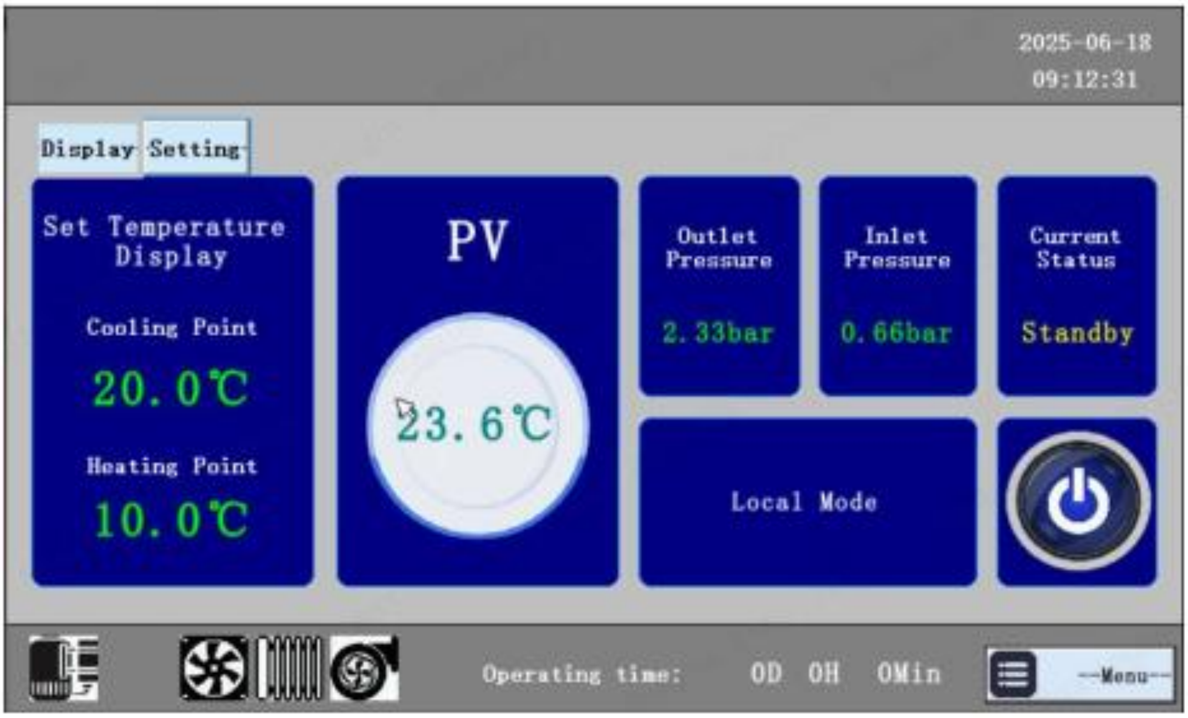
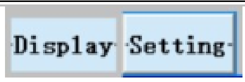
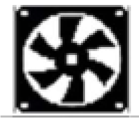
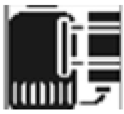
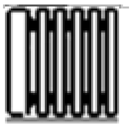


Fig. 9-9 Home interface

The buttons at the control panel have the following function:

Table 9-1　Icon of interface

S/N	Icons	Description
1		Enter “user identification”interface to select user, project or manufacturers.
2		Output, input, historical fault, and system version information can be queried.
3		In case of system fault,flashes in the main interface, and click to query the current fault information. In case of normal operation, the icon is hidden.
4		Used to indicate the running state of the fan.
5		Used to indicate the running state of the press.
6		Used to indicate the running state of the electric heating.
7		Used to indicate the running state of the pump.
8		Start / stop unit key.

（2） Fault interface

In case of fault, the button  is flashed in the main interface, now click the button to mute the sound and enter the following interface:

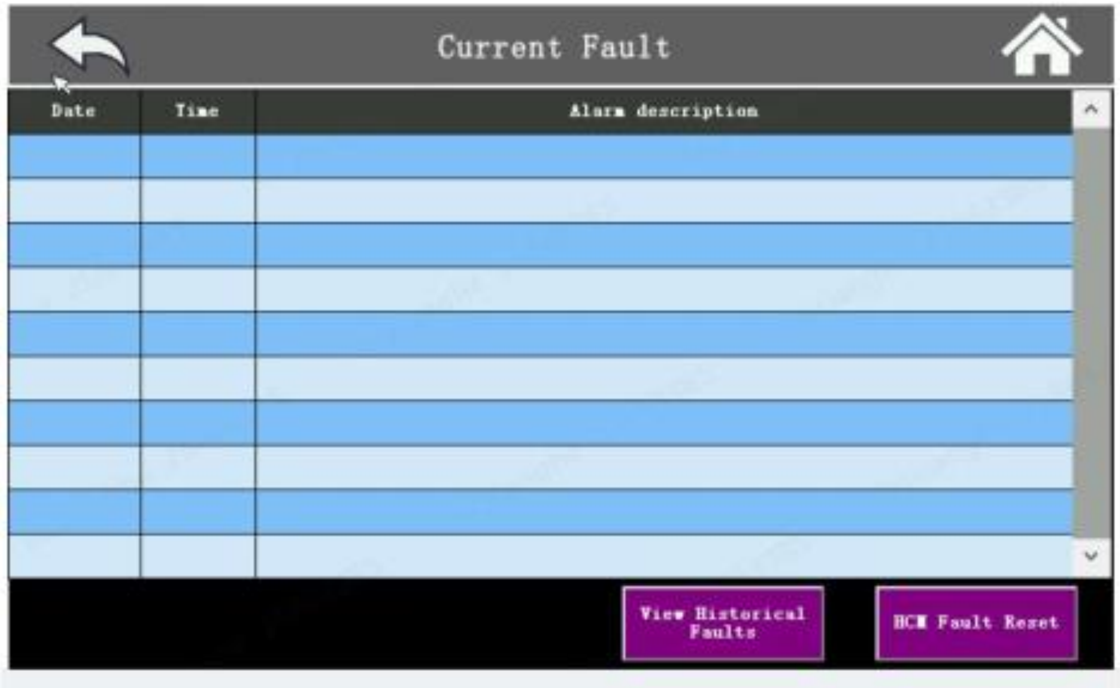


Fig. 9-10 Fault Interface

(3) Menu user interface

Fig. 9-11 Menu user interface

In the main interface, click 

into the menu interface. In the menu interface, click the  into the data query interface, click  into the user menu interface, click  to return to the main interface.

(4) Query interface

1. Data inquiry

View the current probe temperature and pressure value of the system.

Fig. 9-12 Data inquiry

View the current operating information of the electric expansion system, compressors, fans and pumps.

Fig. 9-13 Operating information

2. Input inquiry

Fig. 9-14 Input inquiry

When the light is gray  , the corresponding switch input is invalid.

When the light is green  , the corresponding switch input is valid.

3. Output inquiry

Fig. 9-15 Output inquiry

When the lamp is gray  , it means that the corresponding relay has no output.

When the light is green  , indicating that the corresponding relay is output.

4. Remote key parameters inquiry

Fig. 9-16 Remote key parameters inquiry

It is used to query some local parameters of the liquid cooler issued by remote communication, which is convenient for checking the use of communication. Please refer to the protocol for specific parameters.

This interface only shows some key parameters.

9.2 System Maintenance

In the whole life, abnormal operation must be avoided, and regular maintenance must be carried out according to the specifications to ensure the safe and reliable operation of the system and obtain the best performance.

Comprehensive annual preventive maintenance of energy storage containers is the minimum requirement. It is recommended that preventive maintenance of the system be carried out twice a year, a basic semi-annual inspection and an annual general inspection. Maintenance objects include all equipment and components in energy storage containers, including but not limited to battery packs, liquid cooling units, air conditioners, control cabinets, bus cabinets, etc.

Where there are other provisions on preventive maintenance in laws and regulations, such provisions shall prevail.

If you need to open the battery compartment during the process, the door operation is as shown in the following figure. It should be noted that the hatch is heavy and at least two people are needed to open the door. After the door is opened, fix it with door panel or brace to prevent the door from moving.

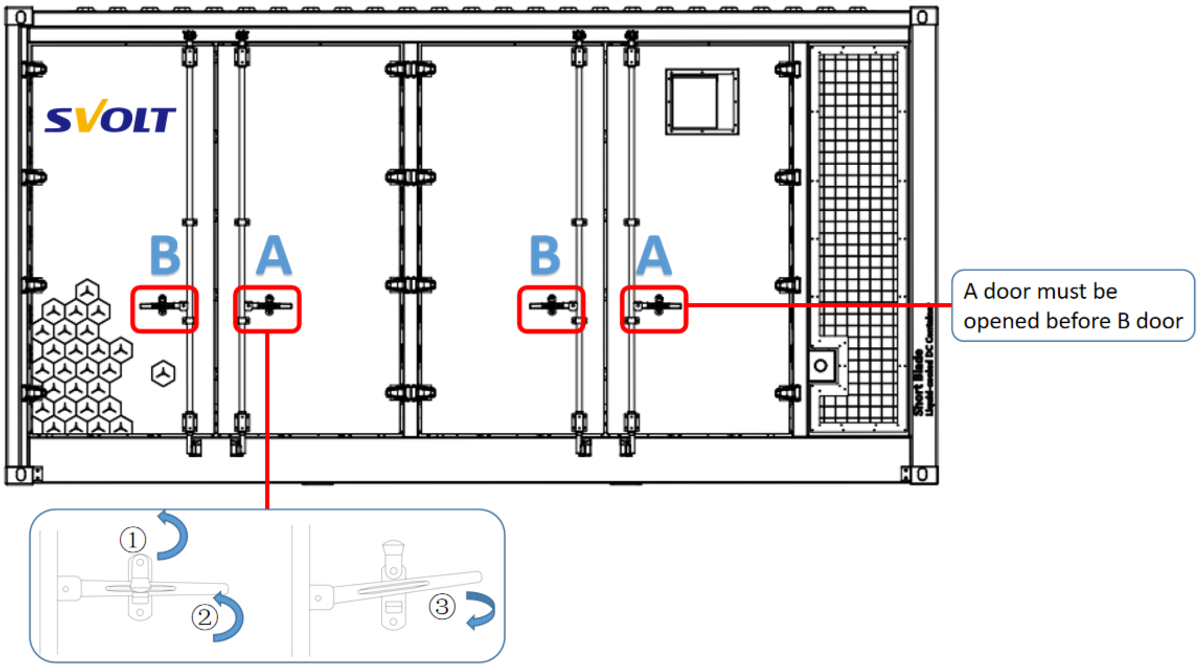
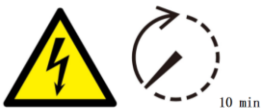


Fig. 9-17 Container Battery Door

9.2.1 Electrical maintenance

Electrical maintenance of bus control cabinet and indirect wiring of equipment should be carried out once every six months.



Delayed operation!

- The system is not running, and there may still be dangerous voltage inside. After disconnecting the power supply, there may be undischarged charge on the internal bus, so it is necessary to let the energy storage converter stand for more than 10 minutes before internal maintenance. Check with voltmeter before maintenance to ensure that the power supply is turned off and in a safe state.

- Do not wear easily conductive objects such as rings and watches during electrical maintenance operations.

1) Equipment cleaning

a) When cleaning the equipment, the equipment should be cut off and the equipment with heat dissipation holes should be well shielded;

b) Use a vacuum cleaner to suck out dust, debris, etc. It is forbidden to use compressed air to blow the equipment;

c) Check whether the wiring is loose after cleaning, and repair it in time if there are problems.

2) Wiring inspection

a) Check whether the cable joints are blackened at the terminals and whether the contacts have burning marks, and the existing problems need to be repaired in time;

b) Check whether there is discoloration in the appearance of copper bar, and timely repair is needed if there are problems;

c) Check whether the cable and heat shrinkable pipe are damaged, and the existing problems need to be repaired in time;

3) Insulation and grounding tests

a) Test the insulation of the system, check the insulation of equipment and cables, and repair the existing problems in time;

b) Measure the grounding, whether the grounding resistance is in the normal range, if there is any problem, it needs to be repaired in time

4) Power-on test

a) Whether the sound of relay suction is normal, If abnormal, It need to be repaired in time.

b) LED indicator lights, relays, switching power supplies and other devices are normally lit after power-on, if abnormal, It need to be repaired in time.

c) Use a multimeter to measure whether the voltage is within the normal range, and if there are problems, it needs to be repaired in time;

5) UPS Battery Maintenance Requirements

a) If the energy storage product is not under operated for a long time, please charge the UPS battery first and charge the battery every 3 months for no less than 10 hours each time; In high temperature environment, when the battery is not charged and discharged for two consecutive months, it needs to be charged once, each time shall not be less than 10 hours.

b) When the battery has not been charged or discharged for 3 months, it should be charged once for not less than 10 hours each time.

c) Batteries should not be replaced individually, and the instructions of the battery supplier should be followed when replacing them as a whole.

d) UPS batteries normally have a life of 3 to 5 years and must be replaced if they are found to be in poor condition. Battery replacement must be performed by a professional and contact your supplier if necessary.

If you need to replace UPS, stop charging and discharging in advance and shut down the power, following steps:

- Disconnect UPS load mini-empty QFA1 ~ QFA5 in control cabinet,
- Press the UPS shutdown button for a long time to shut down the UPS;
- Disconnect the UPS load micro empty QFZ in the control cabinet,
- Disconnect the mini-open QF2 ~ QF7 in the control cabinet;
- Use multimeter and visual inspection to check whether each component is electrically neutral.
- Remove the screw from the front fixing plate;
- Gently pull out the UPS;
- Unplug the AC input and output cable plugs on the back of UPS, and the fixed wiring harness should not be misplaced and moved, and the wiring should be used when replacing UPS;
- Unplug the battery input cable plug on the back of UPS, and the fixed wiring harness should not be moved in dislocation, and the wiring should be used when replacing UPS;
- Unplug the communication line plug on the back of UPS, and fix the wiring harness without dislocation, and use the wiring when replacing UPS;
- Remove the UPS;
- Insert half of the new UPS into the control cabinet;
- Insert the AC input/output cable plug of UPS into the new UPS;
- Insert the battery input cable plug and communication cable plug of UPS into the new UPS;

- The new UPS has been plugged in;
- Fix the new UPS with screws;
- Complete the replacement of UPS.

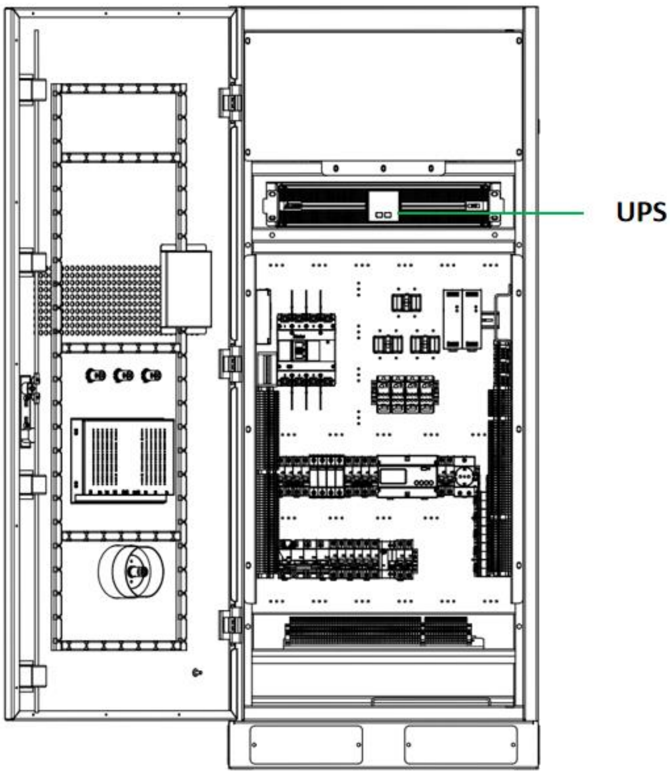


Figure 9-18 UPS Location Diagram

6) Auxiliary electrical maintenance

Check the working status of Chiller, lamps and other electrical appliances in this product. If there is any problem, replace it in time. Please follow the spare parts list for replacing product specifications. The replacement process must be operated by professionals. If necessary, please contact your supplier.

9.2.2 Battery system maintenance

(1) Normal operation system

Batteries in the system shall be maintained in accordance with the following schedule to prevent battery unbalance and control system calculation deviation, ensuring that at least one charge and discharge cycle of full and empty is met every month.

① Option 1 (when the system SOC is low)

Step 1: Discharge the battery system to the off state with rated power, and let it standby for 1 hour.
Step 2: Charge the battery system to the off state with rated power, and let it standby for 1 hour.
Step 3: Discharge the battery system to the SOC required for operation, and continue to put it into operation according to the operation requirements of the project.

② Option 2 (when the system SOC is high)

Step 1: Charge the battery system to the off state with rated power, and let it standby for 1 hour.
Step 2: Discharge the battery system to the off state with rated power, and let it standby for 1 hour.
Step 3: Charge the battery system to the SOC required for operation, and continue to put it into operation according to the operation requirements of the project.

(2) Idle system

Recommended battery storage capacity range: 30% to 50%.
If the battery system is idle for a long time and is not used, the system should be charged and discharged once every 3 ~ 6 months. The operation method is the same as that introduced in the previous section to ensure the power and performance of the system.
A full charge is required to perform a state of charge (SOC) calibration when the system is restored from idle mode to operational mode. To prevent SOC deviation during storage from affecting operation.

(3) Spare battery

If a spare battery is equipped as a spare part, the spare battery should be maintained every 12 months. Maintenance methods refer to the above section to ensure the power and performance of the system.
BMS equalization function can't work during storage, and due to the long maintenance time interval of spare batteries, there may be consistency problems, so it is necessary to consider monomer equalization of batteries with abnormal consistency. When the battery is emptied, the consistency of the battery should be evaluated according to the OCV curve table. If there is inconsistency between single cells, the SOC imbalance (i.e. the difference between the SOC corresponding to the average voltage and the SOC corresponding to the lowest cell voltage) is

greater than 5%, and equalization maintenance should be carried out first (equipment used: battery pack-level charging and discharging machine and equalization machine). After the balancing work is completed, continue the unfinished maintenance steps.

(4) Battery module replacement

Battery module replacement refer to below content.

① Preparation phase

1. Ensure that all tools and batteries are ready, tooling and cranes are in place, and sliders are prepared according to site conditions. Personal protective equipment is complete. When operating, protective measures must be taken, such as wearing insulating gloves or shoes.

② Power down

Power down the system before replacement. Please refer to "6.3 Power down Operation" for system power down steps.

③ Open the door

Open the hatch to the locked position and secure the hatch.

④ Remove a cable

Remove the communication harness which connects the high voltage box and the battery PACK to be replaced.

Remove the positive and negative cables Which connects the high voltage box and the battery PACK to be replaced.

Remove the power cable between PACK.

⑤ Drainage

Visually check the appearance of PACK and close the inlet and outlet branch pipe valves on the left and right sides of the cluster where PACK is located.

Please refer to "9.2.4 Thermal Management System Maintenance" for drainage procedures.

⑥ Remove PACK

Move the lifting platform tooling to the front of the hatch door, and adjust the height of the fork to align with the PACK.

Pull out the PACK to the fork tooling, and pay attention to the safety of operation because PACK is heavy.

Figure 9-19 Tooling disassembly and assembly PACK

⑦ **PACK hoist**

PACK lifting and transfer: Fix the top safety clip on the crane hook, and the hook connects the side hanging hole of PACK for lifting and transportation.

Figure 9-20 Schematic diagram of PACK lifting point

⑧ **PACK replacement**

Operate the above steps in reverse sequence, lift the new PACK to the position to be replaced, and push the new PACK into the empty slot to complete the replacement.

After replacement, the liquid cooling pipeline, cable harness, etc. shall be installed and restored in reverse order according to the above steps, and the liquid injection shall be completed. Refer to "9.2. 3 Thermal Management System Maintenance" for liquid injection steps.

The wiring diagram of single-row 3-cluster battery is as follows:

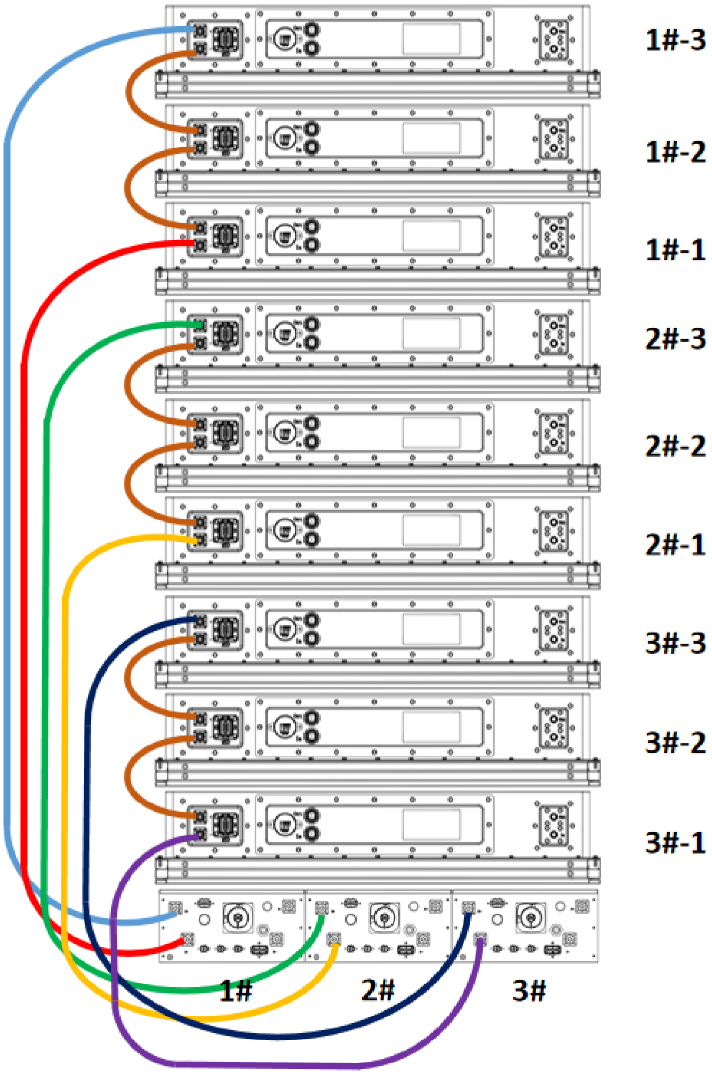


Fig. 9-21 Schematic diagram of single row battery cluster

(5) Replacement of high-volt box

Replacement steps:

Step 1: Turn the control cabinet and the high-volt box to the "OFF" position;

Step 2: Unplug DC cable, power cable and communication cable;

Step 3: Unscrew the fixing screw (as shown in the red box in the figure);

Step 4: Pull out the high-pressure box forcibly.

Step 5: Replace the new high-volt box and secure it, and install the recovered and removed cable harness.

Tools: Tighten gun, sleeve (10mm, 13mm), torsion gun QXXD5AT040PS08 (8-40N. m).

Fig. 9-22 Position of High Pressure Box

Fig. 9-23 Fixed Position of High Pressure Box

9.2.3 Fire protection system maintenance

(1) Periodic maintenance shall include the following

- There is no abnormal change in the appearance of the equipment;

- Signs are in good condition;
- All components of the equipment shall be free from collision deformation and other mechanical damage, the surface shall be free from corrosion, the protective coating shall be intact, the nameplate shall be clear, the position of the manual operation device shall be correct, and the protective cover, lead seal and safety sign shall be complete;
- There should be no abnormal alarm information of fire communication on the display screen.
- Fire extinguishing agent storage container and all system components, such as container valve, connecting pipe, valve actuator, nozzle, signal feedback device, etc. No mechanical damage such as collision deformation, no corrosion on the surface, intact protective coating, clear nameplate and signboard, and intact protective device, sealing device and safety sign of manual operation device.
- The bracket and hanger between equipment, fire extinguishing agent delivery pipeline and storage unit shall be fixed without loosening.
- Connecting pipes shall be free from deformation, cracks and aging. If necessary, send it to the statutory quality inspection agency for testing or replacement.
- Each nozzle hole should be free of blockage.
- When the fire extinguishing agent conveying pipeline is damaged and blocked, sealing test and blowing shall be carried out according to relevant regulations.

If the fire simulation alarm test begins, ensure that the starting lines of all detectors are disconnected from the fire extinguisher.

Table 9-2 Fire protection system Maintenance List

S/N	Object	Check items	RC

1	Aerosol fire extinguishing device	1. Check whether the aerosol fire extinguishing device body is damaged. (Visual inspection, including: if there is serious wear, rusting and corrosion, or other factors that may affect its functionality such as broken nozzle stickers) 2. Check whether the aerosol fire extinguishing device has expired. 3. Check whether the fixing bracket of aerosol fire extinguishing device is loose. 4. Check the battery compartment structure protected by aerosol fire extinguishing system to eliminate the adverse effects that may lead to the change of fire extinguishing volume.	6 months
2	Fire-fighting main engine	1. After the power supply is turned on, there should be no fault indicator light on the fire alarm control panel. 2. Disconnect the activation line of aerosol fire extinguishing device, simulate fire alarm signal or activate manually, and observe whether the linkage of equipment is normal.	6 months
3		1. Check the appearance for signs of deformation, rust and aging. 2. Check whether the installation is safe.	6 months
4		1. Check the production month/year mark on each battery. If the replacement date specified by the manufacturer is exceeded, the battery needs to be replaced. 2. Check whether the battery connection is tight. 3. Check whether the terminal block is corroded, whether the shell is deformed, whether the battery is cracked or leaked, etc. If the battery is found to have corrosion, deformation, liquid leakage and other phenomena, it is necessary to replace the battery. 4. Check the battery voltage.	6 months
5	Smoke detector	1. Check whether the working state is normal.	6 months
6	Temperature detector	1. Check whether the working state is normal.	6 months
7	Gas detector	1. Check the appearance for deformation, rust, aging, etc. 2. Check whether the installation is safe.	6 months
8	Sound-light alarm	1. Check the appearance for signs of deformation, rust and aging. 2. Check whether the installation is safe.	6 months
9	Alarm bell	1. Check the appearance for deformation, rust, aging, etc. 2. Check whether the installation is safe.	6 months
10	Tight stop	1. Check the appearance for deformation, rust, aging, etc. 2. Check whether the installation is safe.	6 months
11	Manual release	1. Check the appearance for signs of deformation, rust and aging. 2. Check whether the installation is safe.	6 months
12	Electric shutter Explosion-proof fan	1. Check the appearance for deformation, rust, aging, etc. 2. Check whether the installation is safe.	6 months

(2) Extinguishing agent should be changed periodically

The service life of aerosol is 15 years. In the process of use, if aerosol is found to have one of the following conditions, it should be replaced after following the manufacturer's advice:

- Serious corrosion or damage or doubts about its safety and reliability.
- Not used for more than 15 years.
- Abnormal appearance.

9.2.4 Thermal management system maintenance

(1) Maintenance instructions of liquid cooler

- It is strictly forbidden to let the liquid cooler equipment run without coolant;
- It is strictly forbidden to let personnel contact and maintain the equipment when the equipment is running;
- It is strictly forbidden for non-professionals to disassemble.
- The unit is strictly forbidden to be squeezed by heavy objects.
- Use a vacuum cleaner every 6 months to clean the condenser, radiator and air inlet and outlet of the equipment (without water).

Table 9-3 Thermal management system Maintenance List

S/N	Object	Check items	RC
1	The reliability of power cables and signal cables on the wiring panel	The electrical cables and signal cables are not loose	6 months
2		The electrical cables and signal cables are free from aging, damage, abnormal heating and other abnormalities	6 months
3		There is no dust at the wiring panel	6 months
4	Maintain the normal operation of the air switch	It will automatically disconnect when there is an abnormality in the circuit (such as a short circuit)	6 months
5	Appearance of the unit	The unit is clean, dust-free and free of dirt	6 months
6	Reliability of fan operation	The fan is free of dust and there are no foreign objects blocking the air outlet	6 months
7		The fan blades are undamaged, and the fan rotates smoothly without any abnormal noise	6 months
8	Condenser cleaning	The condenser is free from dust and foreign object blockages	6 months

9	Coolant	The concentration complies with the range requirements. The PH and the concentrations of each electrolyte meet the requirements. No dirt, sediment or algae are produced	6 months

(2) Coolant maintenance instructions

The coolant has a life span. It is recommended to replace the coolant every five years, otherwise the quality of the coolant will drop below any of the following indexes.

- The PH value is between 7.5 and 8.5;
- Chloride concentration is greater than 60ppm;
- Color: The coolant is obviously darkened;
- Appearance: Turbid coolant containing obvious impurities such as precipitated particles or flocculent impurities.

Table 9-4 Instructions for coolant replacement

Coolant type	Evaluation index	RC	Tools
50% ethylene glycol	The PH value is between 7.5 and 8.5	5 years	Water pump, hose, hose clamp, straight screwdriver Note: Please contact the supplier for replacement.
	Color: The coolant is obviously darkened		
	Appearance: Turbid coolant containing obvious impurities such as precipitated particles or flocculent impurities		

PH value test steps:

Step 1: Shut down the system, disconnect the AC power supply and DC switch;

Step 2: After waiting for ten minutes, open the battery door.

Step 3: Open the drain valve at the end of the main pipe and gently turn the valve (do not open it completely) to release about 10mL of coolant.

Step 4: Test the coolant sample with a pH test paper or pH meter to obtain the current pH value of the coolant. When the coolant pH value is less than 7.3, it is necessary to replace and maintain the coolant (system drainage + system replenishment).

Coolant can only use the model provided by approved suppliers. It is not recommended to mix various types of coolants. If you need coolants, please contact the manufacturer.

Safety considerations for replacing liquid coolant

- Any personal injury caused by misoperation should be dealt with immediately, and emergency first aid or medical care must be obtained immediately.
- Please strictly abide by the safety regulations in your area to prevent accidents. The safety precautions outlined in this manual do not cover all safety matters and are only supplementary to local safety regulations.
- No modification of the equipment beyond the specification is allowed without prior consent and consent.
- Persons and children with limited cognitive and coordination abilities are not allowed to operate, maintain, clean or use equipment as toys.
- Do not install or remove the power cord live. Please turn off the power switch before installing and disassembling the power cord.
- Please carry out maintenance operation after turning off the power supply, and do not turn on the power supply during operation. For some operations that require operation while running (such as water chillers), make sure all equipment is wired correctly before switching on the power supply.
- Protective measures must be taken during operation, such as wearing insulating gloves or shoes.
- The coolant in liquid cooled energy storage system is toxic. Avoid oral administration or prolonged skin contact. If you accidentally enter your eyes, please rinse with clear water immediately and seek medical advice immediately.
- Coolant recovery and treatment
- Do not discard coolant and packaging materials at will. Please refer to the relevant laws and regulations of the region for disposal methods.

(3) Description of battery cluster drainage

When only one column PACK of the battery system needs to be replaced, a single column branch drainage can be carried out.

The operation steps are as follows:

1. The power failure of the system is completed;
2. Close the inlet and outlet globe valves (open when the valve is parallel to the pipeline and close when it is perpendicular to the pipeline);

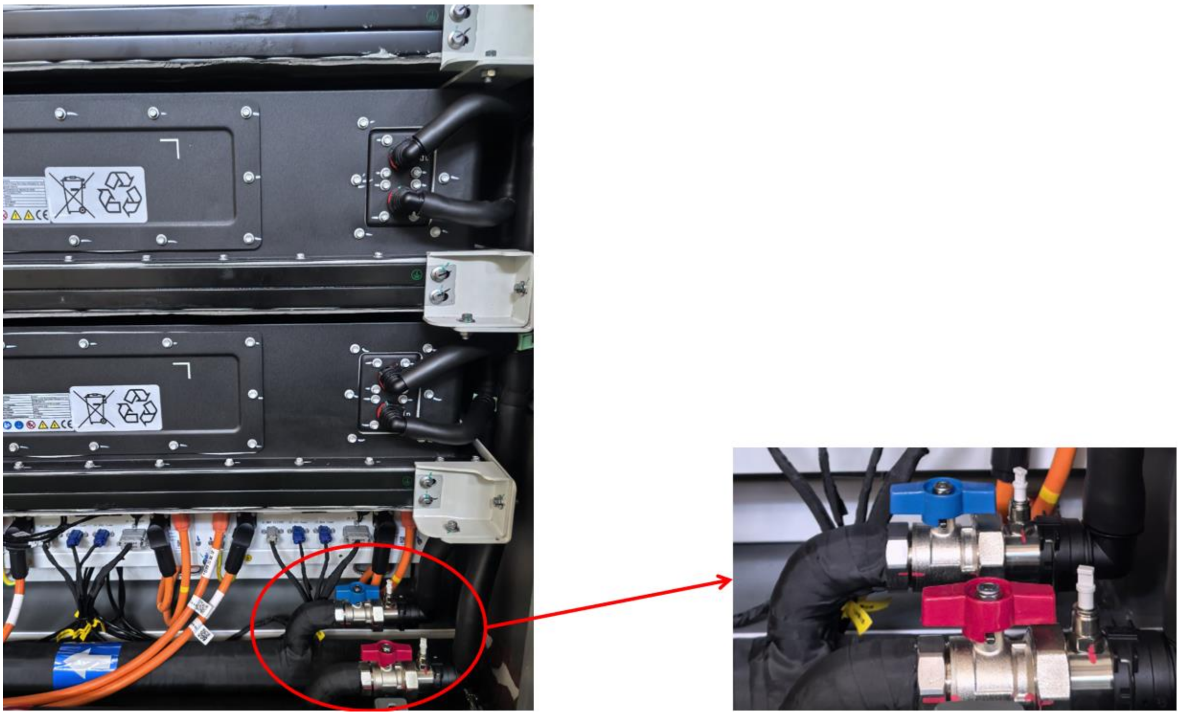


Fig. 9-24 Single-row branch line stop valve

3. Open the automatic exhaust valve located at the highest point of the connecting pipeline;
4. Connect one end of the drain pipe to the external reservoir, and then connect the other end of the drain pipe to the lowest drain port of the liquid cooling pipe (pay attention to the connection sequence, because incorrect sequence will cause coolant to flow outside the reservoir);
5. If no liquid flows out of the drainage tube within 30 seconds, it is considered that the system drainage is completed;
6. After drainage, close the automatic/manual exhaust valve at the highest point of the connecting pipeline, disconnect the connection between the pipeline and the drainage valve.
7. Single branch drainage operation is completed.

(4) Description of whole system drainage

When replacing the whole system coolant or replacing a large number of packs, the whole system needs to be drained.

1. The power failure of the system is completed;
2. Open the chiller compartment and remove the maintenance cover plate at the bottom of the chiller with a screwdriver;
3. One end of the drain pipe is connected to the drain outlet of the water machine, and the other end is inserted into the coolant storage barrel;
4. Open the ball valve at the drain outlet of the water machine and start draining;
5. If no liquid flows out of the drain pipe within 30 seconds, it is considered that the system drainage is completed;
6. Remove the drain tool.

(5) Description of system liquid injection

Liquid injection is generally used for the first operation of coolant to the liquid-cooling unit. The operation method is as follows:

1. Open the liquid cooling unit compartment and use a screwdriver to remove the maintenance cover plate at the bottom of the liquid cooling unit.
2. Check the status of each valve in the system.
   a. Confirm that the main valve of the pipeline leading from the refrigeration machine to the battery is open.
   b. Confirm that one exhaust valve at the upper end of the pipeline is in the open position.
   c. Confirm that the valve corresponding to one drain outlet is closed.
3. Connect the outlet pipe of the liquid injection pump to the liquid injection port of the liquid-cooled unit, and connect the inlet pipe of the external liquid injection pump to the storage tank of the coolant.
4. After opening the front lower panel, open the ball valve of the built-in liquid injection port.
5. Connect the power supply of the chiller to put the entire machine in standby mode.
6. Turn on the liquid injection pump and fill it with coolant.
7. Observe the fluid infusion pressure. Stop the fluid injection when the fluid infusion pressure reaches 1.0bar.
8. Open the upper cover of the liquid replenishment tank and add coolant to it. When the liquid level in the liquid injection tank reaches three quarters of the sight glass level (if the tank is equipped with a sight glass), stop adding coolant to the tank and open the ball valve of the liquid

injection tank.

9. Close the ball valve on the liquid injection port of the chiller, and then remove the external liquid replenishment pipeline.

10. Set the chiller to self-circulation mode and let it run for half an hour. When the external flow and pressure stabilize, it can operate normally. After operation, the pressure range of the return water port is 0.5 to 1.0bar, and the pressure range of the outlet water port is 2.0 to 2.5bar.

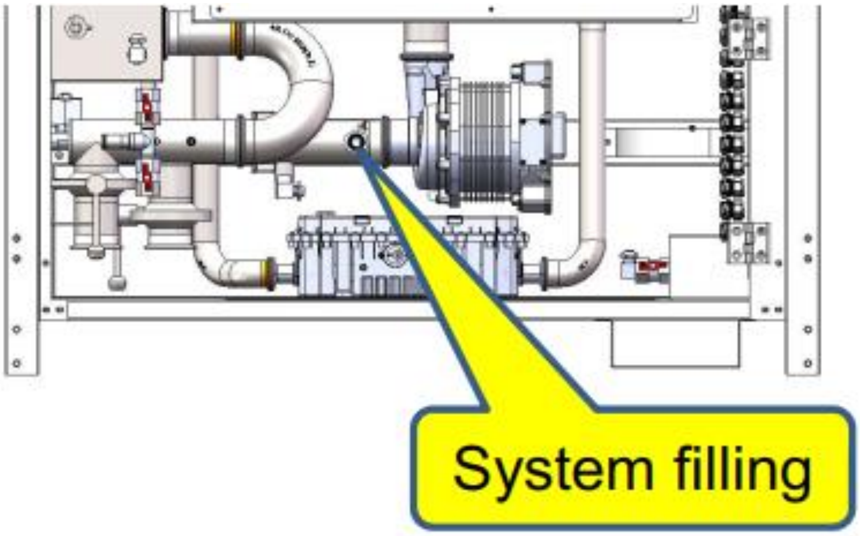


Fig. 9-25 Liquid injection port

(6) Description of system liquid replenishment

Fluid replenishment refers to the process of adding coolant when it is insufficient during the operation of the liquid-cooling unit. The operation steps are as follows:

1. When the low liquid alarm occurs in the liquid replenishment tank, the liquid replenishment pipeline connected to the coolant storage tank should be linked to the liquid replenishment port for liquid addition. The fluid replacement steps refer to the previous section and are the same as the fluid injection method.

2. Stop replenishing the liquid when it reaches the highest liquid level mark.

Note: It is normal for both gas and liquid to flow out of the exhaust valve simultaneously. If only liquid continues to flow out, you need to close the lower valve of the exhaust valve, manually unscrew the exhaust valve, and rinse the interior with clean water 2 to 3 times. After rinsing, reinstall it.

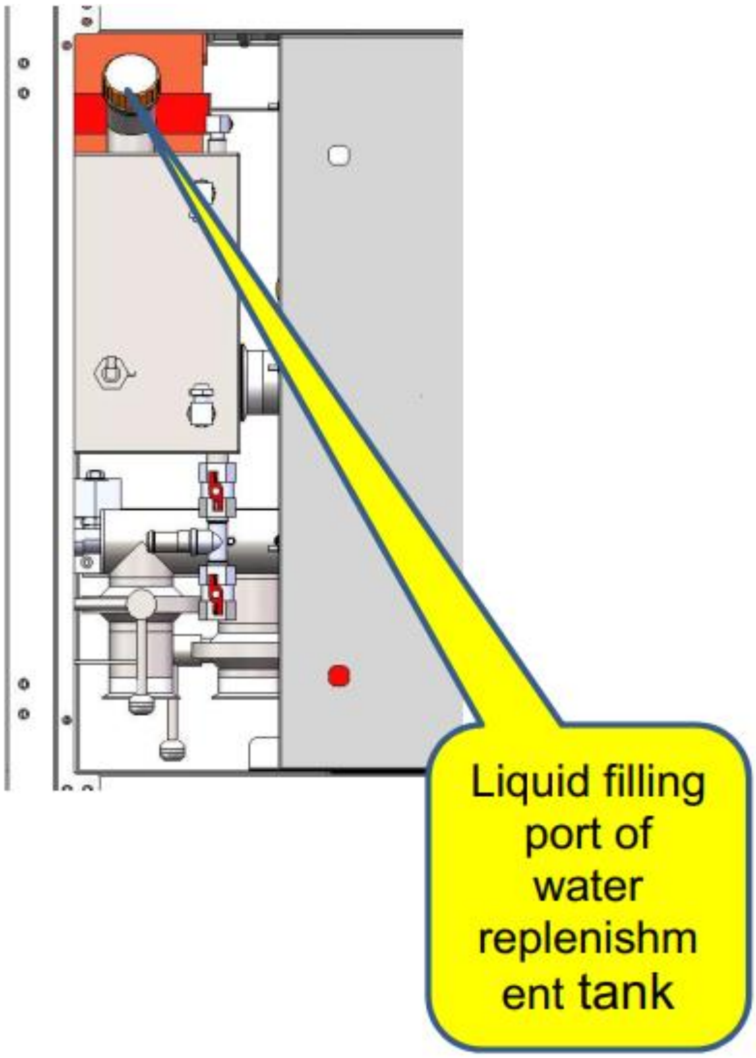


Fig. 9-26 Liquid injection port

9.2.5 Container maintenance

(1) Coating and makeup measures

Check for external damage and select the appropriate procedure according to the degree of damage.

Damage Level 1: Surface dirt caused by water stains and dust can be removed.

① Wet the cleaning cloth (or other scrubbing tools) with water and scrub the dirty parts on the surface.

② If dirt cannot be washed with water, scrub with 97% alcohol until the surface returns to normal. (Or try to use non-corrosive detergents commonly used locally)

Table 9-5 Scrubbing Materials (Case 1)

S/N	Maintenance material	Source
1	Cleaning cloth	Customer prepared
2	water	Customer prepared
3	Alcohol or other non-corrosive detergent	Customer prepared

Damage Level 2: Surface dirt and damaged surface, unable to clean.

① Sand the blistered or scratched paint surface with sandpaper to make its surface smooth.

② Wet the cleaning cloth with water or 97% alcohol and scrub the damaged parts to remove the surface stains again.

③ After the surface is dry, paint and repair the scratched parts with a soft brush, and paint them as evenly as possible.

Fig. 9-27 Schematic diagram of polishing and painting

Table 9-6 Grinding Paint Replenishing Materials (Case 2)

S/N	Maintenance material	Source
1	Cleaning cloth	Customer prepared
2	water	Customer prepared
3	Alcohol or other non-corrosive detergent	Customer prepared
4	Emery paper	Customer prepared
5	Brush	Customer prepared
6	paint	Customer prepared

Damage degree 3: Primer is damaged and substrate is exposed.

① Sand damaged parts with sandpaper to remove rust and other burrs and make the surface smooth.

② Wet the cleaning cloth with water or 97% alcohol and scrub damaged parts to remove surface stains and dust again.

③ After the surface is dried, spray the parts with the substrate exposed with zinc primer for protection. Ensure that the spraying completely covers the bare substrate.

④ After the primer is dry, repair the damaged parts with a soft brush, and brush the paint evenly.

Fig. 9-28 Schematic diagram of polishing and painting

Table 9-7 Grinding paint materials (Case 3)

S/N	Maintenance material	Source
1	Cleaning cloth	Customer prepared
2	water	Customer prepared
3	Alcohol or other non-corrosive detergent	Customer prepared
4	Emery paper	Customer prepared
5	Brush	Customer prepared
6	paint	Customer prepared
7	Zinc primer	Customer prepared

9.2.6 System Inspection

The system should make inspection plan according to the requirements of the station, and make inspection records according to the requirements.

Daily inspection items shall be implemented according to the following points:

Table 9-8 Recommended regular inspection

S/N	Check items	RC
1	It is necessary to monitor the input, output voltage and current and running state of the energy storage container in real time, And fixed-point observation, found abnormal work or abnormal voltage and current need to be maintained in time.	Daily
2	Listen for abnormal noise inside and outside the energy storage container.	Daily
3	Smell whether there is peculiar smell inside and outside the energy storage container.	Daily
4	Read the internal temperature of the system and observe that the temperature is within the normal range.	Daily

Regular inspection items shall be implemented according to the following points:

Table 9-9 Recommended Periodic Inspection

S/N	Check items	RC
1	Check the appearance of energy storage container equipment for signs of damage, rust, paint peeling or oxidation. Clean the dirty surface with water or alcohol, and repair the damaged surface paint.	6 months
2	Check that the ventilation, ambient temperature, humidity, dust and other environments around the energy storage container equipment meet the requirements. Whether there are flammable materials around.	6 months
3	Check the phenomenon of aging and damage of cable. If it occurs, it is necessary to increase corresponding insulation measures or replace cable.	6 months
4	Check that the connection bolts are free of aging and burning marks, and shake them by hand to confirm that they are in a tightened state.	6 months
5	Check whether the external sealant of the energy storage container body is aged and damaged, and replenish the sealant in time. (It is recommended to check every three months)	3 months
6	Check whether the sealing performance of the sealing strip on the door of the energy storage container declines. If there is any damage, please replace it immediately. (It is recommended to check one day after the rain stops, at least once every three months)	3 months
7	Check whether the door lock hinge of the energy storage container opens and closes smoothly. If necessary, lubricate the door lock hole and hinge properly. (It is recommended to do it every three months)	3 months
8	Check whether the air inlet and outlet of container and liquid cooling unit are blocked. If necessary, use brush or cotton cloth to remove dust and dirt from the unit, or replace the air filter screen. (Pay attention to the operation after 10 minutes of power failure)	3 months
9	Check whether the condenser of liquid cooling unit is blocked by dust and foreign matter, and whether the fins are seriously bent and deformed.	6 months
10	Check whether the fan is running normally, whether the fan blades are damaged or deformed, and whether there is any abnormal noise.	6 months
11	Check the cluster frame and internal equipment for any damage or deformation.	6 months
12	Check the bottom of the battery compartment for water. Whether there is accumulated dust in the cabin.	3 months
13	Troubleshoot long-term uninterruptible power supplies that need to be charged every six months.	6 months
14	Check whether the grounding is good and reliable, free from corrosion and looseness.	6 months
15	Check all fasteners for signs of loosening.	6 months
16	Check that all cable entrances at the bottom of the container are well sealed.	6 months

17	Check whether warning signs, labels and nameplates are clear and dirty. Replace them if necessary.	6 months
18	Check whether the coolant of liquid cooling unit is sufficient, and whether there is dirt, precipitation and algae.	6 months

It should be noted that maintenance inspection needs to meet the following conditions:

- Avoid opening the container door in high humidity such as rain, snow or fog, and ensure that the seals around the container door will not curl when the door is closed.
- Make sure that all MSD in the rack are disconnected before maintenance
- When maintaining get start, make sure that the locking structure of the plug is unlocked before operate, so as to avoid damage to the battery pack caused by forced unplugging.

10. Scope of Supply

10.1 Scope of Supply

The supplier（SVLOT） provides the standard Battery Container. Power cables, communication cables, etc. between the Battery Container and other equipment are not within the scope of SVLOT's supply.

Num	product name	product model	Brand	unit	amount	remark
1	Battery Container	5.16MWh	SVLOT	PCS	200	30% SOC shipment

Attention

Due to product version upgrades or other reasons, the content of this document may be updated from time to time. The updated content will not be notified separately.

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